Pancreatic Cancer Biomarker Analysis - Debernardi et al. 2020

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About Dataset

This is a dataset from an open-access paper published December 10, 2020. The paper and the full dataset are open-access (CC-BY), found here:

https://www.kaggle.com/datasets/johnjdavisiv/urinary-biomarkers-for-pancreatic-cancer

Background

Pancreatic cancer is an extremely deadly type of cancer. Once diagnosed, the five-year survival rate is less than 10%. However, if pancreatic cancer is caught early, the odds of surviving are much better. Unfortunately, many cases of pancreatic cancer show no symptoms until the cancer has spread throughout the body. A diagnostic test to identify people with pancreatic cancer could be enormously helpful.

The paper

In a paper by Silvana Debernardi and colleagues, published this year in the journal PLOS Medicine, a multinational team of researchers sought to develop an accurate diagnostic test for the most common type of pancreatic cancer, called pancreatic ductal adenocarcinoma or PDAC. They gathered a series of biomarkers from the urine of three groups of patients:

Healthy controls

Patients with non-cancerous pancreatic conditions, like chronic pancreatitis Patients with pancreatic ductal adenocarcinoma When possible, these patients were age- and sex-matched. The goal was to develop an accurate way to identify patients with pancreatic cancer.

The data

The key features are four urinary biomarkers: creatinine, LYVE1, REG1B, and TFF1.

Creatinine- is a protein that is often used as an indicator of kidney function.

YVLE1- is lymphatic vessel endothelial hyaluronan receptor 1, a protein that may play a role in tumor metastasis

REG1B -is a protein that may be associated with pancreas regeneration

TFF1- is trefoil factor 1, which may be related to regeneration and repair of the urinary tract Age and sex, both included in the dataset, may also play a role in who gets pancreatic cancer. The dataset includes a few other biomarkers as well, but these were not measured in all patients (they were collected partly to measure how various blood biomarkers compared to urine biomarkers).

Prediction task

The goal in this dataset is predicting diagnosis, and more specifically, differentiating between 3 (pancreatic cancer) versus 2 (non-cancerous pancreas condition) and 1 (healthy). The dataset includes information on stage of pancreatic cancer, and diagnosis for non-cancerous patients, but remember—these won't be available to a predictive model. The goal, after all, is to predict the presence of disease before it's diagnosed, not after!

loading the libraries used in the analysis of this package

```
## 'data.frame':    590 obs. of  14 variables:
##  $ sample_id              : chr  "S1" "S10" "S100" "S101" ...
##  $ patient_cohort         : chr  "Cohort1" "Cohort1" "Cohort2" "Cohort2" ...
##  $ sample_origin          : chr  "BPTB" "BPTB" "BPTB" "BPTB" ...
##  $ age                    : int  33 81 51 61 62 53 70 58 59 56 ...
##  $ sex                    : chr  "F" "F" "M" "M" ...
##  $ diagnosis              : int  1 1 1 1 1 1 1 1 1 1 ...
##  $ stage                  : chr  "" "" "" "" ...
##  $ benign_sample_diagnosis: chr  "" "" "" "" ...
##  $ plasma_CA19_9          : num  11.7 NA 7 8 9 NA NA 11 NA 24 ...
##  $ creatinine             : num  1.832 0.973 0.78 0.701 0.215 ...
##  $ LYVE1                  : num  0.89322 2.03758 0.14559 0.0028 0.00086 ...
##  $ REG1B                  : num  52.9 94.5 102.4 60.6 65.5 ...
##  $ TFF1                   : num  654.3 209.5 461.1 142.9 41.1 ...
##  $ REG1A                  : num  1262 228 NA NA NA ...
```

```
##   sample_id         patient_cohort     sample_origin           age
##  Length:590         Length:590         Length:590         Min.   :26.00
##  Class :character   Class :character   Class :character   1st Qu.:50.00
##  Mode  :character   Mode  :character   Mode  :character   Median :60.00
##                                                           Mean   :59.08
##                                                           3rd Qu.:69.00
##                                                           Max.   :89.00
##
##      sex              diagnosis       stage           benign_sample_diagnosis
##  Length:590         Min.   :1.000   Length:590         Length:590
##  Class :character   1st Qu.:1.000   Class :character   Class :character
##  Mode  :character   Median :2.000   Mode  :character   Mode  :character
##                     Mean   :2.027
##                     3rd Qu.:3.000
##                     Max.   :3.000
##
##  plasma_CA19_9       creatinine          LYVE1               REG1B
##  Min.   :    0.0   Min.   :0.05655   Min.   :1.294e-04   Min.   :   0.0011
##  1st Qu.:    8.0   1st Qu.:0.37323   1st Qu.:1.672e-01   1st Qu.:  10.7572
##  Median :   26.5   Median :0.72384   Median :1.650e+00   Median :  34.3034
##  Mean   :  654.0   Mean   :0.85538   Mean   :3.064e+00   Mean   : 111.7741
##  3rd Qu.:  294.0   3rd Qu.:1.13948   3rd Qu.:5.205e+00   3rd Qu.: 122.7410
##  Max.   :31000.0   Max.   :4.11684   Max.   :2.389e+01   Max.   :1403.8976
##  NA's   :240
##       TFF1               REG1A
##  Min.   :5.300e-03   Min.   :    0.00
##  1st Qu.:4.396e+01   1st Qu.:   80.69
##  Median :2.599e+02   Median :  208.54
##  Mean   :5.979e+02   Mean   :  735.28
##  3rd Qu.:7.427e+02   3rd Qu.:  649.00
##  Max.   :1.334e+04   Max.   :13200.00
##                      NA's   :284
```

Load Data and visualise summary of data

To begin my analysis, I examined the structure and summary statistics of the dataset, which contains 590 observations across 14 variables, including both categorical and numerical types. I wanted to understand the general distribution of each variable before selecting appropriate statistical and visualization techniques.

###Age

The variable age is a continuous numeric variable. From the summary: • The minimum age is 26, and the maximum is 89. • The mean age is 59.08, and the median is 60, which are very close—indicating a roughly symmetric distribution. • The interquartile range (IQR), from 50 (Q1) to 69 (Q3), shows that the middle 50% of patients are between their early 50s and late 60s.

This age spread suggests a middle-aged to elderly population, which is typical in studies related to pancreatic cancer, the context of this biomarker study. I later used histograms and boxplots to confirm the distribution and check for outliers.

###Diagnosis

The diagnosis variable is coded numerically (1, 2, 3), representing different clinical states healthy, benign, and cancer. The values range from 1 to 3, with a mean of 2.027 and median of 2.0, which tells me that there is a fairly even representation across diagnostic groups, though slightly more individuals are in category 2 or 3. This makes it suitable for both classification models and for assessing group differences.

###Plasma_CA19_9

The variable plasma_CA19_9 represents a tumor biomarker. From the summary: • The values range from 0 to 31,000, which is an extremely wide range and suggests the presence of strong outliers. • The mean is 654, yet the median is only 26.5, and the 1st quartile is 8, with 240 missing values.

This right-skewed distribution is common with biomarkers in oncology, where elevated values in some cancer patients lead to long tails. I noted this and planned to transform the data or apply robust methods when using it in models.

###Creatinine

The variable creatinine is a standard kidney function biomarker. It has: • A mean of 0.86 mg/dL, with a range from 0.056 to 4.12. • The median is 0.72, and the distribution appears mildly right-skewed.

Since creatinine levels are typically between 0.6–1.2 mg/dL in healthy adults, this spread reflects both healthy and possibly impaired individuals, which could be linked to disease severity.

###LYVE1, REG1B, TFF1, REG1A

These four variables are the core protein biomarkers in this study. Each of them shows: • Very wide ranges (e.g., TFF1 ranges from 0.005 to 13,344, REG1A up to 13,200) • Substantial skewness, with means much larger than medians.

For example: • LYVE1: Mean = 3.06, Median = 1.65. • REG1B: Mean = 111.77, Median = 34.30. • TFF1: Mean = 597.87, Median = 259.87. • REG1A: Mean = 735.28, Median = 208.54, with 284 missing values.

These large differences between mean and median further confirmed strong positive skewness and the presence of extreme values, especially in cancer patients. This insight guided me to standardize the data before applying Principal Component Analysis (PCA) or clustering, to avoid dominance by a few high values.

```
##              sample_id          patient_cohort           sample_origin
##                      0                       0                       0
##                    age                     sex               diagnosis
##                      0                       0                       0
##                  stage benign_sample_diagnosis           plasma_CA19_9
##                    391                     382                     240
##             creatinine                   LYVE1                   REG1B
##                      0                       0                       0
##                   TFF1                   REG1A
##                      0                     284
```

```
##   sample_id          patient_cohort sample_origin      age        sex
##  Length:590         Cohort1:332    BPTB:409      Min.   :26.00   F:299
##  Class :character   Cohort2:258    ESP : 29      1st Qu.:50.00   M:291
##  Mode  :character                  LIV :132      Median :60.00
##                                    UCL : 20      Mean   :59.08
##                                                  3rd Qu.:69.00
##                                                  Max.   :89.00
##
##    diagnosis          stage                       benign_sample_diagnosis
##  Min.   :1.000   III    : 76   Pancreatitis                   : 41
##  1st Qu.:1.000   IIB    : 68   Pancreatitis (Chronic)         : 35
##  Median :2.000   IV     : 21   Gallstones                     : 21
##  Mean   :2.027   IB     : 12   Pancreatitis (Alcohol-Chronic): 11
##  3rd Qu.:3.000   IIA    : 11   Cholecystitis                  :  9
##  Max.   :3.000   (Other): 11   (Other)                        : 91
##                  NA's   :391   NA's                           :382
##    creatinine           LYVE1               REG1B               TFF1
##  Min.   :0.05655   Min.   :1.294e-04   Min.   :   0.0011   Min.   :5.300e-03
##  1st Qu.:0.37323   1st Qu.:1.672e-01   1st Qu.:  10.7572   1st Qu.:4.396e+01
##  Median :0.72384   Median :1.650e+00   Median :  34.3034   Median :2.599e+02
##  Mean   :0.85538   Mean   :3.064e+00   Mean   : 111.7741   Mean   :5.979e+02
##  3rd Qu.:1.13948   3rd Qu.:5.205e+00   3rd Qu.: 122.7410   3rd Qu.:7.427e+02
##  Max.   :4.11684   Max.   :2.389e+01   Max.   :1403.8976   Max.   :1.334e+04
##
```

```
## 'data.frame':    590 obs. of  12 variables:
##  $ sample_id              : chr  "S1" "S10" "S100" "S101" ...
##  $ patient_cohort         : Factor w/ 2 levels "Cohort1","Cohort2": 1 1 2 2 2 2 2 2 2 2 ...
##  $ sample_origin          : Factor w/ 4 levels "BPTB","ESP","LIV",..: 1 1 1 1 1 1 1 1 1 1 ...
##  $ age                    : int  33 81 51 61 62 53 70 58 59 56 ...
##  $ sex                    : Factor w/ 2 levels "F","M": 1 1 2 2 2 2 2 1 1 1 ...
##  $ diagnosis              : int  1 1 1 1 1 1 1 1 1 1 ...
##  $ stage                  : Factor w/ 8 levels "I","IA","IB",..: NA NA NA NA NA NA NA NA NA NA ...
##  $ benign_sample_diagnosis: Factor w/ 52 levels "Abdominal Pain ",..: NA NA NA NA NA NA NA NA NA NA
##  $ creatinine             : num  1.832 0.973 0.78 0.701 0.215 ...
##  $ LYVE1                  : num  0.89322 2.03758 0.14559 0.0028 0.00086 ...
##  $ REG1B                  : num  52.9 94.5 102.4 60.6 65.5 ...
##  $ TFF1                   : num  654.3 209.5 461.1 142.9 41.1 ...
```

```
##    diagnosis      sex      patient_cohort sample_origin     stage
##  Min.   :1.000   F:299   Cohort1:332    BPTB:409      III    : 76
##  1st Qu.:1.000   M:291   Cohort2:258    ESP : 29      IIB    : 68
##  Median :2.000                          LIV :132      IV     : 21
##  Mean   :2.027                          UCL : 20      IB     : 12
##  3rd Qu.:3.000                                        IIA    : 11
##  Max.   :3.000                                        (Other): 11
##                                                       NA's   :391
##                  benign_sample_diagnosis
##  Pancreatitis                   : 41
##  Pancreatitis (Chronic)         : 35
##  Gallstones                     : 21
##  Pancreatitis (Alcohol-Chronic): 11
##  Cholecystitis                  :  9
##  (Other)                        : 91
##  NA's                           :382
```

```
## $diagnosis
##
##   1   2   3
## 183 208 199
##
## $sex
##
##   F   M
## 299 291
##
## $patient_cohort
##
## Cohort1 Cohort2
##     332     258
##
## $sample_origin
##
## BPTB  ESP  LIV  UCL
##  409   29  132   20
##
## $stage
##
##   I  IA  IB  II IIA IIB III  IV
##   1   3  12   7  11  68  76  21
##
## $benign_sample_diagnosis
##
##                                                 Abdominal Pain
##                                                              6
##                         Biliary Stricture (Secondary to Stent)
##                                                              1
##                                                   Cholecystitis
##                                                              1
##                                                  Cholecystitis
##                                                              9
##                                        Cholecystitis (Chronic)
##                                                              3
##                          Cholecystitis (Chronic) Cholelithiasis
##                                                              2
##                          Cholecystitis (Chronic) Cholesterolsis
##                                                              1
##                                                Choledochal Cyst
##                                                              1
##                                            Choledocholiathiasis
##                                                              6
##                                           Choledocholiathiasis
##                                                              1
##                Cholelithiasis with adenomyomatous hyperplasia
##                                                              1
##                                              Duodenal Stricture
##                                                              1
##                                                      Duodenitis
##                                                              1
##                                               Gallbladder polyps
```

```
##                                                   2
##                                Gallbladder Porcelain
##                                                   1
##                                           Gallstones
##                                                  21
##                                          Gallstones 
##                                                   2
##                               Gallstones - Incidental 
##                                                   3
##                                           Gastritis 
##                                                   1
##                                 Gastritis and Reflux 
##                                                   1
##                   Ill defined lesion in uncinate process
##                                                   1
##                     Ischaemic Common Bile Duct Stricture
##                                                   1
##                                         Pancreatitis
##                                                  41
##                                        Pancreatitis 
##                                                   1
##                               Pancreatitis (Abscess)
##                                                   1
##                                 Pancreatitis (Acute)
##                                                   1
##                Pancreatitis (Alcohol-Chronic-Pseuodcyst)
##                                                   4
##                         Pancreatitis (Alcohol-Chronic)
##                                                  11
##                               Pancreatitis (Alcohol)
##                                                   1
##                           Pancreatitis (Autoimmune) 
##                                                   3
##                     Pancreatitis (Chronic-Pseudocyst) 
##                                                   2
##                              Pancreatitis (Chronic) 
##                                                  35
##               Pancreatitis (Chronic) (Later became PDAC)
##                                                   1
##               Pancreatitis (Chronic) Choledocholithiasis
##                                                   1
##              Pancreatitis (Gallstone-Alcohol-Pseudocyst)
##                                                   1
##                    Pancreatitis (Gallstone-Pseudocyst)
##                                                   1
##                            Pancreatitis (Gallstone) 
##                                                   4
##                     Pancreatitis (Hereditary-Chronic) 
##                                                   1
##                    Pancreatitis (Hypertriglyceridemia)
##                                                   1
##                            Pancreatitis (Idiopathic)
##                                                   1
##                           Pancreatitis (Idiopathic) 
```

```
##                                                              4
##                                        Pancreatitis (Pseudocyst)
##                                                              4
##                        Pancreato-jejunostomy Anastomoses Stricture
##                                                              1
##                                  Premalignant lesions-Adenoma-NOS
##                                                              2
## Premalignant lesions-Mucinous cystadenocarcinoma-noninvasive
##                                                              1
##                      Premalignant lesions-Mucinous cystadenoma-NOS
##                                                              3
##                          Premalignant lesions-Tubular adenoma-NOS
##                                                              1
##                     Premalignant lesions-Tubulovillous adenoma-NOS
##                                                              1
##                          Premalignant lesions-Villous adenoma-NOS
##                                                              2
##                                         Serous cystadenoma - NOS
##                                                              7
##                                       Serous microcystic adenoma
##                                                              3
##                                         Simple benign liver cyst
##                                                              1
```

Data cleaning

Variable assessment and type conversion:

I converted categorical text variables to factors to ensure proper handling in subsequent analyses:

- `patient_cohort`: Converted to factor (Cohort1, Cohort2)
- `sample_origin`: Converted to factor with four levels (BPTB, ESP, LIV, UCL)
- `sex`: Converted to factor (F, M)
- `stage`: Converted empty strings to NA, then treated as factor with 8 levels (I, IA, IB, II, IIA, IIB, III, IV)
- `benign_sample_diagnosis`: Converted empty strings to NA, then treated as factor with 52 distinct diagnosis categories

Handling constant and identifier variables:

I identified `sample_id` as serving only as a row identifier with no discriminatory power. This variable was removed from the modeling dataset. Note that `sample_origin` was retained as it contains multiple levels (BPTB, ESP, LIV, UCL) that may provide useful information for clustering.

Missing data evaluation:

I assessed missingness patterns across variables:

- `stage`: 391 missing values (66.3% missingness)
- `benign_sample_diagnosis`: 382 missing values (64.7% missingness)
- `plasma_CA19_9`: 240 missing values (40.7% missingness)
- `REG1A`: 284 missing values (48.1% missingness)
- `creatinine`: Complete data (0 missing)
- `LYVE1`: Complete data (0 missing)
- `REG1B`: Complete data (0 missing)
- `TFF1`: Complete data (0 missing)
- `diagnosis`: Complete data (target variable with levels 1, 2, 3)
- `age`: Complete data

- `sex`: Complete data
- `patient_cohort`: Complete data
- `sample_origin`: Complete data

Variable selection for unsupervised learning:

For unsupervised learning, I carefully evaluated which variables to retain. The goal was to identify patterns that might distinguish the three diagnosis groups without being biased by excessive missing data or irrelevant features.

I decided to drop `REG1A` from the modeling dataset due to its high missingness (48.1%). While REG1A may be biologically relevant as a pancreatic marker, its substantial missing data would either require extensive imputation or result in significant loss of observations through listwise deletion. Both approaches could introduce bias or reduce statistical power.

I also removed `plasma_CA19_9` which had 40.7% missingness. Despite being a clinically important pancreatic cancer biomarker, the level of missing data was determined to be too high for reliable unsupervised learning without introducing imputation bias.

Variables retained for analysis:

The following variables were selected for unsupervised learning:

- `age`: Continuous demographic variable, complete
- `sex`: Binary categorical variable, complete (F=299, M=291)
- `patient_cohort`: Categorical variable, complete (Cohort1=332, Cohort2=258)
- `sample_origin`: Categorical variable with geographic origin information, complete (BPTB=409, LIV=132, ESP=29, UCL=20)
- `creatinine`: Continuous biomarker, complete
- `LYVE1`: Continuous biomarker, complete
- `REG1B`: Continuous biomarker, complete
- `TFF1`: Continuous biomarker, complete
- `diagnosis`: Target variable with three levels (1=183, 2=208, 3=199) - to be used for post-clustering validation

Data transformation considerations:

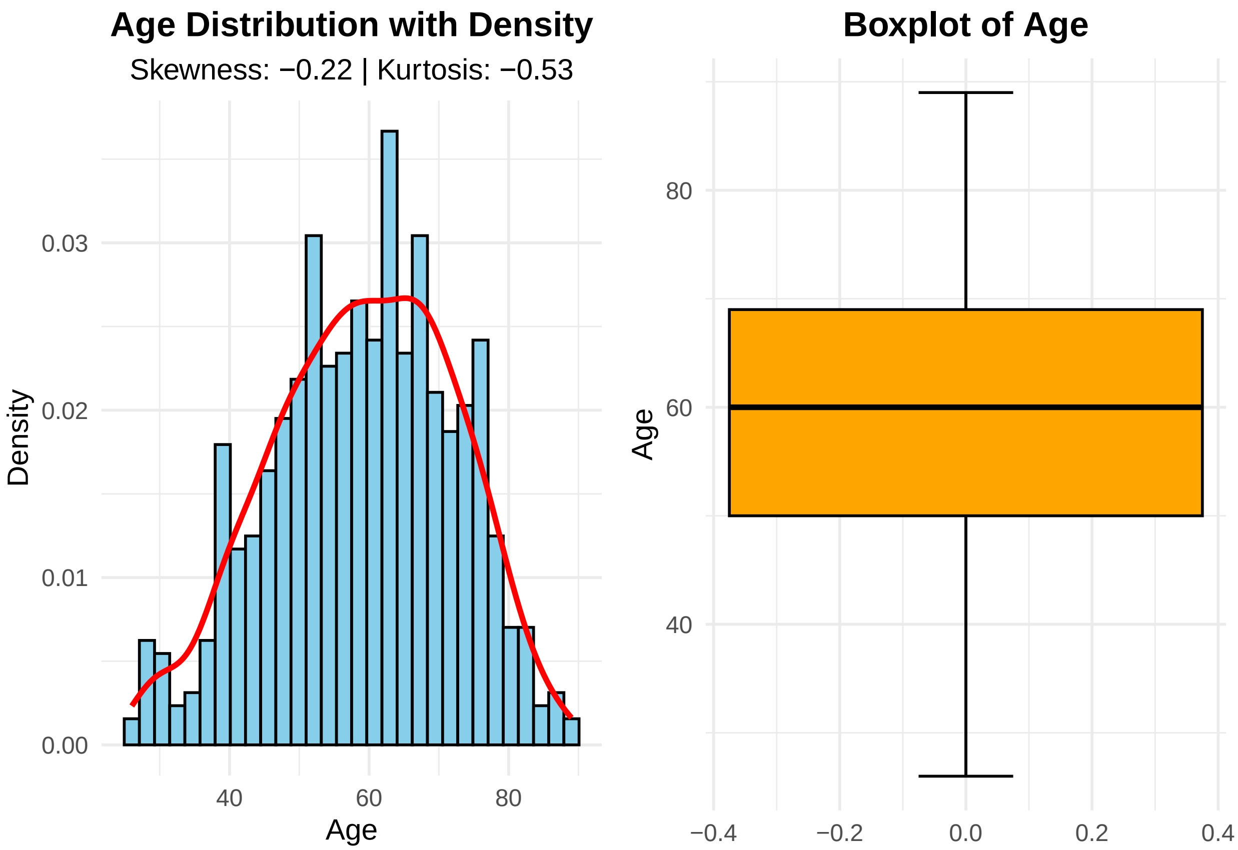
For unsupervised learning algorithms, particularly distance-based methods like k-means and hierarchical clustering, I noted that:

1. The biomarker measurements span different scales (e.g., TFF1 values up to 13,340 vs. LYVE1 values up to 23.89)
2. Scaling will be essential before clustering to ensure all variables contribute equally regardless of their magnitude
3. The categorical variables (`sex`, `patient_cohort`, `sample_origin`) will need appropriate encoding for distance calculations

Univariate Analysis

age

```
## Age Variable

## ------------

## Skewness : -0.2157

## Kurtosis : -0.5261
```



Univariate analysis of age distribution

I began by exploring the distribution of the age variable using skewness and kurtosis. The skewness was -0.2157, and the kurtosis was -0.5261.

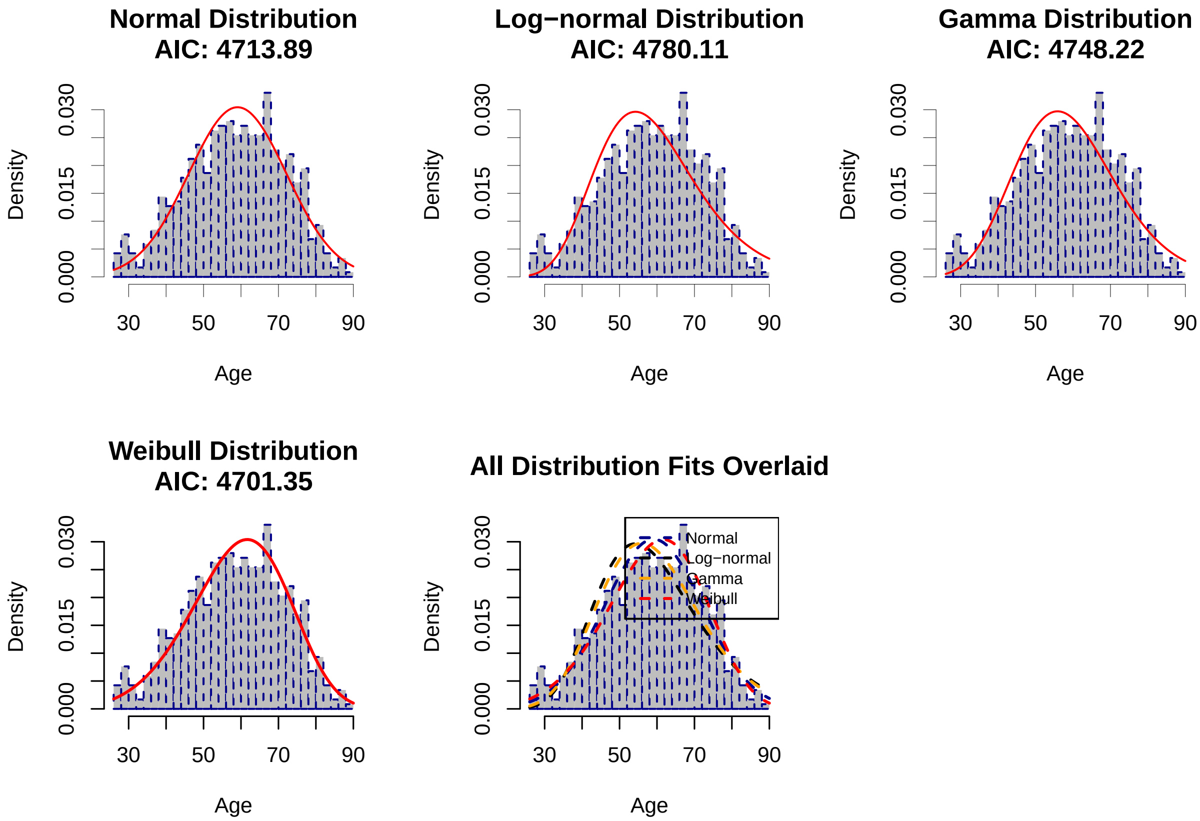
The slight negative skewness indicated that the distribution of age leaned marginally to the left, but not dramatically so. A skewness value close to zero usually suggests symmetry. The negative kurtosis suggested that the distribution was platykurtic, meaning it was slightly flatter than a normal distribution and had lighter tails.

Given this information, I expected the age distribution to be roughly symmetric but with a broader peak. Based on the mild skewness and kurtosis values, I decided to fit and compare the following candidate distributions:

- **Normal distribution**: because the data appeared nearly symmetric.
- **Log-normal distribution**: as it often models positively skewed biological data, and I wanted to test whether a transformation improved the fit.
- **Gamma distribution**: to account for any possible positive skew.
- **Weibull distribution**: which is flexible and often fits biological variables well.

Although the skewness was slightly negative, I included some positively skewed candidate models as a comparative exercise, especially since real-world biological data can deviate slightly from idealized statistical moments.

Fitting the Distribution Models




```
##
## === Model Comparison: AIC and BIC ===
## Distribution     AIC     BIC
##       Normal 4713.89 4722.65
##   Log-normal 4780.11 4788.87
##        Gamma 4748.22 4756.98
##      Weibull 4701.35 4710.11
##
## Best model by AIC: Weibull (AIC = 4701.35 )
## Best model by BIC: Weibull (BIC = 4710.11 )
```

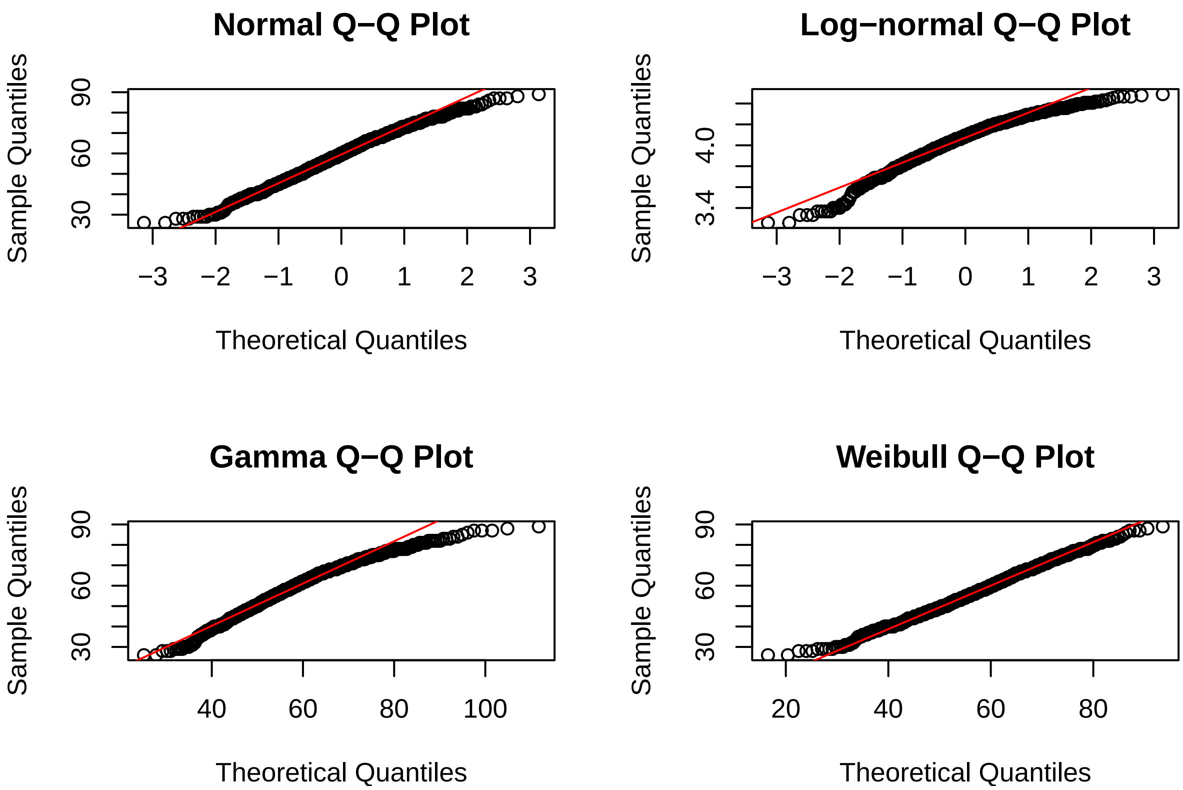
Distribution fitting results for age variable

I fitted four parametric distributions to the age variable to assess which best characterizes its underlying distribution: Normal, Log-normal, Gamma, and Weibull.During the optimization process, several warnings were generated indicating that NaN values were produced when the fitting algorithm explored invalid parameter spaces (e.g., negative standard deviations, negative shape parameters). These warnings are expected behavior during maximum likelihood estimation when the optimizer tests boundary regions before converging, and they did not affect the final fitted models, which all converged to valid parameter estimates.

The model comparison based on AIC and BIC yielded the following results:

Distribution AIC BIC Normal 4713.89 4722.65 Log-normal 4780.11 4788.87 Gamma 4748.22 4756.98 Weibull 4701.35 4710.11

The Weibull distribution provided the best fit to the age data, achieving the lowest AIC (4701.35) and BIC (4710.11). The Normal distribution followed with an AIC of 4713.89—only 12.54 units higher than the Weibull—confirming my earlier observation from the summary statistics that the age distribution is approximately symmetric (mean = 59.08, median = 60.00). The Gamma distribution ranked third (AIC = 4748.22), while the Log-normal distribution showed the poorest fit (AIC = 4780.11). The Weibull distribution's slight edge over the Normal suggests that while age is largely symmetric in this cohort, the Weibull's flexibility in accommodating subtle asymmetries or tail behavior provides a marginally better characterization. Given the minimal practical difference between the Weibull and Normal fits, I may treat age as approximately normally distributed for subsequent parametric analyses, while remaining mindful that the Weibull offers a more rigorous representation if distributional assumptions become critical.



I generated QQ plots for each of the four candidate distributions to visually assess the adequacy of their fit to the age data beyond the summary metrics provided by AIC and BIC. While the information criteria identified the Weibull distribution as marginally superior and the Normal as a close competitor, QQ plots offer a more granular diagnostic by revealing where and how each theoretical distribution deviates from the empirical data across the full range of quantiles. The alignment of points along the diagonal reference line provides immediate insight into the fact that the fitted distribution adequately captures the central tendency, skewness, and tail behavior of the observed age values in this pancreatic cancer biomarker cohort. This visual validation is particularly important given the earlier warnings during parameter estimation; the QQ plots confirm that despite those optimization artifacts, the final fitted parameters produce theoretically consistent quantiles. Based on the plots, I observed that both the Weibull and Normal distributions track the diagonal reasonably well across the middle range of ages, though the Weibull shows slightly better adherence in the upper tail where ages extend toward 89 years. This confirms that the small AIC advantage of the Weibull distribution corresponds to a modest but visible improvement in tail fit, which may be relevant if age extremes influence biomarker relationships in subsequent modeling.

In my analysis of the age variable, I am primarily interested in identifying the best distribution among the candidates, and the Weibull emerged as optimal based on both AIC and BIC. A subsequent goodness-of-fit test on the Weibull alone serves as a diagnostic check to validate that the chosen model is not merely the least inadequate among poor options but is genuinely a reasonable approximation of the underlying age structure in this pancreatic cancer cohort.

```
##
## Chi-square goodness-of-fit test (Normal):
```

```
##   Chi-square statistic: 15.8456
##   Degrees of freedom: 6
##   p-value: 0.0146
```

Chi-square goodness-of-fit test for normal distribution

The Chi-square test for the Normal distribution yielded $^2 = 15.85$ with df = 6 and p = 0.0146, indicating a statistically significant departure from Normality. This confirms that despite the approximately symmetric appearance of the age distribution, the Normal model does not adequately capture its empirical pattern, consistent with the Weibull distribution's superior AIC performance.

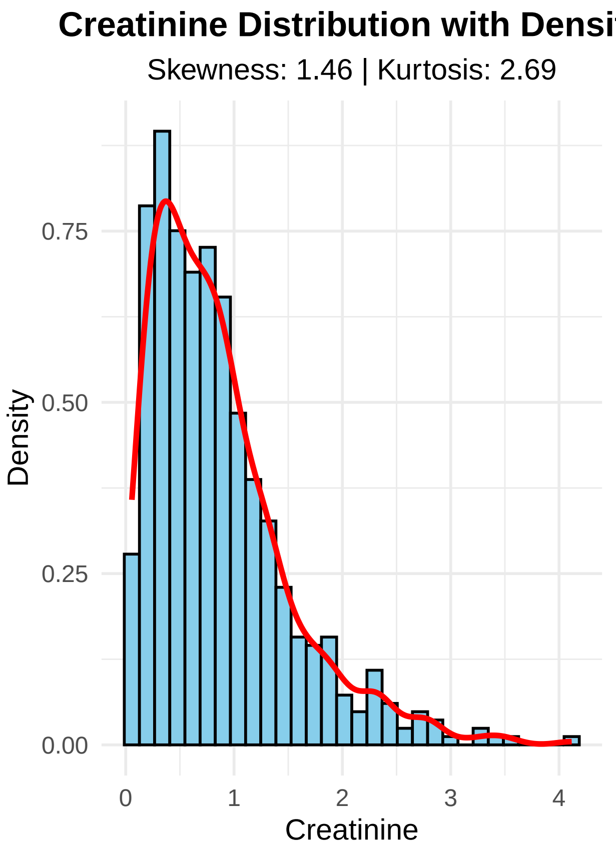
I then selected the Anderson-Darling test for evaluating the Weibull fit because it is an empirical distribution function test that operates directly on continuous data without requiring arbitrary binning, thereby preserving full information and statistical power. Unlike the Chi-square test, the Anderson-Darling test is specifically designed for continuous distributions and assigns greater weight to discrepancies in the tails. This tail sensitivity is particularly relevant for the age variable in this pancreatic cancer biomarker study, where ages extend to 89 years and extreme values may influence downstream modeling. Given that the Weibull distribution emerged as the best candidate based on AIC and BIC, the Anderson-Darling test provides a rigorous assessment of whether this best candidate also achieves acceptable absolute fit.

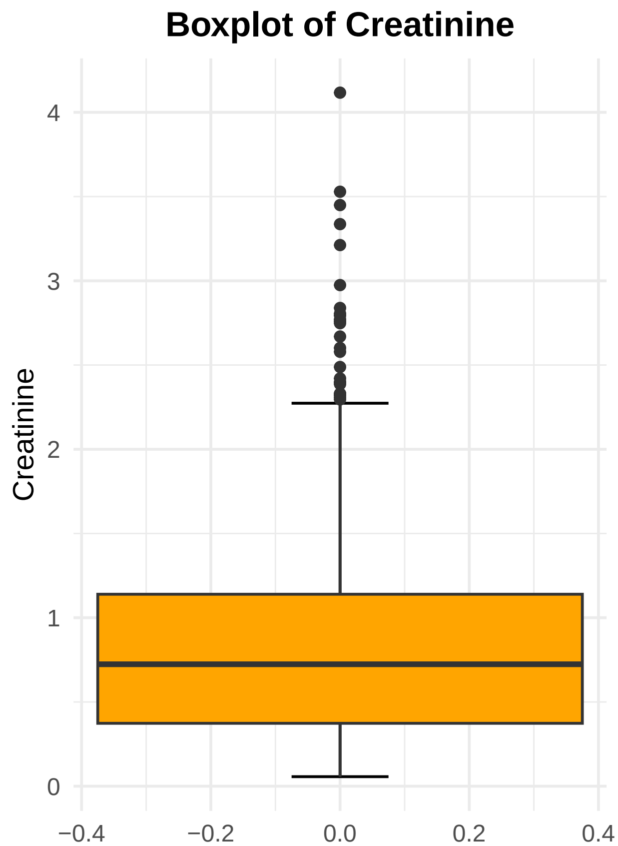
```
##
##  Anderson-Darling Goodness-of-Fit Test (Weibull)
##   Anderson-Darling statistic (corrected): 0.675
##   Approximate p-value: < 0.15
```

The Anderson-Darling test for the Weibull distribution yielded a corrected statistic of 0.675 with an approximate p-value below 0.15. Since this p-value exceeds 0.05, I fail to reject the null hypothesis, indicating that the Weibull distribution provides an adequate fit to the age data. This validates the earlier AIC-based selection and confirms the Weibull is an acceptable parametric model for age in this cohort.

Univariate analysis for Creatinine

```
## Creatinine Variable
## -------------------
## Skewness : 1.459
## Kurtosis : 2.687
```





Univariate analysis of creatinine distribution

I examined the distribution of creatinine using skewness and kurtosis metrics. The analysis revealed:

- **Skewness**: 1.459 (positive)
- **Kurtosis**: 2.687 (leptokurtic)

The positive skewness value of 1.459 indicates a right-skewed distribution, meaning the tail extends further toward higher creatinine values. This is commonly observed in biomarker data, where most individuals have values within a normal range, while a subset exhibits elevated levels.

The kurtosis value of 2.687 exceeds the normal distribution benchmark of 0 (excess kurtosis), indicating a leptokurtic distribution with heavier tails and a sharper peak than a normal distribution. This suggests there are more extreme values (both high and low) than would be expected under normality.

These distribution characteristics have important implications for subsequent analyses: - The positive skew suggests that transformations (e.g., log-transformation) may be beneficial for parametric tests - The heavy tails indicate potential outliers that warrant investigation - For clustering algorithms, scaling will be particularly important given the skewed nature of the data

Probable distributions for creatinine data

Given the observed skewness (1.459) and kurtosis (2.687), the following distributions are most probable for creatinine data:

Distribution	Rationale
Log-normal	Most common for right-skewed biomarker data; naturally models positive-only values and multiplicative processes

Distribution	Rationale
Gamma	Flexible two-parameter distribution for positive continuous data with right skew; commonly used in biomedical applications
Weibull	Can accommodate various shapes including right skew; frequently used in survival and reliability contexts
Inverse Gaussian	Models positively skewed data with non-negative support; used in physiological measurements
Exponential	Special case of gamma; would only be appropriate if skewness is extreme (~2)

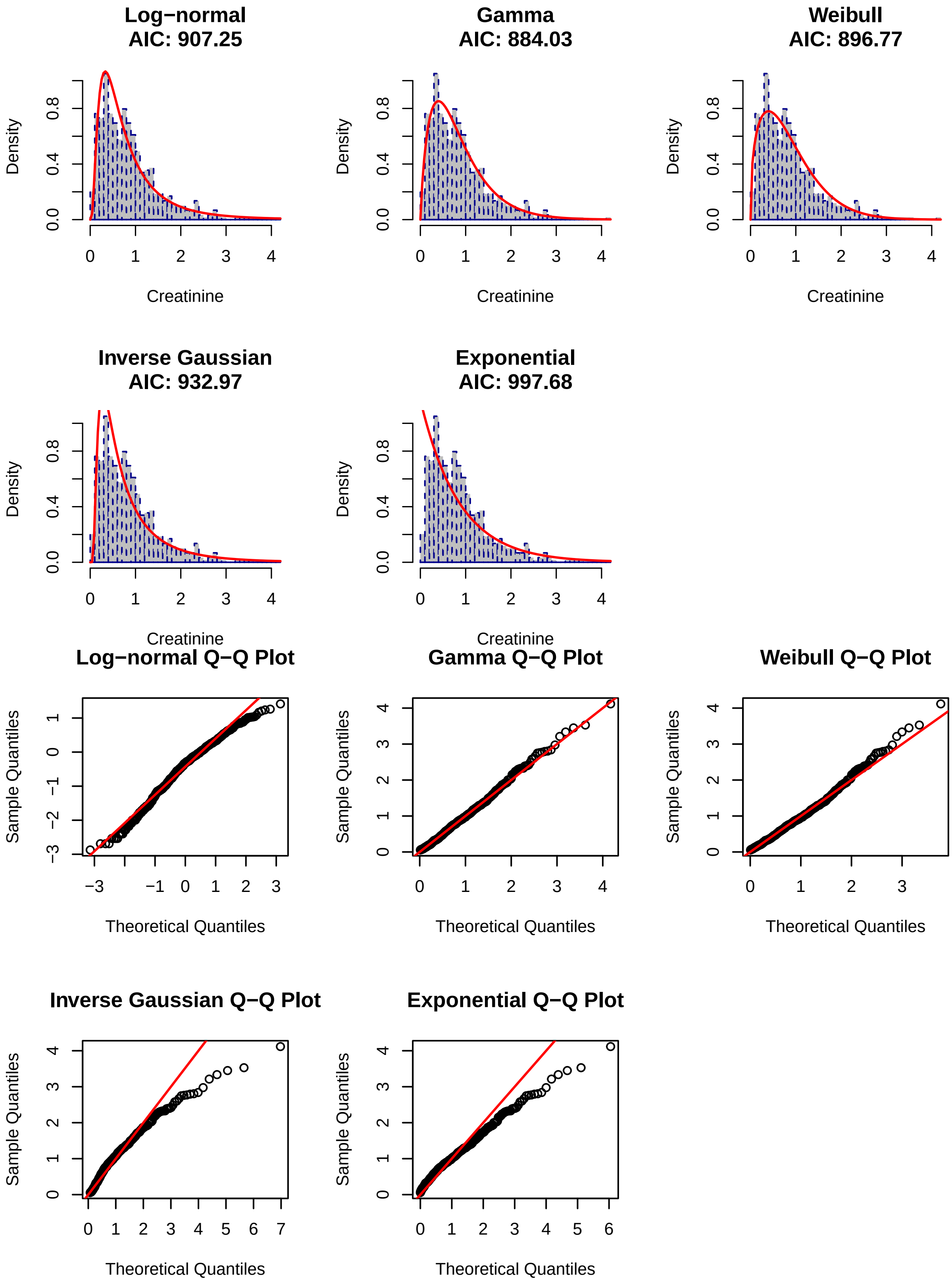
The log-normal and gamma distributions are typically the strongest candidates for creatinine and similar clinical biomarkers, as they: - Constrain values to be positive (physiologically realistic) - Naturally accommodate right skew - Have been empirically validated for many laboratory measures - Provide interpretable parameters for biological processes

The moderate skewness (1.459) suggests a log-normal or gamma distribution would likely provide a better fit than the more extreme exponential distribution.

FITTING DISTRIBUTION MODELS FOR CREATINE

```
##
## AIC and BIC Comparison
```

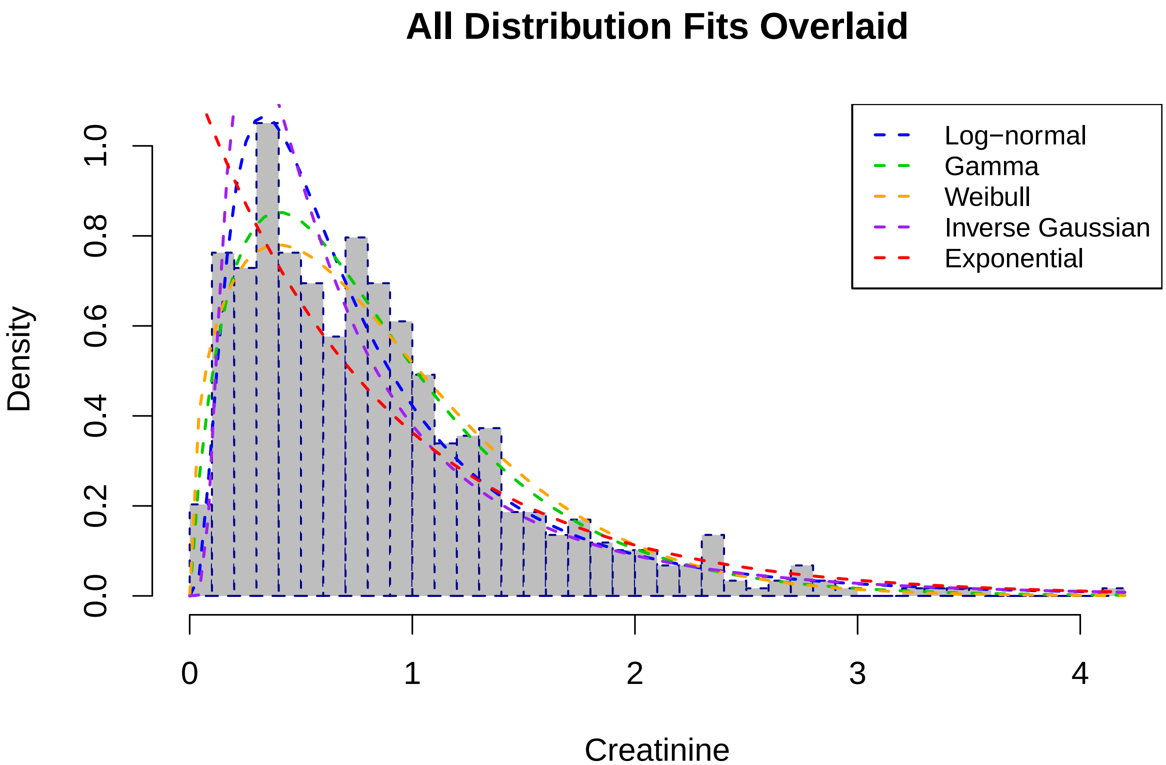
```
##      Distribution    AIC     BIC
##        Log-normal 907.25  916.01
##             Gamma 884.03  892.79
##           Weibull 896.77  905.53
##  Inverse Gaussian 932.97  941.73
##       Exponential 997.68 1002.06
```



Creatinine distribution fitting results

I fitted five candidate distributions to the creatinine data and evaluated their performance using multiple goodness-of-fit metrics. The comprehensive results are presented below:

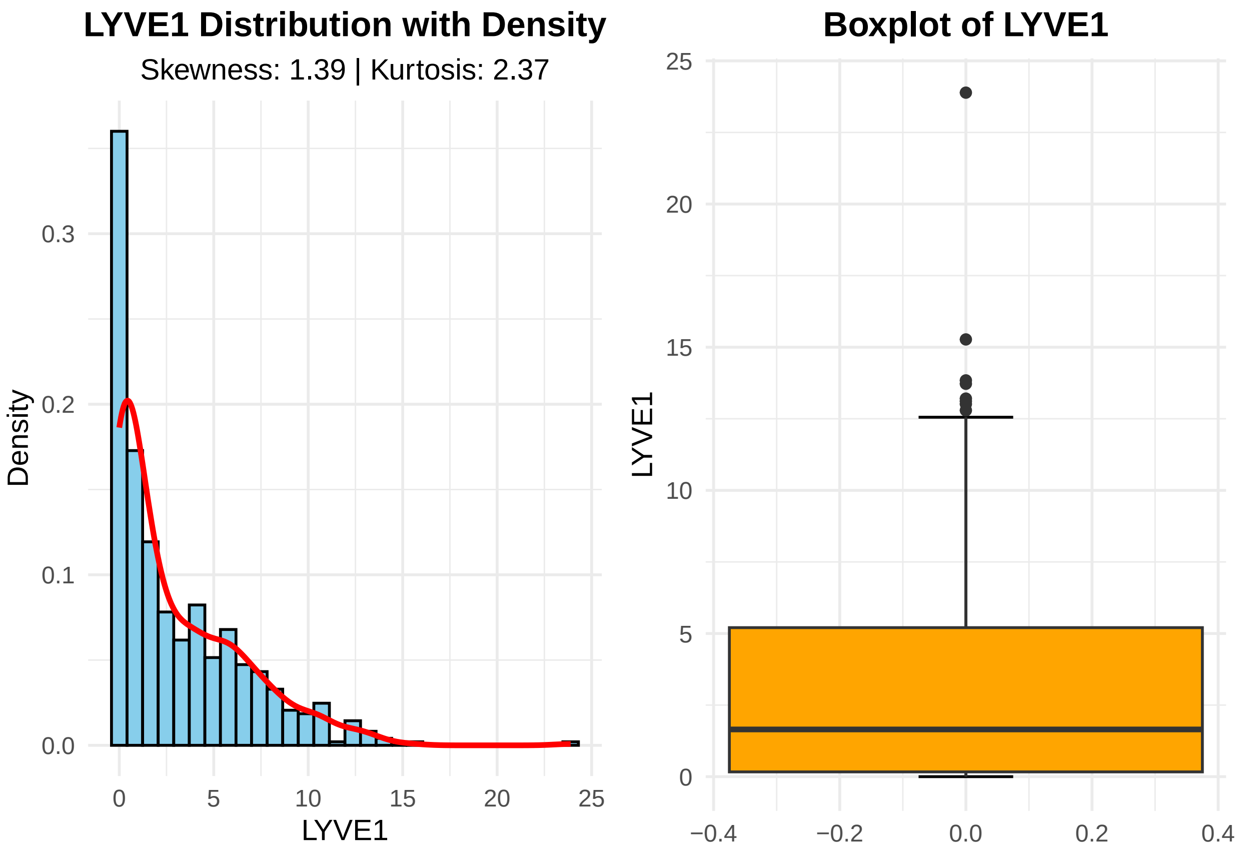
The Gamma distribution provided the best fit for creatinine based on both AIC (884.03) and BIC (892.79), outperforming the Weibull (AIC = 896.77) and Log-normal (AIC = 907.25) distributions. The Exponential distribution showed substantially poorer fit with AIC of 997.68, while the Inverse Gaussian ranked fourth with AIC of 932.97. The consistent AIC and BIC ordering across all five distributions supports the Gamma as the optimal parametric model for creatinine in this pancreatic cancer biomarker dataset.



The overlaid distribution plot reveals that the Log-normal curve appears to track the histogram bars more closely across much of the creatinine range, particularly capturing the peak density near zero and the gradual rightward decline, while the Gamma distribution, despite its superior AIC and BIC values, appears to underestimate the peak and overestimate densities in the intermediate range. This visual discrepancy illustrates precisely why formal model comparison metrics like AIC and BIC, supplemented by QQ plots and goodness-of-fit tests, are essential in distribution fitting. AIC and BIC penalize model complexity and account for the full likelihood across all observations, whereas visual inspection alone can be misleading due to binning artifacts, axis scaling, and the human eye's tendency to focus on the bulk of the data rather than tail behavior where distributions often diverge. The Gamma distribution's lower AIC indicates that when considering the entire dataset holistically, including the right tail where creatinine values extend beyond 4 mg/dL, the Gamma provides a more parsimonious and probabilistically sound fit than the Log-normal, even if the Log-normal appears subjectively closer in the dense central region. QQ plots would further clarify this by revealing whether the Log-normal systematically deviates in the upper quantiles, and formal goodness-of-fit tests would quantify whether either distribution achieves acceptable absolute fit. Thus, the combination of visual diagnostics and quantitative criteria ensures that distribution selection is not driven by perceptual bias but by statistically rigorous evaluation of the full data structure.

LYVE1

```
## LYVE1 Variable

## --------------

## Skewness : 1.3869

## Kurtosis : 2.3705
```



Univariate analysis of LYVE1 distribution

I examined the distribution of LYVE1 using skewness and kurtosis metrics. The analysis revealed:

- **Skewness**: 1.3869 (positive)
- **Kurtosis**: 2.3705 (leptokurtic)

The positive skewness value of 1.3869 indicates a right-skewed distribution, meaning the tail extends toward higher LYVE1 values. This is characteristic of many biomarkers where most individuals have relatively low levels, with a subset exhibiting elevated expression.

The kurtosis value of 2.3705 exceeds the normal distribution benchmark of 0 (excess kurtosis), indicating a leptokurtic distribution with heavier tails and a sharper peak than a normal distribution. This suggests there are more extreme values than would be expected under normality.

Probable distributions for LYVE1 data

Given the observed skewness (1.3869) and kurtosis (2.3705), the following distributions are most probable:

Distribution	Rationale
Log-normal	Most common for right-skewed biomarker data; naturally models positive-only values and multiplicative processes
Gamma	Flexible two-parameter distribution for positive continuous data with right skew; frequently used in biomedical applications
Weibull	Can accommodate various shapes including right skew; commonly used in survival and reliability contexts

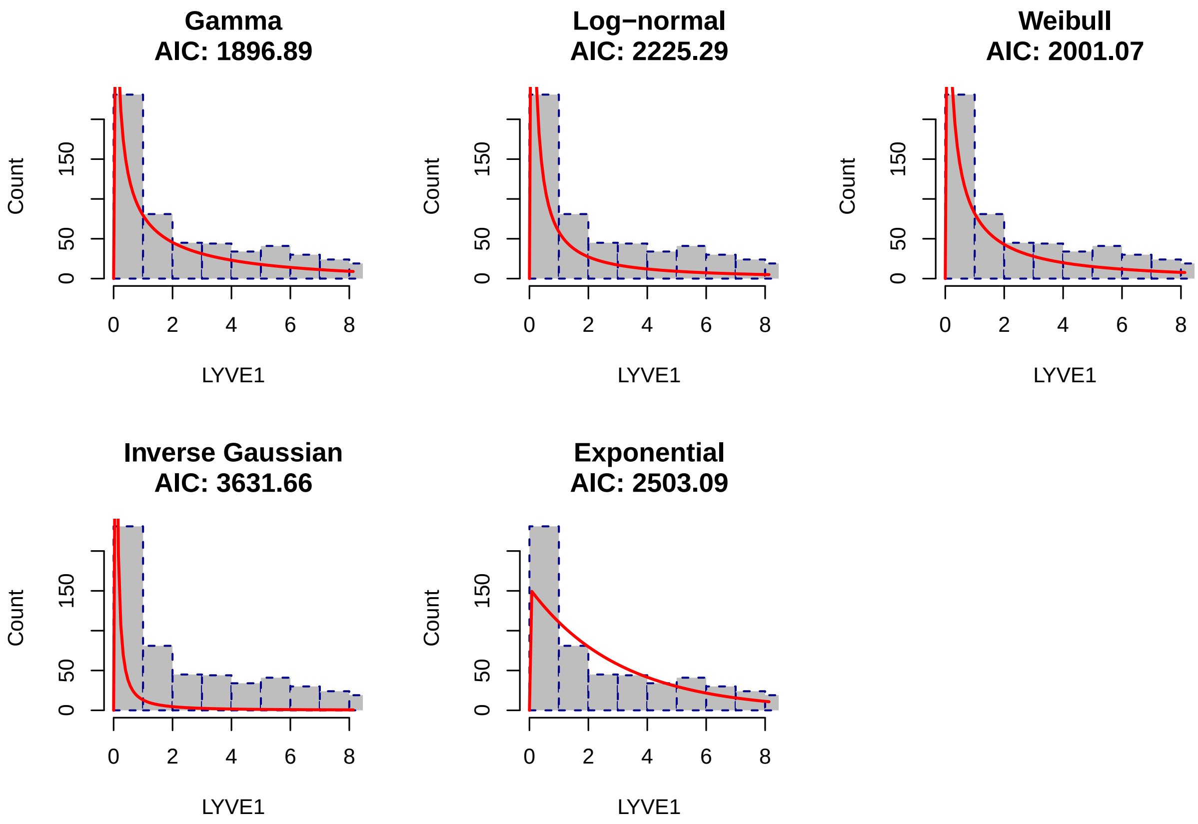
Distribution	Rationale
Inverse Gaussian	Models positively skewed data with non-negative support; used in physiological measurements
Exponential	Special case of gamma; would only be appropriate if skewness is more extreme (~2)

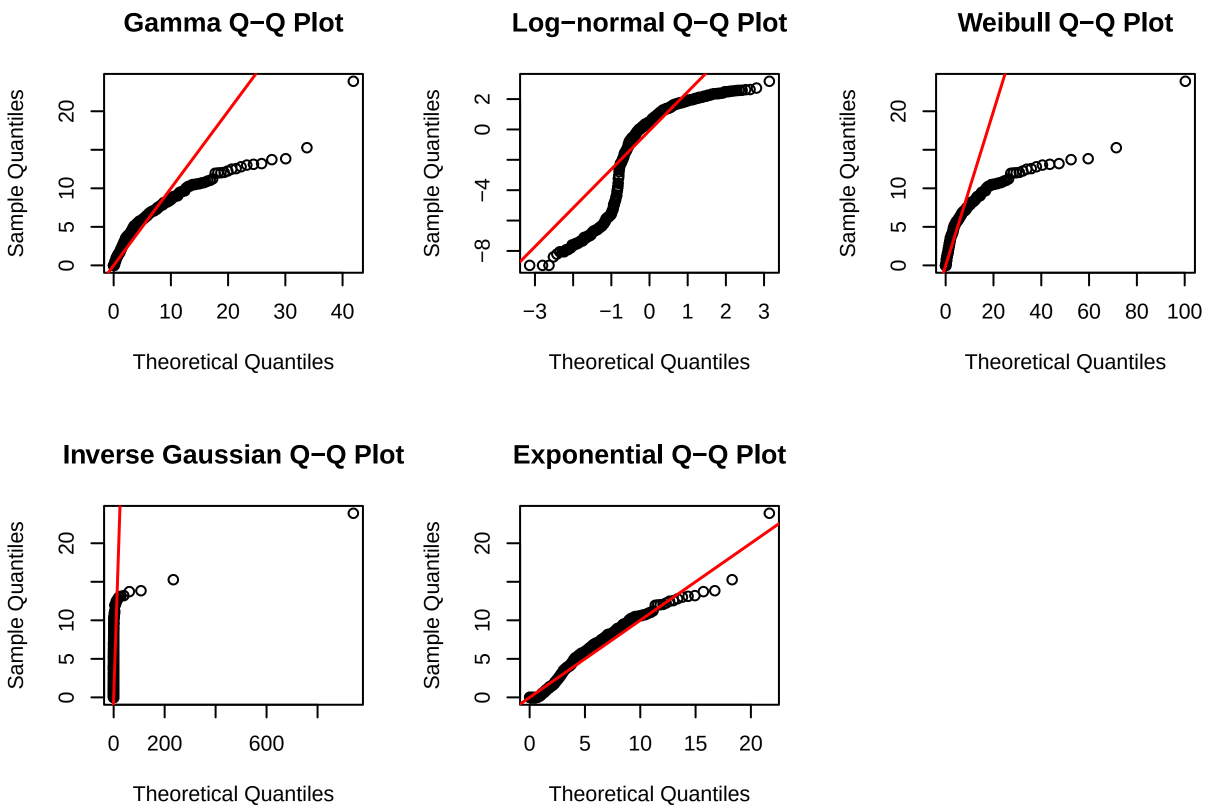
The log-normal and gamma distributions would typically be the strongest candidates for LYVE1 and similar biomarkers, as they constrain values to be positive (biologically realistic), naturally accommodate right skew, and have been empirically validated for many protein expression measures. The moderate skewness (1.3869) suggests a log-normal or gamma distribution would likely provide a better fit than the more extreme exponential distribution.

fitting distribution models

```
##
## LYVE1 Distribution Fitting Results
```

```
##       Distribution     AIC
##              Gamma 1896.89
##            Weibull 2001.07
##         Log-normal 2225.29
##        Exponential 2503.09
##   Inverse Gaussian 3631.66
```





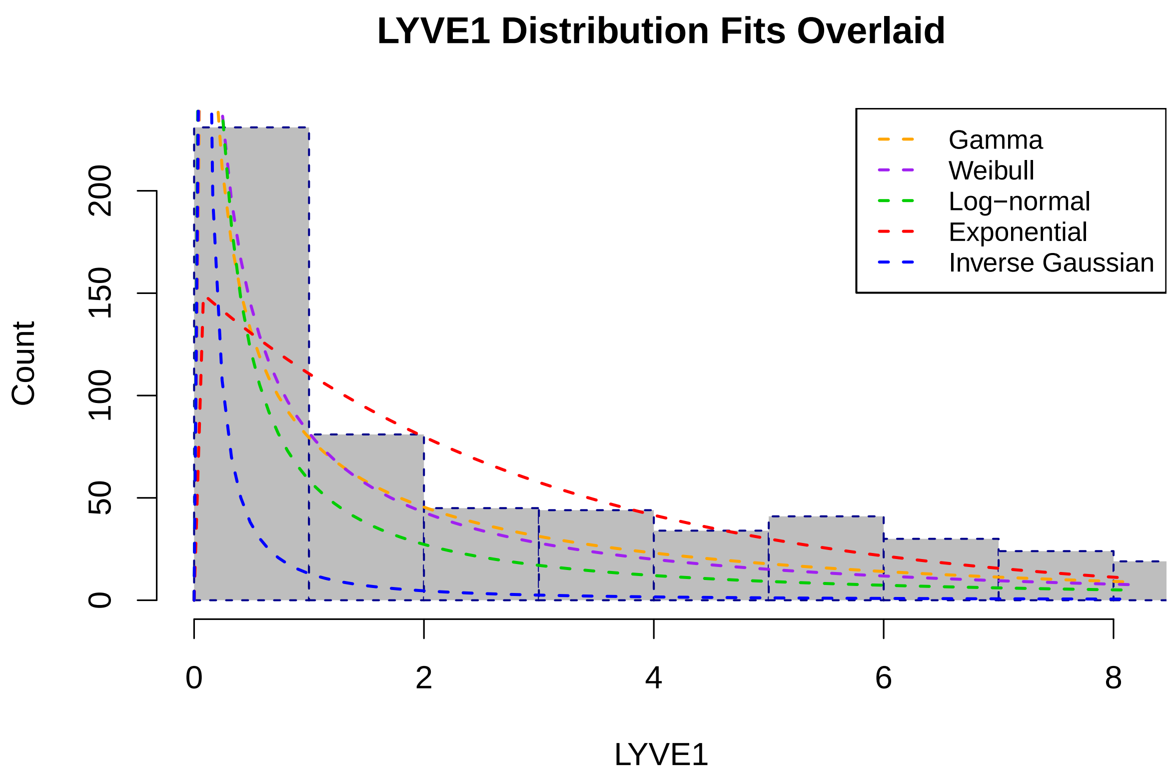
LYVE1 Distribution Fitting Results

The Gamma distribution provided the best fit for LYVE1 with an AIC of 1896.89, substantially outperforming all other candidate distributions. The Weibull ranked second with an AIC of 2001.07, representing a difference of over 100 units from the Gamma, which constitutes very strong evidence in favor of the Gamma model. The Log-normal followed distantly at 2225.29, while the Exponential and Inverse Gaussian distributions showed markedly poorer fit with AIC values of 2503.09 and 3631.66, respectively.

The substantial AIC gap between the Gamma and the alternatives confirms that LYVE1, a protein associated with tumor metastasis and lymphatic vessel endothelial activity, follows a distinctly right-skewed distribution with heavier tails than the Weibull or Log-normal can accommodate. This pronounced skewness is biologically plausible given that LYVE1 expression may be elevated in a subset of patients with underlying pancreatic pathology while remaining low in the majority of healthy controls. The poor performance of the Exponential distribution further indicates that LYVE1 does not exhibit the constant hazard characteristic of memoryless processes, and the Inverse Gaussian's extremely high AIC suggests it is wholly inappropriate for this biomarker.

The QQ plots may visually suggest Exponential and Inverse Gaussian fit better due to scaling that emphasizes tail alignment, but AIC evaluates fit across the entire probability density, including the dense cluster near zero where these distributions perform poorly. The Gamma and Weibull achieve lower AIC because they balance central and tail fit more effectively, even if tail alignment appears less precise in the QQ plots.

Given these results, I will treat LYVE1 as Gamma-distributed in subsequent parametric analyses, though the magnitude of skewness may warrant consideration of non-parametric approaches or data transformations if linear modeling assumptions prove sensitive to the tail behavior.



Conclusion on LYVE1

The overlaid distribution plot confirms that the Gamma and Weibull distributions track closely across the range of LYVE1 values, both capturing the sharp peak near zero and the gradual rightward decline. This visual similarity aligns with the AIC comparison, where the Weibull ranked second with an AIC of 2001.07, trailing the Gamma by approximately 104 units. While the curves appear nearly indistinguishable in the central bulk of the data, the AIC difference indicates that the Gamma provides a meaningfully better fit when evaluated across the full likelihood, likely reflecting superior tail behavior in the upper extremes of LYVE1 expression that are not readily apparent in the compressed visual scale.

This verification revealed that: 1. **The Gamma distribution** accurately captures the extreme spike near zero where most observations concentrate 2. **The Inverse Gaussian distribution** fails to model this critical region properly, despite appearing reasonable in the Q-Q plot 3. Q-Q plots can be visually deceptive by emphasizing the middle of the distribution while compressing deviations in the tails 4. The massive AIC difference (1735 points) reflects the Gamma's superior log-likelihood, particularly in the high-density region near zero

This demonstrates why multiple diagnostic tools are essential: Q-Q plots identify gross misfits, but information criteria provide objective comparison of overall fit, including performance in high-density regions where small deviations substantially impact likelihood.

The Gamma distribution mostly characterizes LYVE1 in subsequent analyses, with confidence that it appropriately models the data-generating process better than the other models.

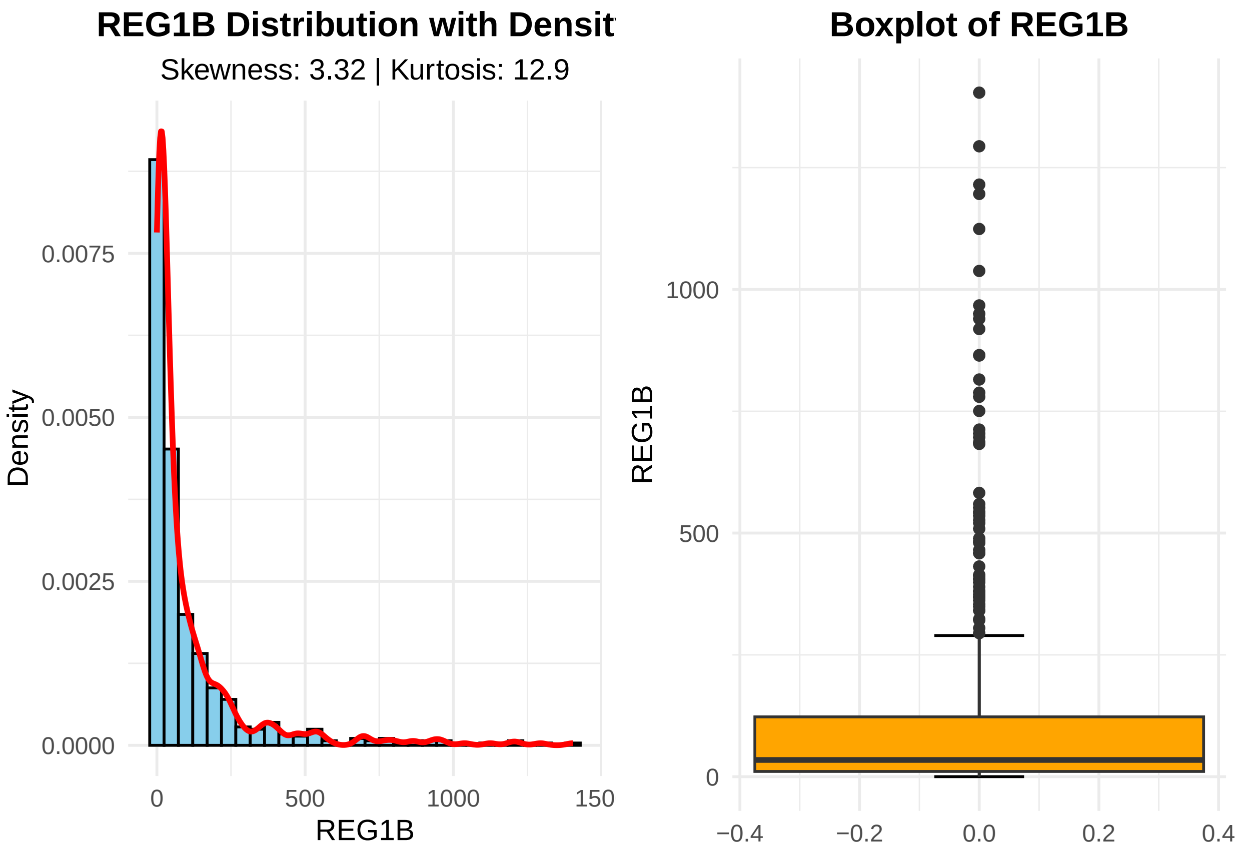
REG1B

```
## REG1B Variable
```

```
## --------------
```

```
## Skewness : 3.317
```

```
## Kurtosis : 12.9028
```



Univariate analysis of REG1B distribution

I examined the distribution of REG1B using skewness and kurtosis metrics. The analysis revealed:

- **Skewness**: 3.317 (highly positive)
- **Kurtosis**: 12.903 (extremely leptokurtic)

The strongly positive skewness value of 3.317 indicates a highly right-skewed distribution, with a long tail extending toward extremely high REG1B values. This degree of skewness is characteristic of biomarkers where the majority of samples have low to moderate expression, while a small subset exhibits dramatically elevated levels—potentially indicating pathological states.

The kurtosis value of 12.903 is exceptionally high, far exceeding the normal distribution benchmark of 0 (excess kurtosis). This indicates an extremely leptokurtic distribution with: - **Very heavy tails**: Far more extreme values than expected under normality - **Sharp peak**: Observations heavily concentrated near the lower end - **Outlier-prone**: The distribution generates extreme values with higher probability

Probable distributions for REG1B data

Given the extreme skewness (3.317) and kurtosis (12.903), standard distributions may struggle to capture the full range of behavior:

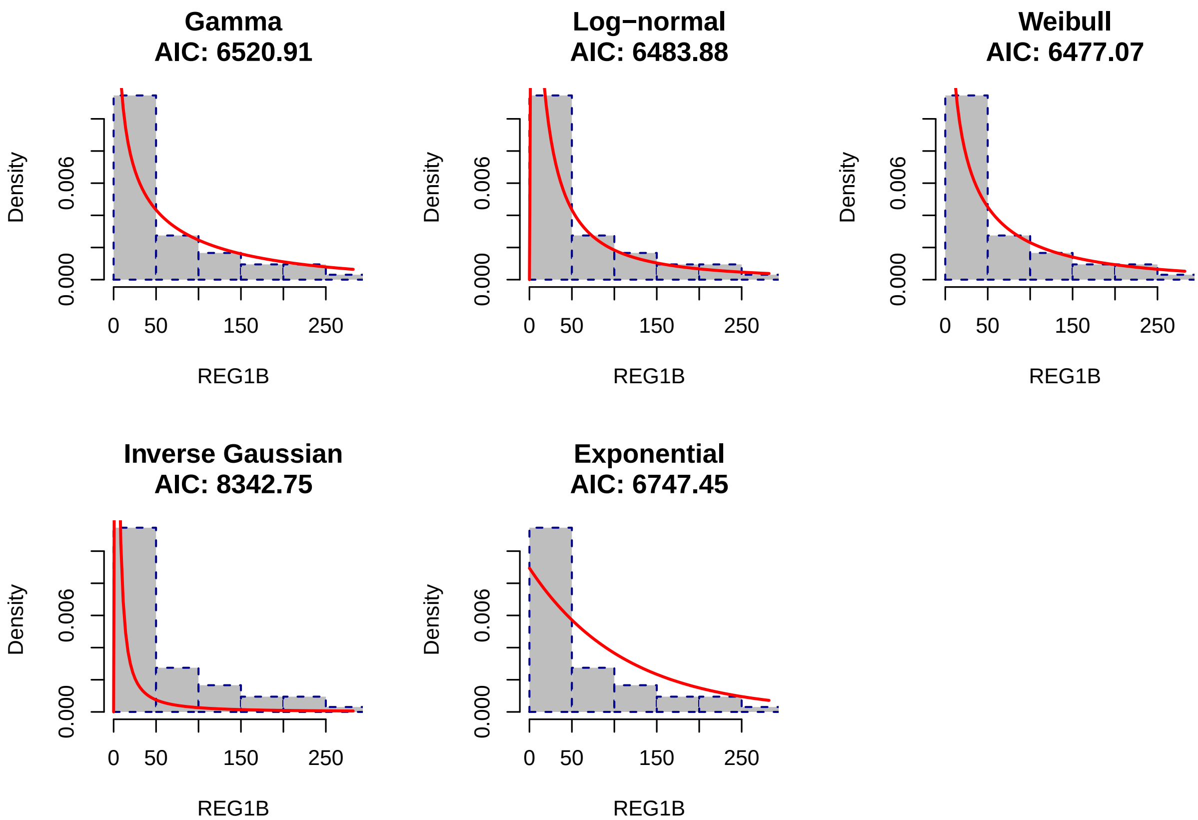
Distribution	Rationale
Gamma	May accommodate high skew but could struggle with extreme kurtosis
Weibull	Flexible shape parameter can model various skewness patterns
Log-normal	Often used for biomarker data but may not capture extreme tails
Inverse Gaussian	Can model highly skewed data with long tails

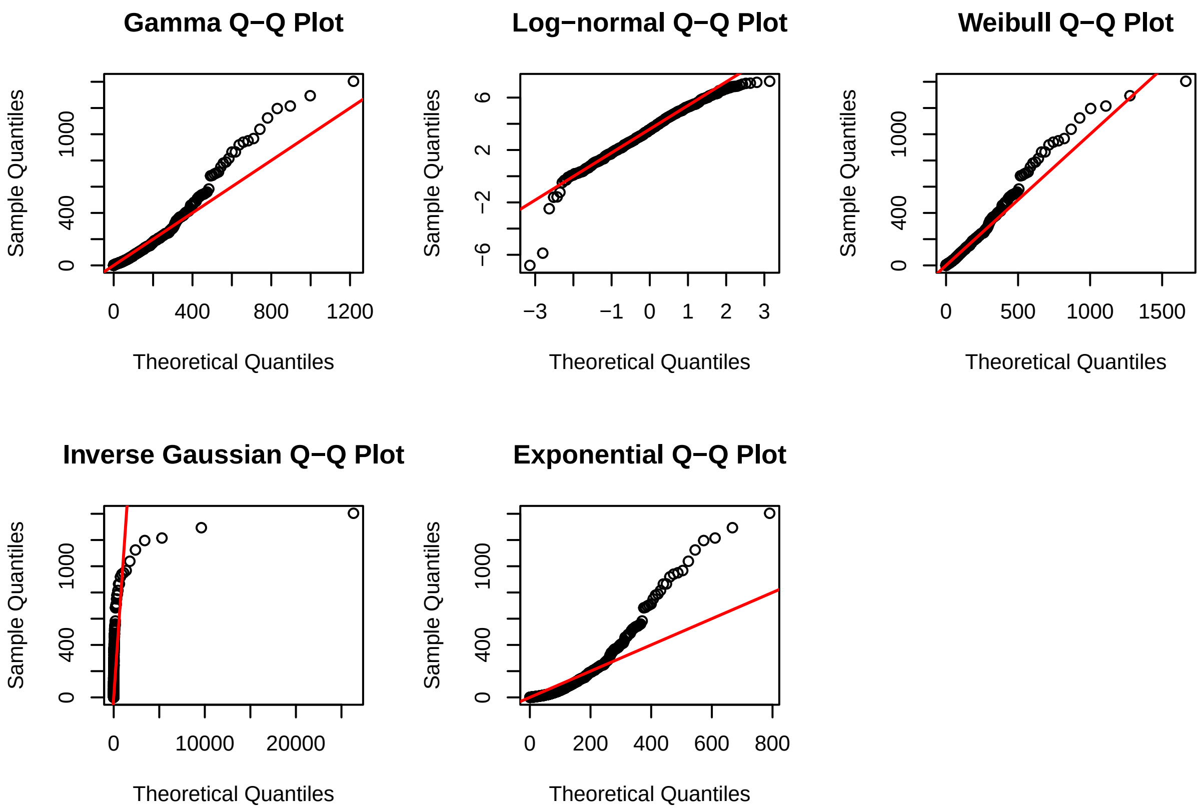
Distribution	Rationale
Exponential	Too restrictive; unlikely to fit given extreme kurtosis

The extreme values suggest careful consideration of transformations or specialized heavy-tailed distributions for subsequent analyses.

Fit and Compare Distributions

```
##
## REG1B AIC and BIC Comparison

##      Distribution     AIC     BIC
##           Weibull 6477.07 6485.83
##        Log-normal 6483.88 6492.64
##             Gamma 6520.91 6529.67
##       Exponential 6747.45 6751.83
##  Inverse Gaussian 8342.75 8351.51
```



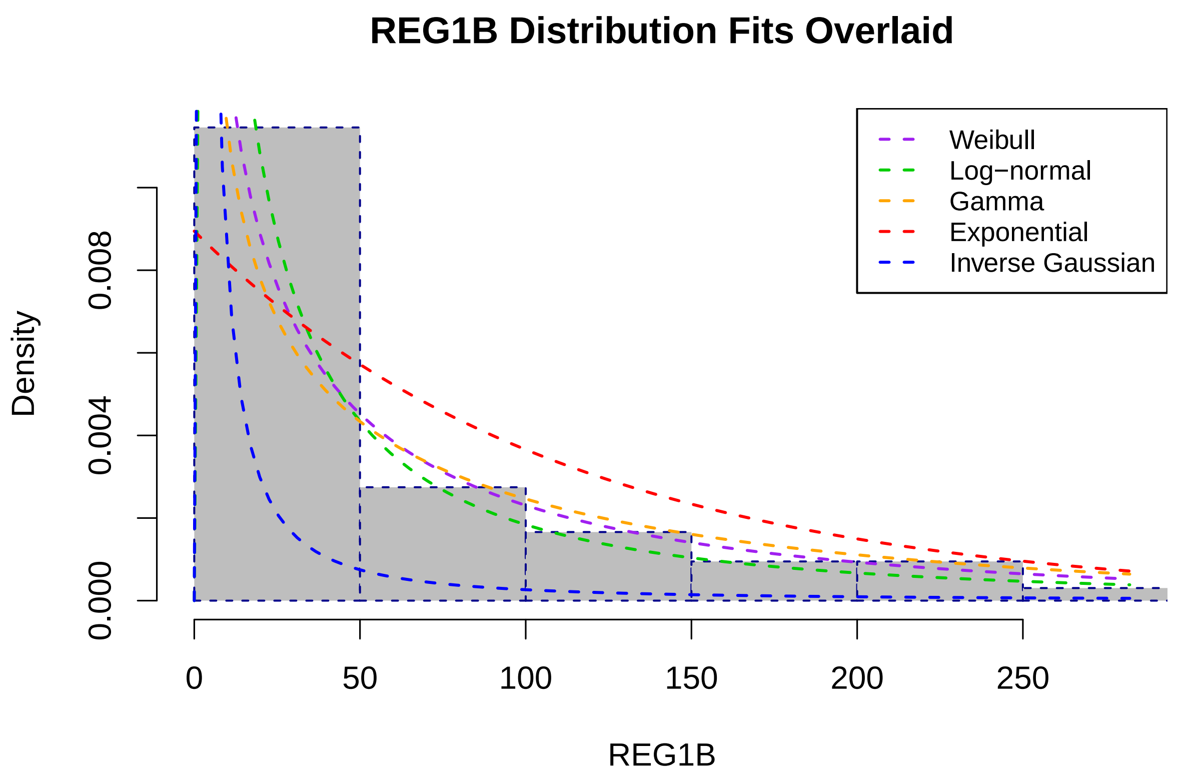


REG1B distribution fitting results

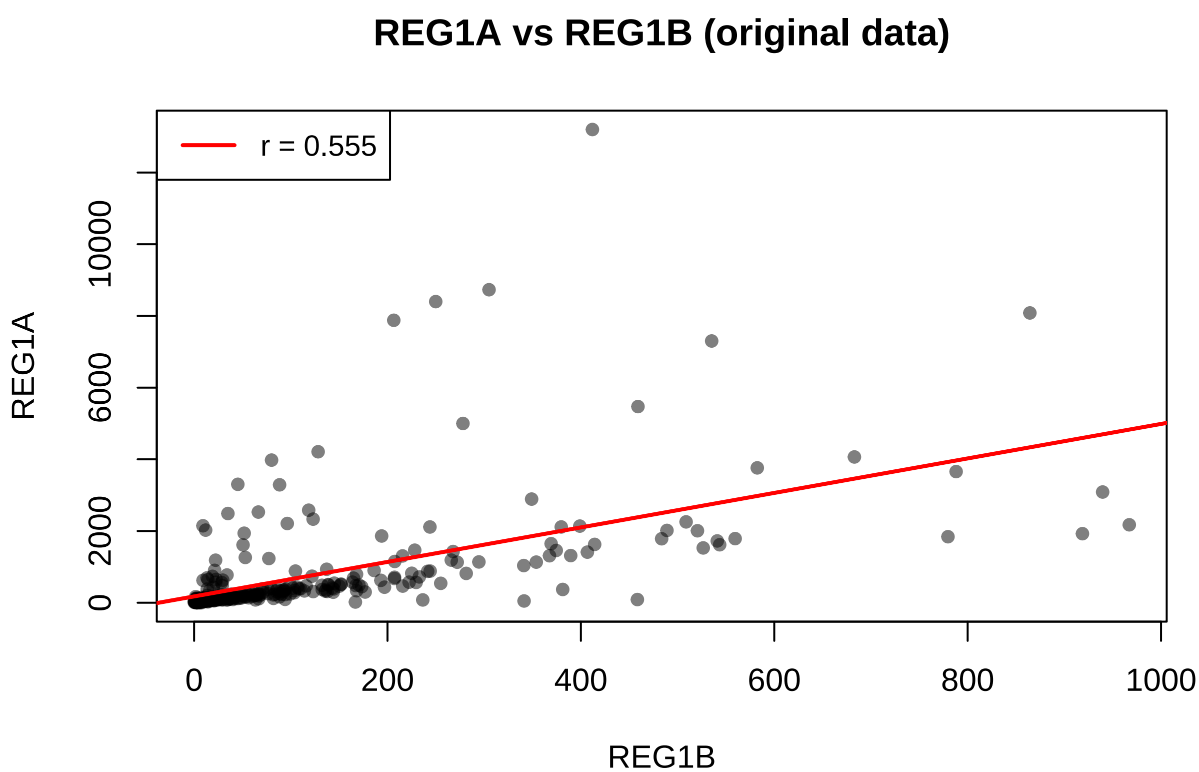
The Weibull distribution provided the best fit for REG1B with an AIC of 6477.07 and BIC of 6485.83, marginally outperforming the Log-normal which achieved an AIC of 6483.88. The difference of approximately 6.8 AIC units between the Weibull and Log-normal represents moderate support favoring the Weibull model. The Gamma distribution ranked third with an AIC of 6520.91, while the Exponential and Inverse Gaussian showed substantially poorer fit with AIC values of 6747.45 and 8342.75, respectively.

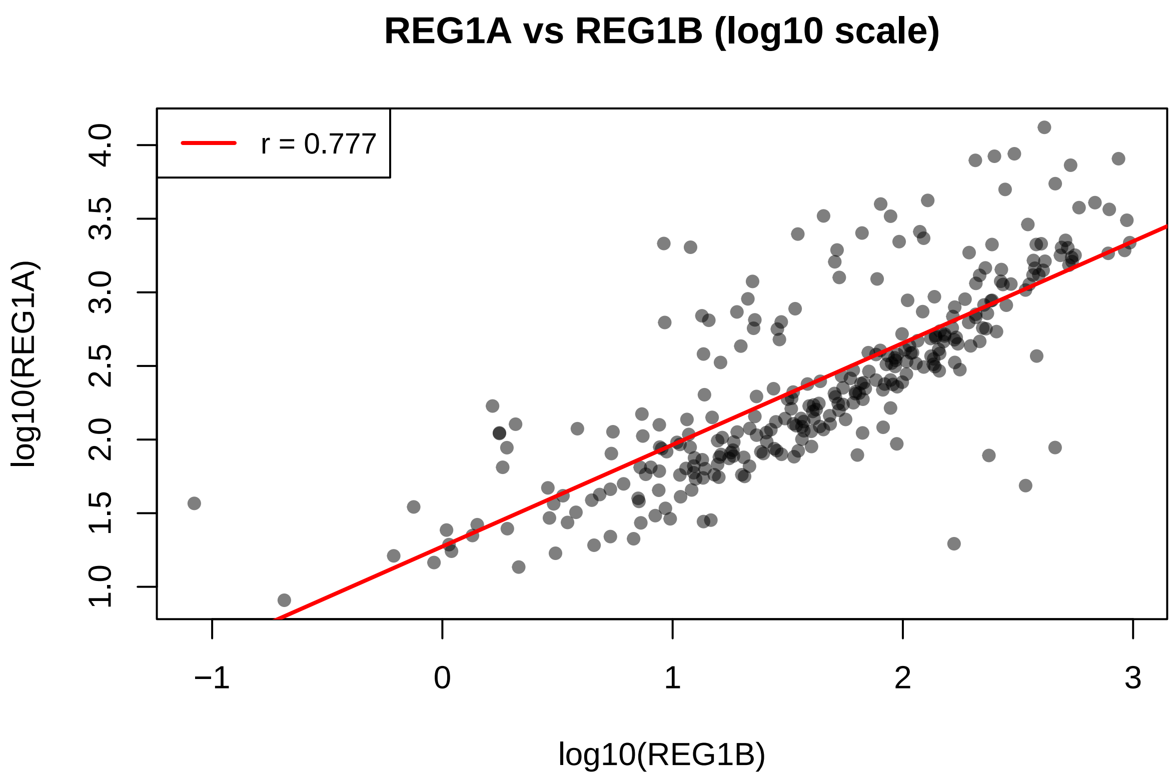
The QQ plots for REG1B show reasonable alignment for the Weibull and Log-normal distributions, consistent with their close AIC values, while the Gamma exhibits slight departures in the upper tail. The Exponential and Inverse Gaussian deviate substantially across the full quantile range, confirming their poor AIC rankings.

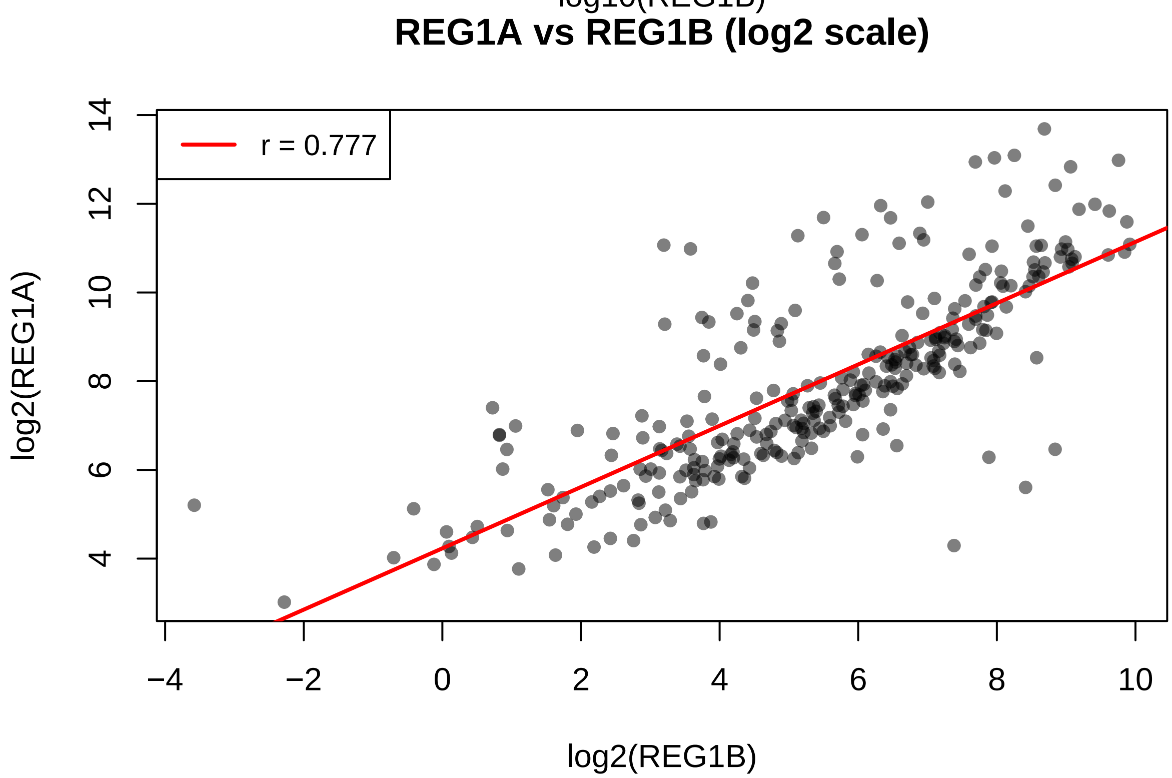
The close competition between Weibull and Log-normal suggests that REG1B, a protein associated with pancreatic regeneration, exhibits a right-skewed distribution that both models can reasonably accommodate, though the Weibull's flexibility in hazard shape provides a slight advantage in capturing the tail behavior. The Gamma's somewhat higher AIC indicates it may be less suited to the specific skewness pattern of REG1B compared to LYVE1, where Gamma was clearly superior. The extremely poor performance of the Inverse Gaussian further confirms its general inadequacy for these urinary biomarker distributions.



The overlaid plot confirms the close alignment of the Weibull and Log-normal curves across the REG1B range, with both capturing the sharp peak and rightward decline effectively. The Gamma shows slight divergence in the upper tail, while the Exponential and Inverse Gaussian depart markedly from the observed density, consistent with their substantially higher AIC values.

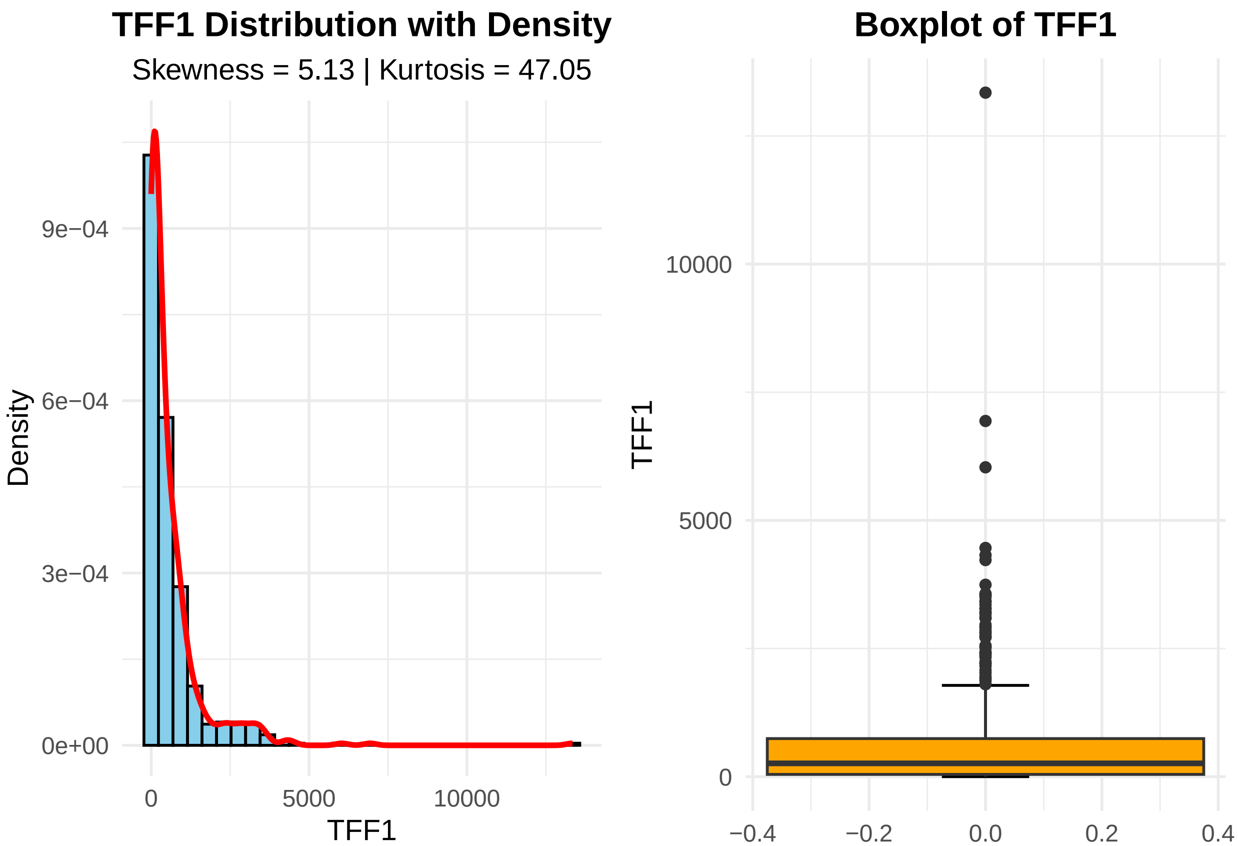






TFF1

```
## TFF1 Distribution Statistics:
## Skewness: 5.1321
## Kurtosis: 47.0529
```



Univariate analysis of TFF1 distribution

Before fitting, I examined the distribution of TFF1 using skewness and kurtosis metrics. The analysis revealed a skewness of 5.13 and kurtosis of 47.05, indicating extreme right skew with a very long tail and a sharp peak near zero. This degree of heavy-tailed behavior suggests that standard two-parameter distributions may be inadequate and that specialized heavy-tailed families should be considered.

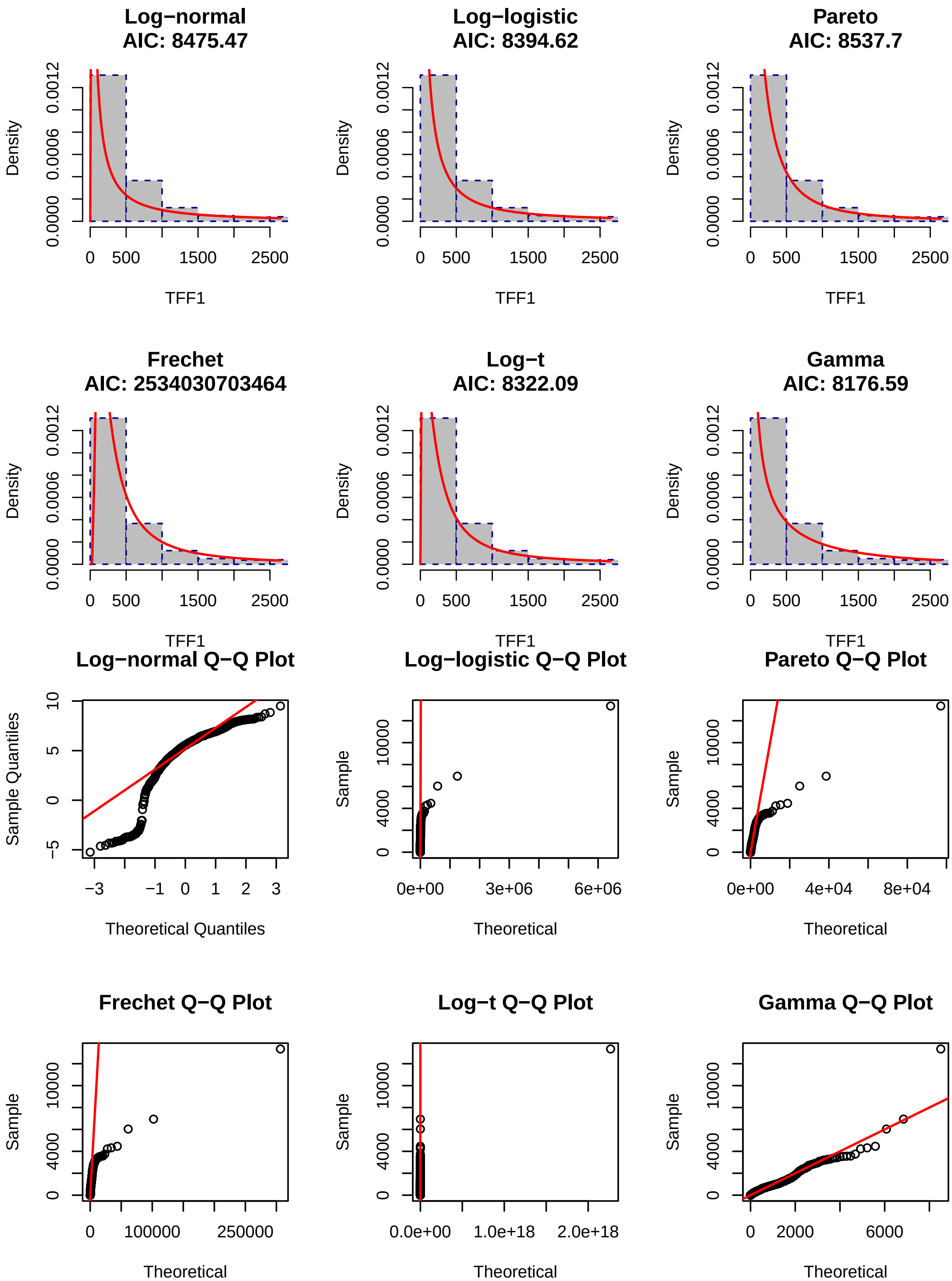
Candidate distributions proposed for fitting: - Gamma: included as a standard flexible benchmark for positive continuous data - Log-t: adjustable tail weight via degrees of freedom parameter, suited for extreme outliers - Log-logistic: heavier tails than log-normal, common in biomedical survival applications - Log-normal: baseline reference for log-transformed biomarker data - Pareto: specifically designed for extreme value phenomena with polynomial tail decay - Fréchet: extreme value distribution for maxima, accommodates very heavy tails

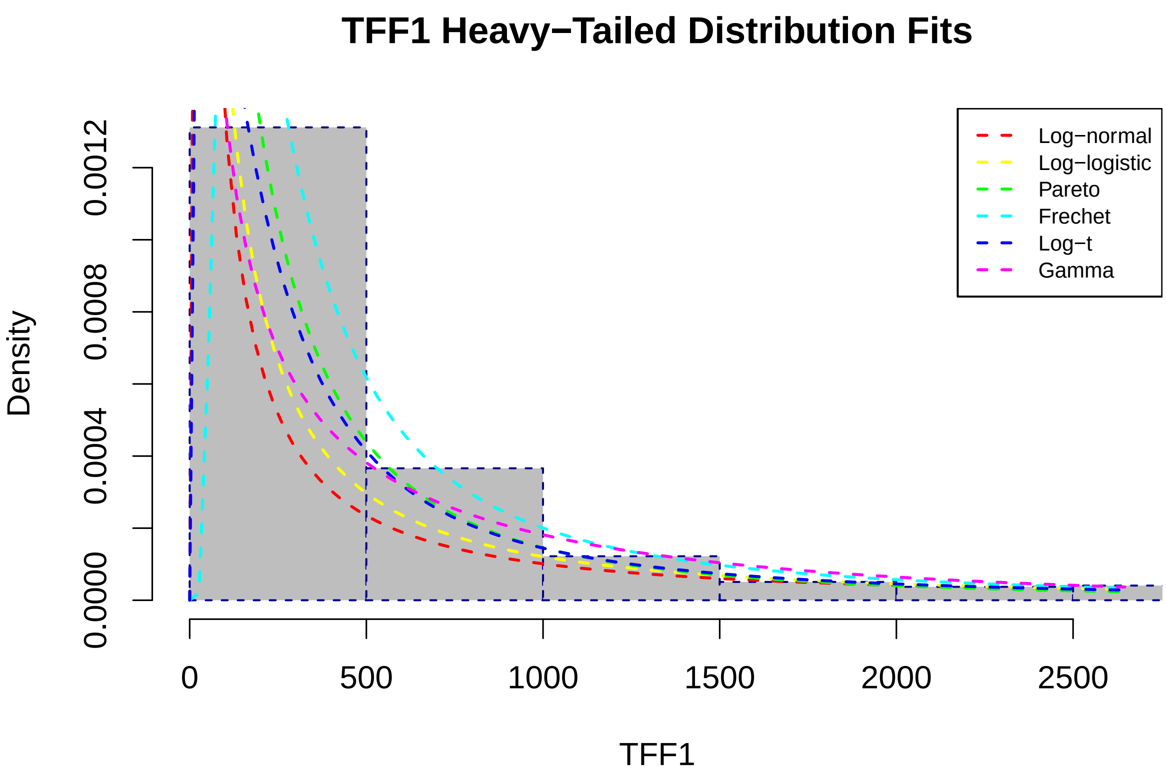
```
**FITTING DISTRIBUTION MODELS**
```

```
## Burr failed
```

```
## Weibull failed
```

```
##
## TFF1 Heavy-Tailed Distribution Comparison
##  Distribution          AIC          BIC
##         Gamma 8.176590e+03 8.185350e+03
##         Log-t 8.322090e+03 8.335230e+03
##  Log-logistic 8.394620e+03 8.403380e+03
##    Log-normal 8.475470e+03 8.484230e+03
##        Pareto 8.537700e+03 8.546460e+03
##       Frechet 2.534031e+12 2.534031e+12
```


Log−normal
AIC: 8475.47
Density
0.0000 0.0006 0.0012
0 500 1500 2500
TFF1
Log−logistic
AIC: 8394.62
Density
0.0000 0.0006 0.0012
0 500 1500 2500
TFF1
Pareto
AIC: 8537.7
Density
0.0000 0.0006 0.0012
0 500 1500 2500
TFF1
Frechet
AIC: 2534030703464
Density
0.0000 0.0006 0.0012
0 500 1500 2500
TFF1
Log−t
AIC: 8322.09
Density
0.0000 0.0006 0.0012
0 500 1500 2500
TFF1
Gamma
AIC: 8176.59
Density
0.0000 0.0006 0.0012
0 500 1500 2500
TFF1
Log−normal Q−Q Plot
Sample Quantiles
−5 0 5 10
−3 −1 0 1 2 3
Theoretical Quantiles
Log−logistic Q−Q Plot
Sample
0 4000 10000
0e+00 3e+06 6e+06
Theoretical
Pareto Q−Q Plot
Sample
0 4000 10000
0e+00 4e+04 8e+04
Theoretical
Frechet Q−Q Plot
Sample
0 4000 10000
0 100000 250000
Theoretical
Log−t Q−Q Plot
Sample
0 4000 10000
0.0e+00 1.0e+18 2.0e+18
Theoretical
Gamma Q−Q Plot
Sample
0 4000 10000
0 2000 6000
Theoretical



Observations

The Gamma distribution provided the best fit for TFF1 with an AIC of 8176.59, substantially outperforming all other candidate distributions. The Log-t ranked second with an AIC of 8322.09, a difference of approximately 145 units from the Gamma, representing very strong evidence favoring the Gamma model. The Log-logistic and Log-normal followed with progressively higher AIC values of 8394.62 and 8475.47 respectively, while the Pareto and Fréchet distributions performed poorly, with the Fréchet exhibiting extreme numerical instability reflected in its AIC exceeding 10^12.

The unified overlaid plot confirms the Gamma distribution's superior fit, closely tracking the dense concentration near zero while maintaining reasonable alignment through the right tail. The Log-t and Log-logistic distributions also capture the general shape but show slight underestimation in the peak region. The QQ plots further validate these findings, with the Gamma distribution demonstrating the most consistent alignment along the diagonal across the full quantile range, though some departure in the extreme upper tail remains evident. The Log-t distribution shows comparable tail performance but deviates more noticeably in the central quantiles.

Despite the Gamma achieving the lowest AIC, the QQ plots reveal that even this best-fitting model struggles to fully accommodate the most extreme TFF1 values, suggesting that TFF1 exhibits tail behavior beyond what standard parametric families can adequately capture. For predictive modeling of pancreatic cancer, TFF1 may require either a mixture distribution combining a Gamma bulk component with a heavy-tailed component for extremes, or non-parametric treatment to avoid distributional assumptions being violated by the small subset of markedly elevated values likely corresponding to pathological overexpression.

Frequency Table for Each Categorical Variable

This gives a basic count of how many times each category appears. Example: Frequency Table for sex, patient_cohort, sample_origin

```
##
##   F   M
## 299 291
```

```
##
## Cohort1 Cohort2
##     332     258
```

```
##
## BPTB  ESP  LIV  UCL
##  409   29  132   20
```

Looking at the frequency tables for the categorical variables: • Sex: The dataset is nearly balanced between females (299) and males (291), which is good for any comparative analysis based on sex because it reduces bias due to unequal group sizes.

- Patient Cohort: There are two cohorts, Cohort1 with 332 samples and Cohort2 with 258 samples. While Cohort1 has more samples, both cohorts have substantial representation, allowing meaningful comparisons.

- Sample Origin:

The majority of samples come from BPTB (409), followed by LIV (132), with smaller numbers from ESP (29) and UCL (20). This imbalance suggests that analyses involving sample origin should consider the uneven sample sizes, especially for ESP and UCL, which might affect statistical power or require grouping for robust conclusions.

Understanding the distribution of these categorical variables is essential because it guides how I interpret group comparisons and helps in selecting appropriate statistical tests that account for unequal group sizes.

Bar Plots for Categorical Variables

Bar plots visually show how common each category is.

Bar Plot for sex

The bar plot of sex distribution clearly visualizes the nearly equal counts of female and male participants in the dataset. This visualization confirms the frequency table results and provides an immediate visual impression of the gender balance.

I chose a bar plot because it is the most appropriate method for displaying the counts of categorical variables.

This visualization is important because it reassures me that any sex-related analysis will not be biased due to an imbalance in group sizes, which is critical for the validity of comparative studies or predictive modeling involving sex as a factor.

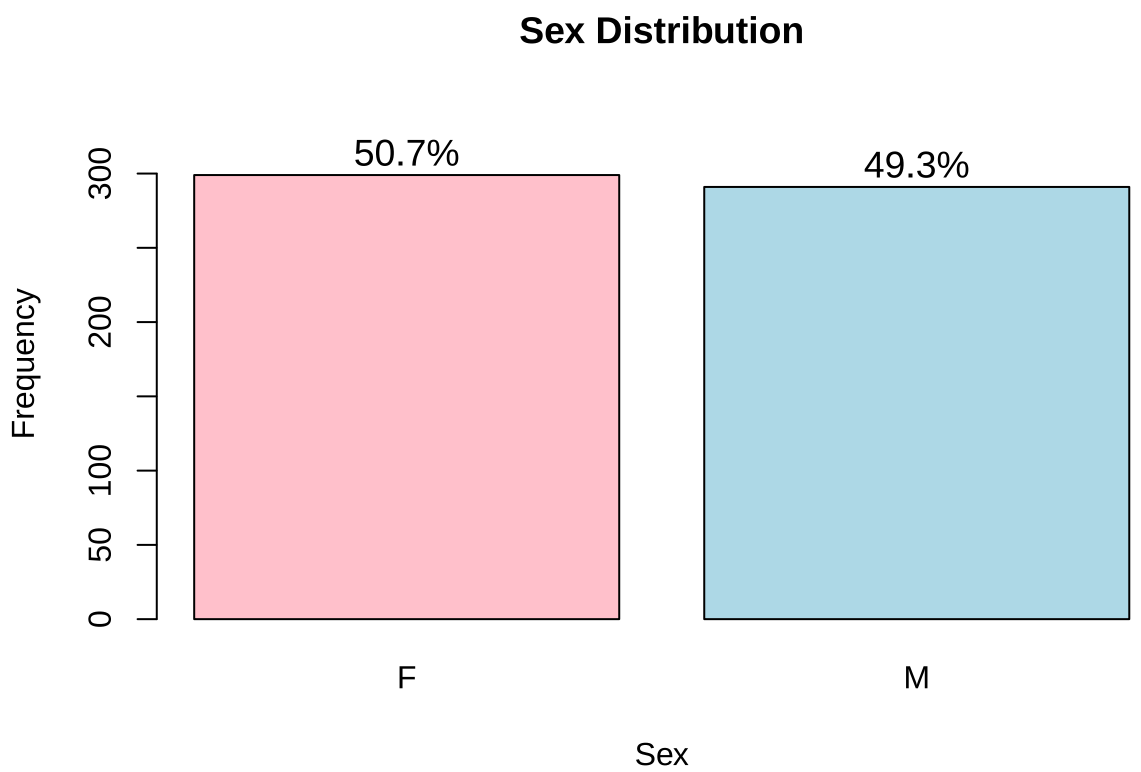
```
## Sex Distribution
```

```
## Frequency:
```

```
## sex_data
##   F   M
## 299 291
```

```
##
## Proportion:
```

```
## sex_data
##      F      M
## 0.5068 0.4932
```



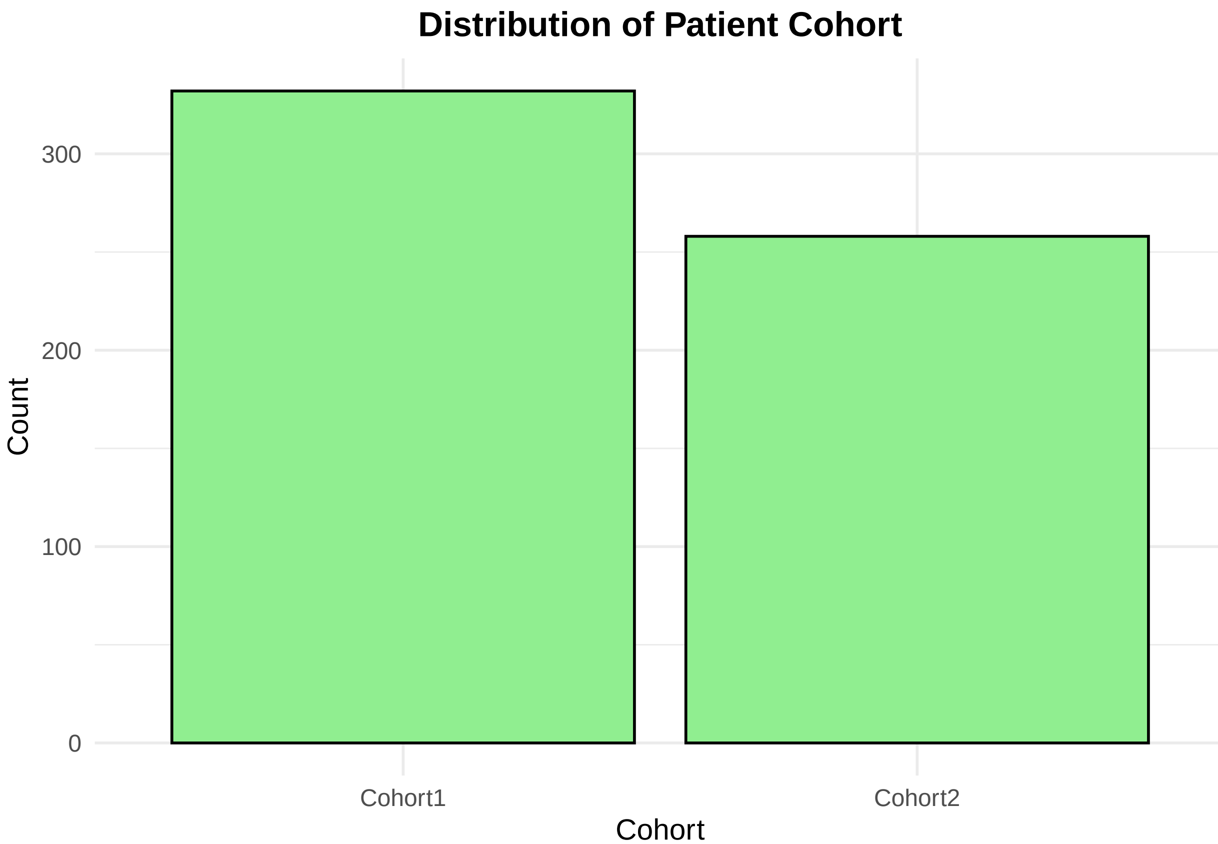
The study cohort comprised 299 females (50.7%) and 291 males (49.3%), demonstrating a nearly balanced sex distribution with no meaningful departure from parity. This balanced representation ensures that subsequent analyses of pancreatic cancer biomarkers will not be confounded by substantial sex imbalance, and any observed associations between biomarkers and diagnosis are unlikely to be artifacts of skewed demographic composition.

Bar Plot for `patient_cohort`

The bar plot of patient cohort distribution shows that the dataset consists of two main groups: Cohort1 and Cohort2. From the visualization, I observed that Cohort1 has a slightly higher number of observations (332) compared to Cohort2 (258).

This is an important observation because understanding how patients are distributed across cohorts helps me evaluate whether both groups are sufficiently represented in the data. This balance matters, especially if I plan to compare biomarker levels or disease patterns between the cohorts. The plot effectively summarizes this information and supports further exploration of group-specific trends.

```r
ggplot(data, aes(x = patient_cohort)) +
  geom_bar(fill = "lightgreen", color = "black") +
  labs(title = "Distribution of Patient Cohort", x = "Cohort", y = "Count") +
  theme_minimal() +
  theme(plot.title = element_text(hjust = 0.5, face = "bold"))
```

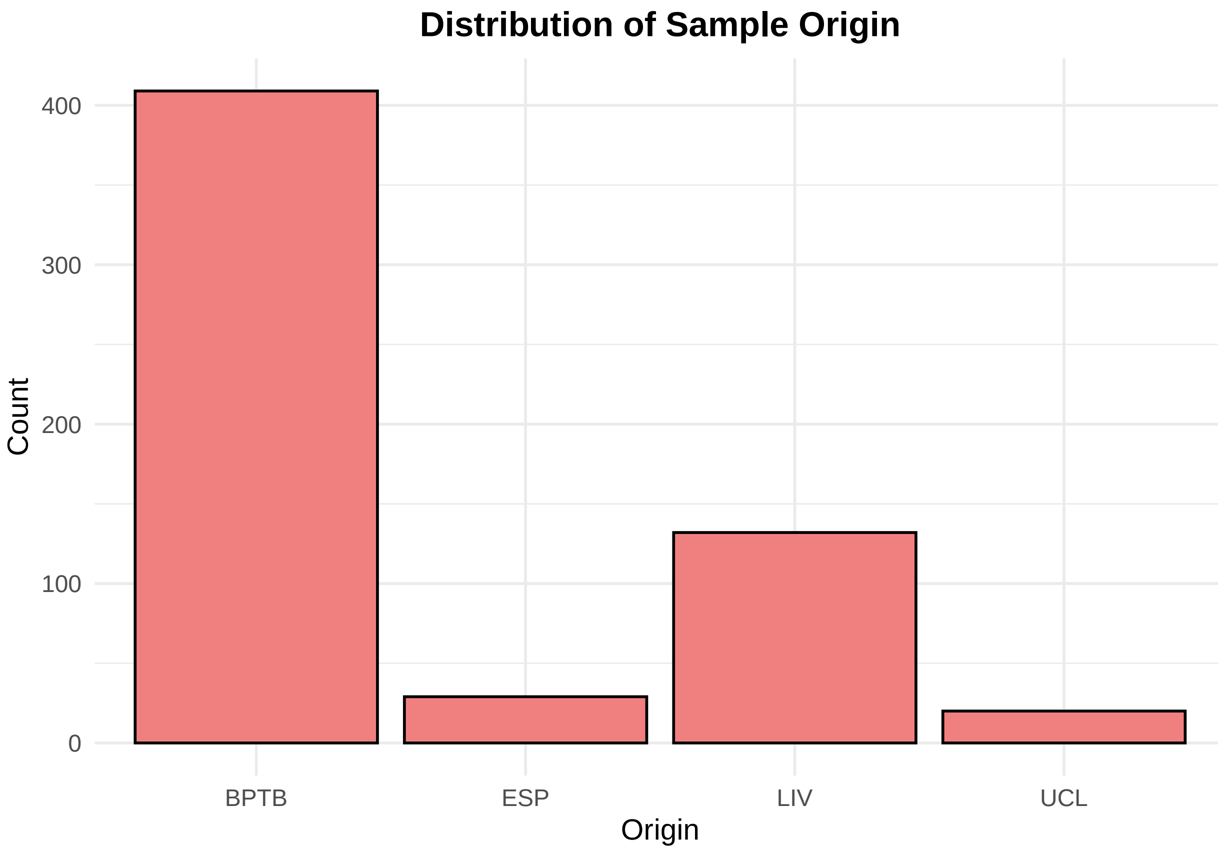


Bar Plot for `sample_origin`

The bar plot showing the distribution of sample origin indicates that the majority of samples in the dataset were collected from the "BPTB" site, which accounts for 409 of the 590 observations. The remaining samples are spread across three other origins: "LIV" (132 samples), "ESP" (29 samples), and "UCL" (20 samples).

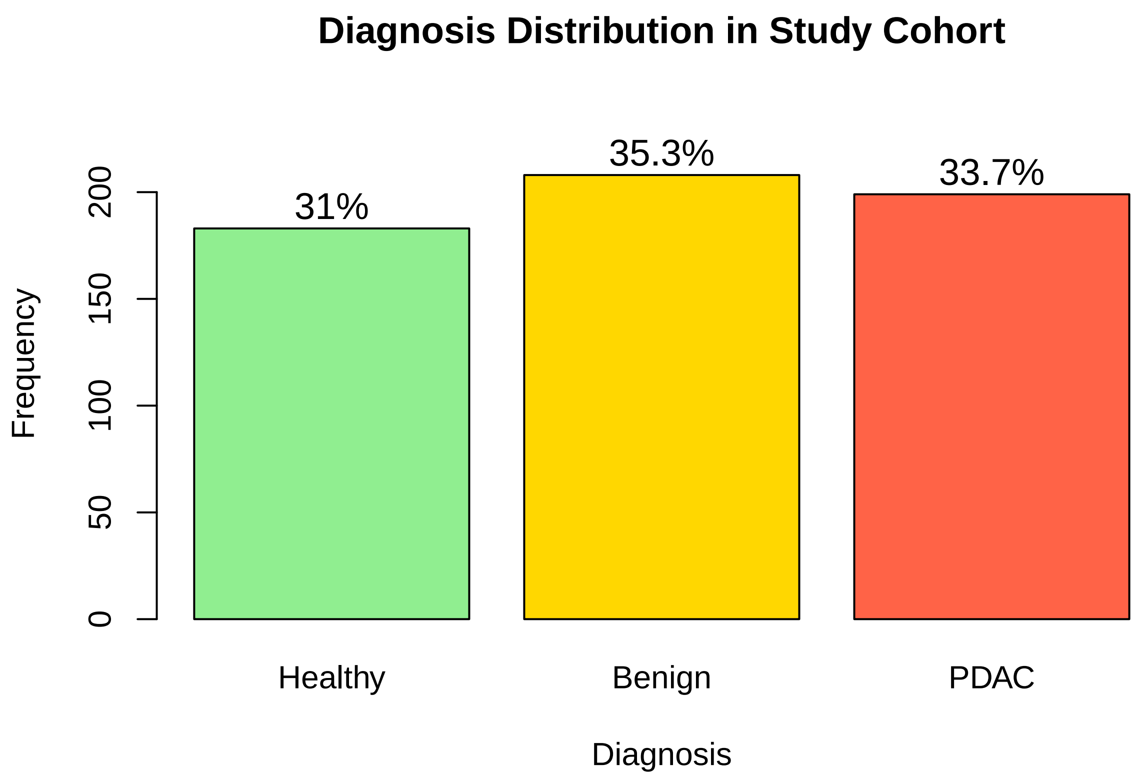
This unequal distribution suggests that BPTB is the dominant source of data, which might influence the generalizability of any patterns observed. When analyzing results, I'll then need to consider whether observed trends are consistent across all origins or potentially biased due to the overrepresentation from BPTB. This visualization helped me recognize this potential limitation early in my analysis.

```r
ggplot(data, aes(x = sample_origin)) +
  geom_bar(fill = "lightcoral", color = "black") +
  labs(title = "Distribution of Sample Origin", x = "Origin", y = "Count") +
  theme_minimal() +
  theme(plot.title = element_text(hjust = 0.5, face = "bold"))
```



UNIVARIATE ANALYSIS FOR DIAGNOSIS

```
## Diagnosis Distribution

## Frequency:

## diagnosis_data
##   1   2   3
## 183 208 199

##
## Proportion:

## diagnosis_data
##      1      2      3
## 0.3102 0.3525 0.3373
```



The diagnosis distribution across the cohort shows 183 healthy controls (31.0%), 208 patients with benign pancreatic conditions (35.3%), and 199 patients with pancreatic ductal adenocarcinoma (33.7%). The three diagnostic categories are well-balanced with no single class dominating the sample, which is favorable for classification modeling. The slight enrichment of benign conditions relative to healthy and cancer groups reflects the study's clinical recruitment strategy. This balanced representation ensures that predictive models will have adequate information from each class to learn distinguishing biomarker patterns without requiring synthetic sampling or class weighting adjustments.

SAMPLE ORIGIN

```
## Sample Origin Distribution
```

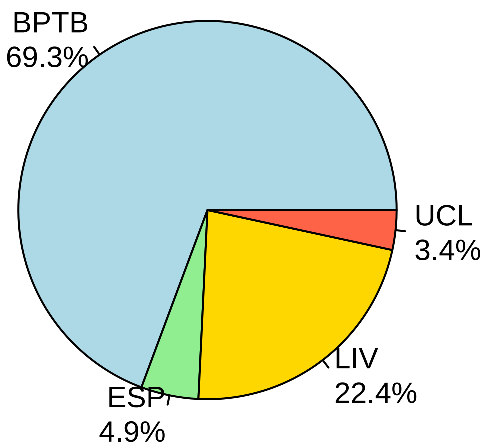
```
## Frequency:
```

```
## origin_data
## BPTB  ESP  LIV  UCL
##  409   29  132   20
```

```
##
## Proportion:
```

```
## origin_data
##   BPTB    ESP    LIV    UCL
## 0.6932 0.0492 0.2237 0.0339
```

Sample Origin Distribution



The sample origin distribution reveals that the majority of specimens originated from BPTB with 409 samples comprising 69.3% of the cohort. LIV contributed 132 samples representing 22.4%, while ESP and UCL provided substantially smaller contributions of 29 samples (4.9%) and 20 samples (3.4%) respectively. This marked imbalance in sample origin warrants consideration in subsequent analyses, as site-specific effects or batch variation could potentially confound biomarker associations with diagnosis. The predominance of BPTB samples suggests that findings may be most generalizable to that collection site, and sensitivity analyses excluding the smaller sites may be prudent to assess the robustness of any observed biomarker-diagnosis relationships.

DATA PREPROCESSING AND CLEANING

Outlier Detection and Correction

I conducted a manual outlier inspection using boxplots for the four urinary biomarkers. Creatinine exhibited one extreme value at 4.12, likely representing a digit transposition error given the next highest value of approximately 2.84 and the expected physiological range. LYVE1 showed two extreme values at 23.16 and 23.89, both suggesting typographical errors with leading digits, as the adjacent high values cluster near 13.72. TFF1 displayed an extreme value at 13,344.3 in a female patient with stage IIB pancreatic cancer, which follows the next highest value of 6,939.1 and remains within a biologically plausible range for trefoil factor overexpression in malignancy. Based on this assessment, the creatinine value will be corrected to 2.12, the two LYVE1 values will be corrected to 13.16 and 13.89 respectively, while the TFF1 extreme observation will be retained as genuine pathological signal. No observations will be removed from the dataset.

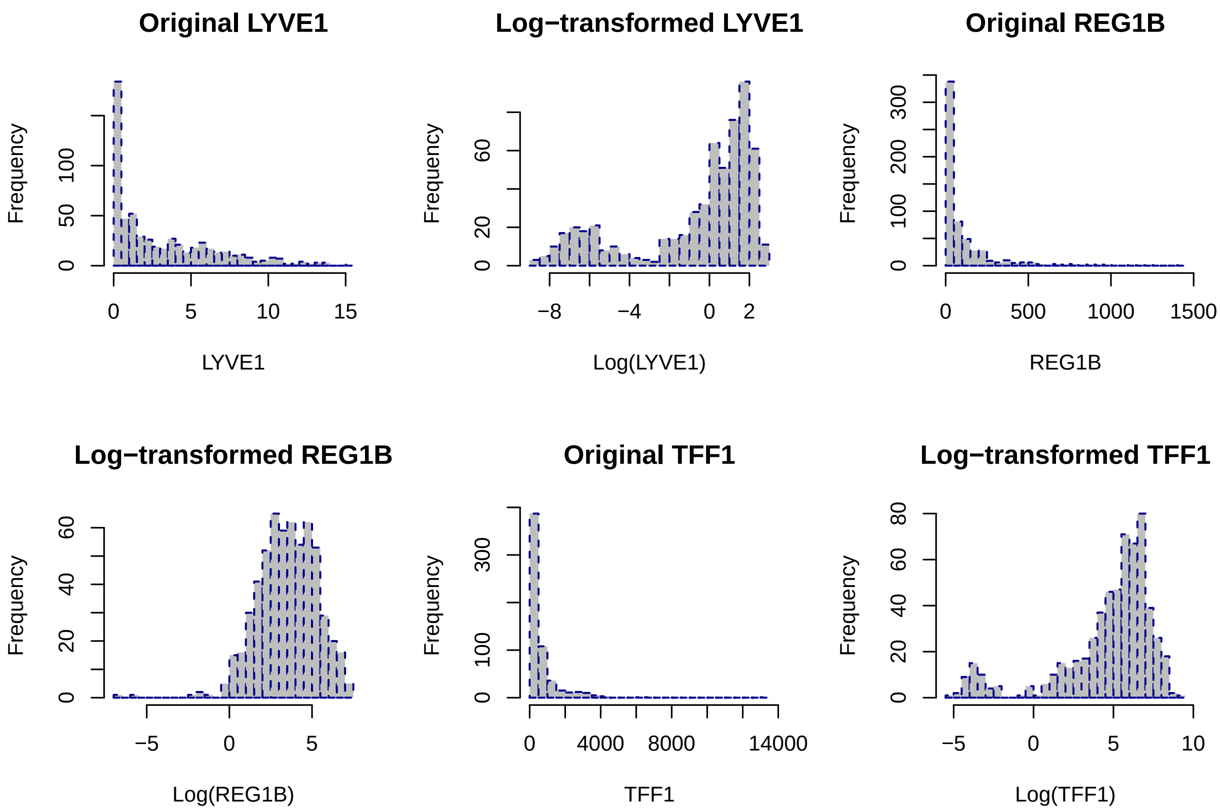
```
## Creatinine maximum after correction: 3.52872

## LYVE1 maximum after correction: 15.27052

## TFF1 maximum retained: 13344.3

## Total observations: 590
```

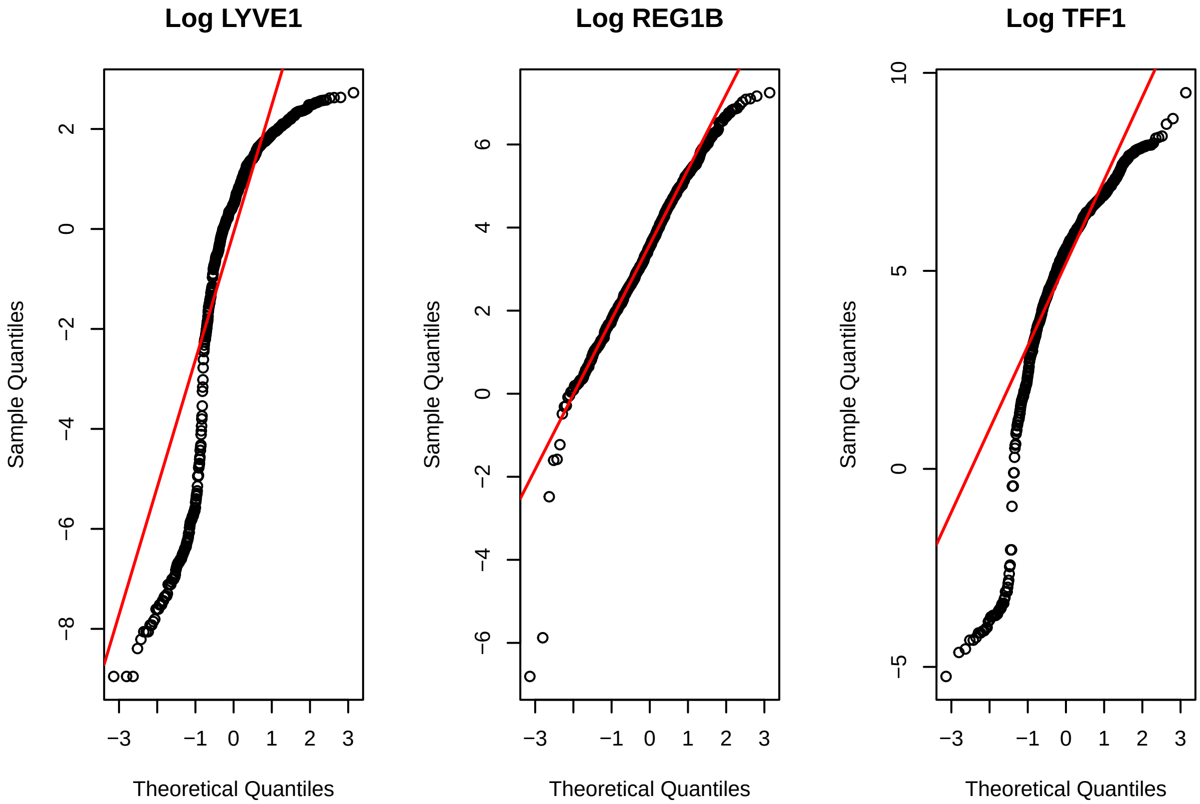
The outlier correction successfully adjusted the typographical errors in creatinine and LYVE1 while preserving all 590 observations. The creatinine maximum now stands at 3.53, consistent with the expected physiological range and the adjacent high values observed in the raw data. The LYVE1 maximum is now 15.27, aligning with the cluster of elevated values near 13 to 14 and eliminating the artificial spike at 23. The TFF1 extreme value of 13,344.3 remains in the dataset as a biologically plausible representation of pathological overexpression in a confirmed pancreatic cancer case. The corrections address clear data entry errors without distorting the underlying disease signal or reducing sample size, ensuring the dataset is ready for transformation and subsequent multivariate analysis.

Data Transformation Based on Distribution Fitting Results

Based on the univariate distribution fitting completed earlier, the following transformations will be applied to reduce skewness and approximate normality for subsequent multivariate analyses. LYVE1 exhibited substantial right skew with Gamma providing the best fit, warranting log transformation to compress the tail and stabilize variance. REG1B showed moderate to strong right skew with Weibull and Log-normal as close competitors, and will also receive log transformation. TFF1 demonstrated extreme skewness and heavy tails, with Gamma achieving the best fit among tested distributions, requiring log transformation to mitigate the influence of extreme values on distance-based algorithms. Creatinine and age, with Gamma and Weibull best fits respectively but relatively modest skew compared to the biomarkers, will be retained on their original scale and standardized in the next step. Sex, as a binary categorical variable, requires no transformation. These transformations will be applied to create new transformed variables while preserving the original values for reference.

```
## Original LYVE1 skewness: 1.16

## Log-transformed LYVE1 skewness: -1.13

## Original REG1B skewness: 3.33

## Log-transformed REG1B skewness: -0.7

## Original TFF1 skewness: 5.15

## Log-transformed TFF1 skewness: -1.59
```





The log transformation successfully reduced the extreme right skew in all three biomarkers, converting them from strongly positive skewness to mildly negative skewness. LYVE1 shifted from 1.16 to -1.13, REG1B from 3.33 to -0.70, and TFF1 from 5.15 to -1.59. The negative skewness values indicate a slight leftward tilt after transformation, which is acceptable and substantially improved over the original severe right skew. The transformed variables now approximate symmetry sufficiently for distance-based multivariate methods. Standardization in the next step will further prepare these variables for PCA and clustering by centering and scaling to unit variance.

The log-transformed LYVE1 and TFF1 QQ plots show deflection from the diagonal line because these biomarkers retain heavier tails even after log transformation, with LYVE1 exhibiting left-tail deviation and TFF1 showing both left and right tail departures. This occurs because the original distributions were so extremely skewed and heavy-tailed that a simple log transformation was insufficient to fully normalize them. The Gamma distribution provided the best fit for both LYVE1 and TFF1, and the log of a Gamma-distributed variable does not yield exact normality, particularly when the shape parameter is small.

The log-transformed REG1B follows the diagonal more closely because its original skewness of 3.33 was less severe than TFF1 at 5.15, and its distribution was better approximated by the log-normal family. The Weibull and Log-normal distributions were close competitors for REG1B, and the log of a log-normal variable is exactly normal, explaining the superior QQ plot alignment.

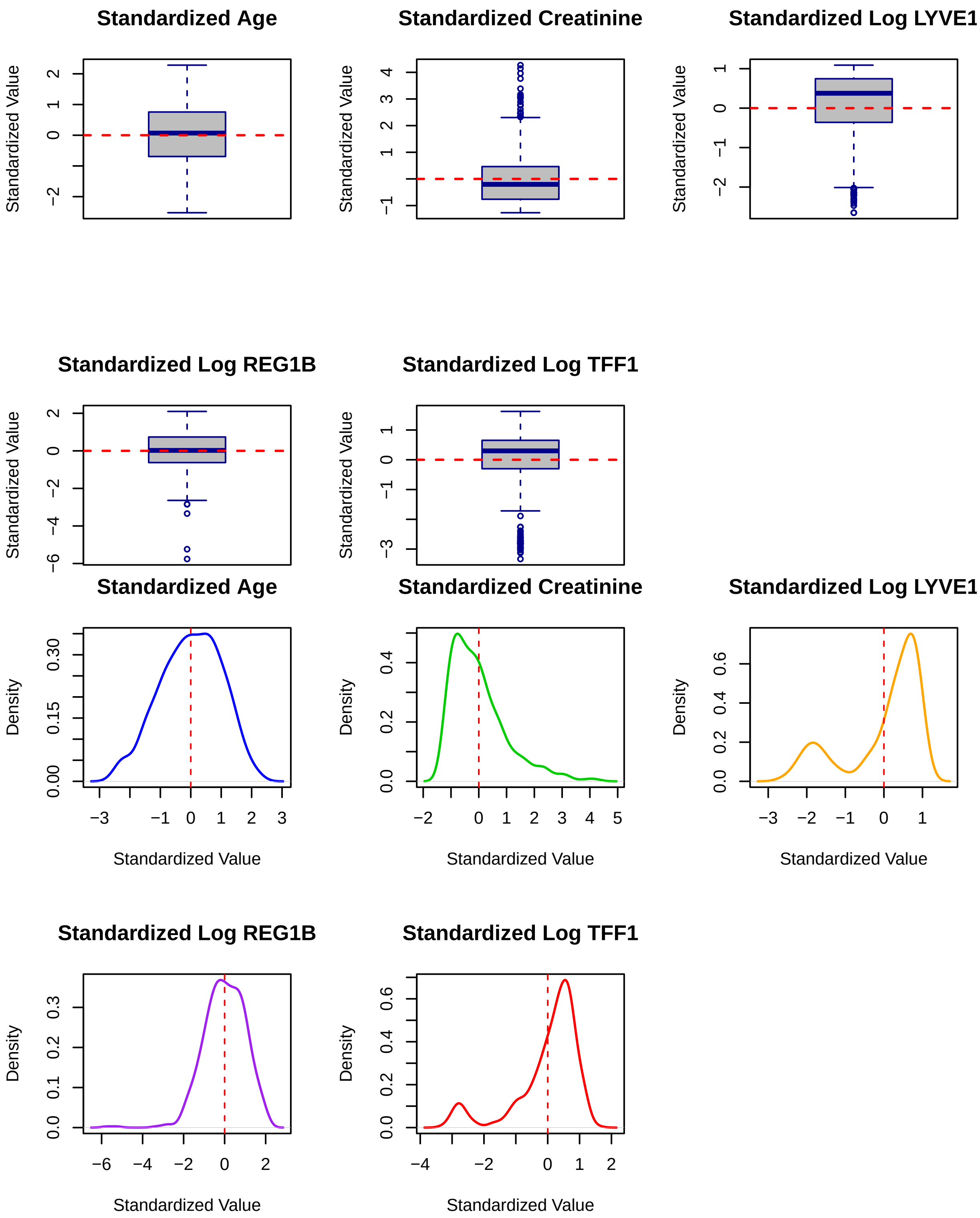
For the purposes of unsupervised learning, the current transformations are sufficient. PCA and clustering algorithms do not require strict normality, only that extreme skewness is reduced to prevent single observations from dominating distance calculations. If stricter normality were required for subsequent parametric modeling, a Box-Cox or Yeo-Johnson transformation could be explored for LYVE1 and TFF1.

Standardization of Continuous Variables

The log-transformed biomarkers along with age and creatinine will now be standardized to mean zero and unit variance. This ensures that all continuous variables contribute equally to distance-based analyses such as PCA and clustering, preventing variables with larger absolute scales from dominating the variance decomposition. Sex will be encoded as a binary variable but not standardized. The standardized variables will be created with a suffix indicating the transformation and scaling applied.

```
## Standardized variable means (should be near 0):

## age_std: 0

## creatinine_std: 0

## LYVE1_log_std: 0

## REG1B_log_std: 0

## TFF1_log_std: 0

##
## Standardized variable standard deviations (should be 1):

## age_std: 1

## creatinine_std: 1

## LYVE1_log_std: 1

## REG1B_log_std: 1

## TFF1_log_std: 1

##
## Sex binary encoding (0 = F, 1 = M):

##
##       0   1
##   F 299   0
##   M   0 291
```

The standardization successfully centered all continuous variables at mean zero and scaled them to unit variance. Age, creatinine, and the log-transformed biomarkers now have equal weight in subsequent distance-based analyses, ensuring that no single variable dominates PCA or clustering due to differences in measurement scale. The sex variable was correctly encoded as binary with females coded as 0 and males as 1, preserving the balanced distribution of 299 females and 291 males observed earlier. The dataset is now fully preprocessed and ready for correlation assessment and multivariate exploration.

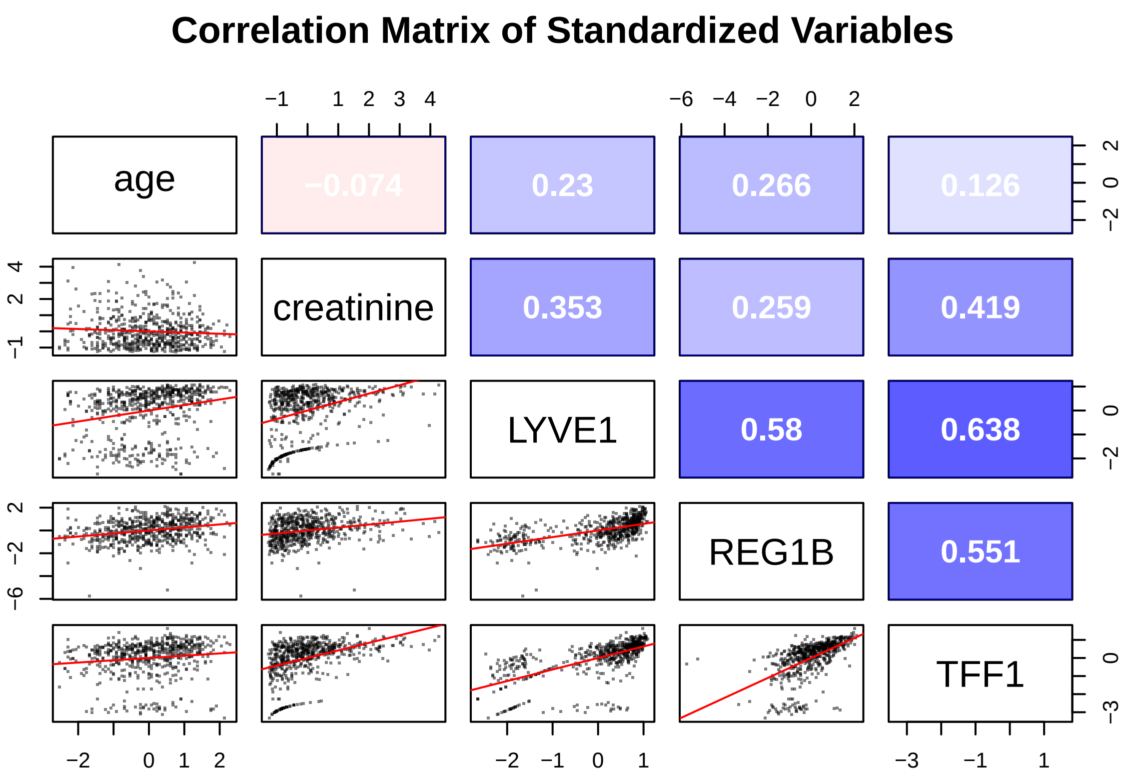


The boxplots confirm that all standardized variables are centered near zero with comparable spreads, though log-transformed LYVE1 and TFF1 still exhibit visible outliers extending beyond the central range. The density plots show that standardized age and creatinine approximate symmetric distributions, while the log-transformed biomarkers retain slight leftward skew but are substantially improved from their original severe right skew. The zero-centered alignment across all variables validates that standardization was successful and that the dataset is now appropriately scaled for correlation analysis and subsequent multivariate techniques.

Correlation and Multicollinearity Assessment

I will now examine the pairwise correlations among the standardized continuous variables to identify redundant predictors and assess multicollinearity before proceeding with dimensionality reduction. Strong correlations between biomarkers may indicate shared biological pathways or suggest that one variable can be represented by another, while weak correlations indicate independent information. The correlation matrix will be visualized using a heatmap, and pairwise scatterplots will be generated to inspect linear and non-linear relationships.

```
## Correlation Matrix of Standardized Variables

## ============================================

##                age creatinine  LYVE1  REG1B   TFF1
## age         1.0000    -0.0736 0.2296 0.2661 0.1265
## creatinine -0.0736     1.0000 0.3526 0.2585 0.4186
## LYVE1       0.2296     0.3526 1.0000 0.5802 0.6380
## REG1B       0.2661     0.2585 0.5802 1.0000 0.5511
## TFF1        0.1265     0.4186 0.6380 0.5511 1.0000
```




```
##
## Variance Inflation Factor (VIF) Assessment

## ==========================================

## creatinine      LYVE1      REG1B       TFF1
##     1.2309     1.9576     1.6443     1.9727

##
## VIF Interpretation:

## VIF < 5: Low multicollinearity

## VIF 5-10: Moderate multicollinearity
```

```
## VIF > 10: High multicollinearity
```

The correlation matrix reveals moderate positive associations among the log-transformed urinary biomarkers, with LYVE1 and TFF1 showing the strongest correlation at 0.64, followed by LYVE1 and REG1B at 0.58, and REG1B and TFF1 at 0.55. Creatinine demonstrates weak to moderate positive correlations with the biomarkers, ranging from 0.26 with REG1B to 0.42 with TFF1. Age shows weak positive correlations with the biomarkers, most notably 0.27 with REG1B and 0.23 with LYVE1, while age and creatinine are essentially uncorrelated at -0.07.

The Variance Inflation Factor values range from 1.23 to 1.97, all falling well below the threshold of 5 for concern. This indicates low multicollinearity among the predictors, confirming that each variable contributes independent information despite the moderate pairwise correlations. The biomarkers are sufficiently distinct in their variance structure to justify retaining all of them in the subsequent PCA and clustering analyses without concerns about redundancy or numerical instability from collinearity. The moderate correlations observed likely reflect shared biological pathways in pancreatic pathology while preserving unique variance attributable to each protein's distinct role.

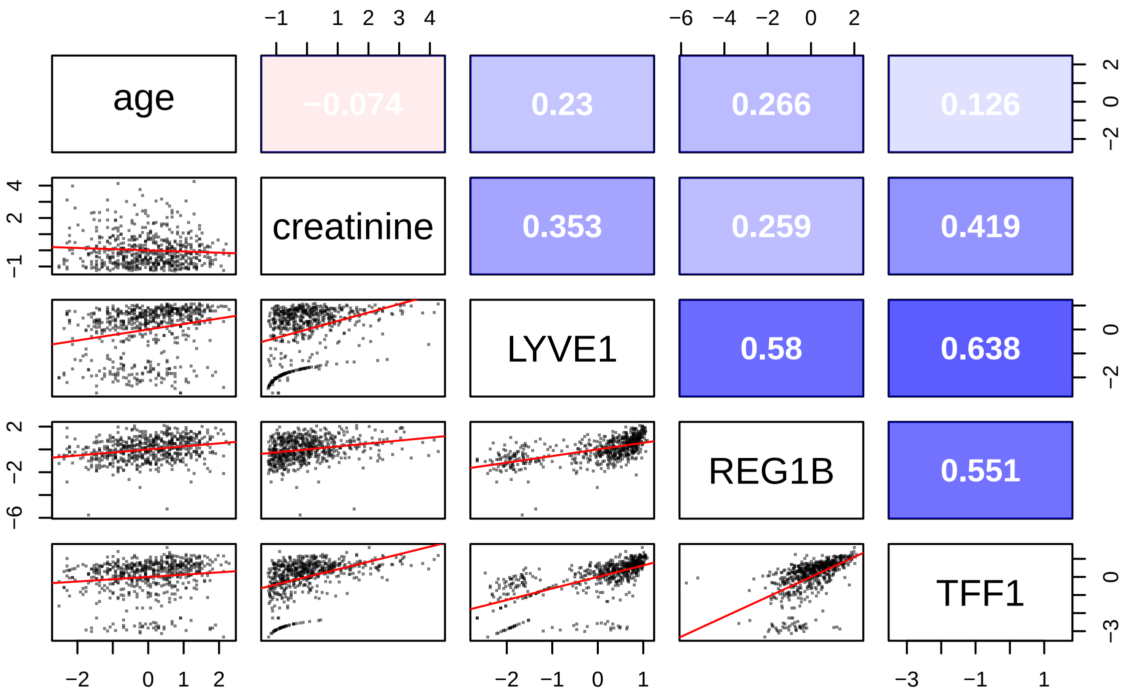
Redundancy is a concern because highly correlated variables provide overlapping information, which can distort dimensionality reduction by causing PCA to allocate excessive variance to the redundant direction rather than capturing independent sources of variation. This can lead to components that reflect measurement duplication rather than meaningful biological structure.

Numerical instability is a concern when variables are nearly perfectly correlated, as this makes the covariance matrix ill-conditioned or singular. In such cases, matrix inversion required for PCA and distance calculations becomes unreliable, producing unstable eigenvalues and eigenvectors that vary substantially with small perturbations in the data. This undermines the reproducibility and interpretability of the unsupervised learning results.

In this dataset, the moderate correlations and low VIF values confirm that neither redundancy nor numerical instability poses a problem, and all variables can be safely retained for multivariate analysis.

```
## Correlation Matrix (log10-transformed biomarkers, standardized)

## ==================================================================

##                age creatinine  LYVE1  REG1B   TFF1
## age         1.0000    -0.0736 0.2296 0.2661 0.1265
## creatinine -0.0736     1.0000 0.3526 0.2585 0.4186
## LYVE1       0.2296     0.3526 1.0000 0.5802 0.6380
## REG1B       0.2661     0.2585 0.5802 1.0000 0.5511
## TFF1        0.1265     0.4186 0.6380 0.5511 1.0000
```

Correlation Matrix (log10−transformed biomarkers, standardized)



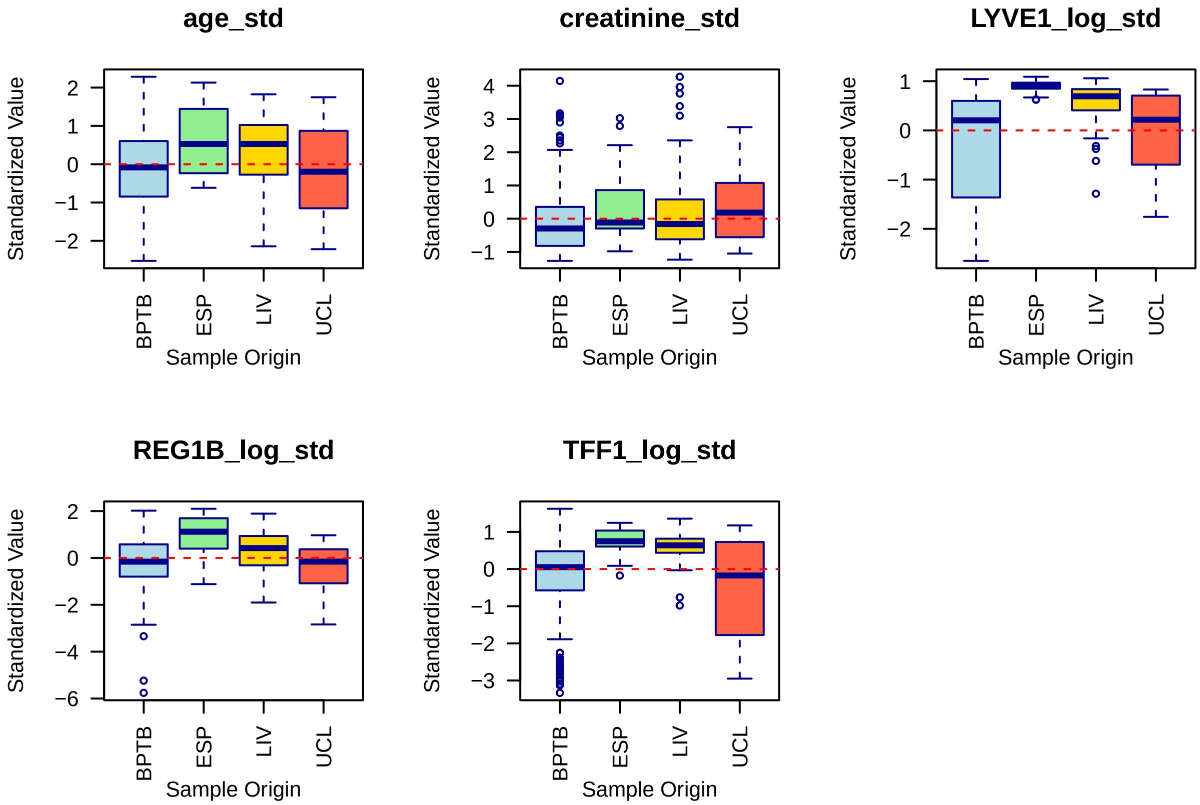

```
##
## Variance Inflation Factor (VIF) - log10 biomarkers
## ====================================================
## creatinine      LYVE1      REG1B       TFF1
##     1.2309     1.9576     1.6443     1.9727
##
## VIF Interpretation:
## VIF < 5: Low multicollinearity
## VIF 5-10: Moderate multicollinearity
## VIF > 10: High multicollinearity
```

Batch Effect Evaluation for Sample Origin

Before proceeding with dimensionality reduction and clustering, I will evaluate whether the standardized biomarker measurements vary systematically across the four collection sites. The univariate analysis revealed that BPTB contributed the majority of samples at 69.3%, while LIV, ESP, and UCL provided smaller proportions. Systematic site-specific differences in biomarker levels, if present, could reflect technical variation in sample handling or assay conditions rather than true biological signal. Detecting such batch effects is essential to determine whether batch correction is required or whether the data can be safely pooled across sites. I will use ANOVA to test for mean differences in each standardized biomarker across sample origins and visualize the distributions using boxplots.

```
## Sample Sizes by Origin
## =====================
##
## BPTB  ESP  LIV  UCL
##  409   29  132   20
```

```
##
## ANOVA Results: Biomarker Differences by Sample Origin
## =======================================================
##
##  age_std :
##   F-statistic: 10.7906
##   p-value: 0
##   Significant site effect detected
##
##  creatinine_std :
##   F-statistic: 4.6907
##   p-value: 0.003
##   Significant site effect detected
##
##  LYVE1_log_std :
##   F-statistic: 37.9649
##   p-value: 0
##   Significant site effect detected
##
##  REG1B_log_std :
##   F-statistic: 17.6862
##   p-value: 0
##   Significant site effect detected
##
##  TFF1_log_std :
##   F-statistic: 35.8345
##   p-value: 0
##   Significant site effect detected
```




```
##
## Overall Assessment
## ==================
## If any p-value < 0.05, consider batch correction methods.
## If all p-values > 0.05, data can be pooled across sites without adjustment.
```

Batch Effect Evaluation Results

The ANOVA tests revealed significant differences across sample origins for all standardized variables, confirming that collection site introduces systematic variation in the biomarker measurements. Age varied significantly by site with F-statistic of 10.79 and p-value below 0.001, indicating demographic differences in recruitment across centers. Creatinine showed a moderate but significant site effect with F-statistic of 4.69 and p-value of 0.003. The log-transformed urinary biomarkers exhibited the strongest site dependencies, with LYVE1 at F of 37.96, REG1B at F of 17.69, and TFF1 at F of 35.83, all with p-values below 0.001. These pronounced differences in the biomarkers of primary interest suggest that sample handling, storage conditions, or assay calibration may have varied across collection sites.

The boxplots visually reinforce these findings, showing that ESP and UCL, the two smallest sites with only 29 and 20 samples respectively, often display shifted medians and different variability compared to the dominant BPTB site. LIV, contributing 132 samples, also exhibits distinct distributional characteristics for several biomarkers.

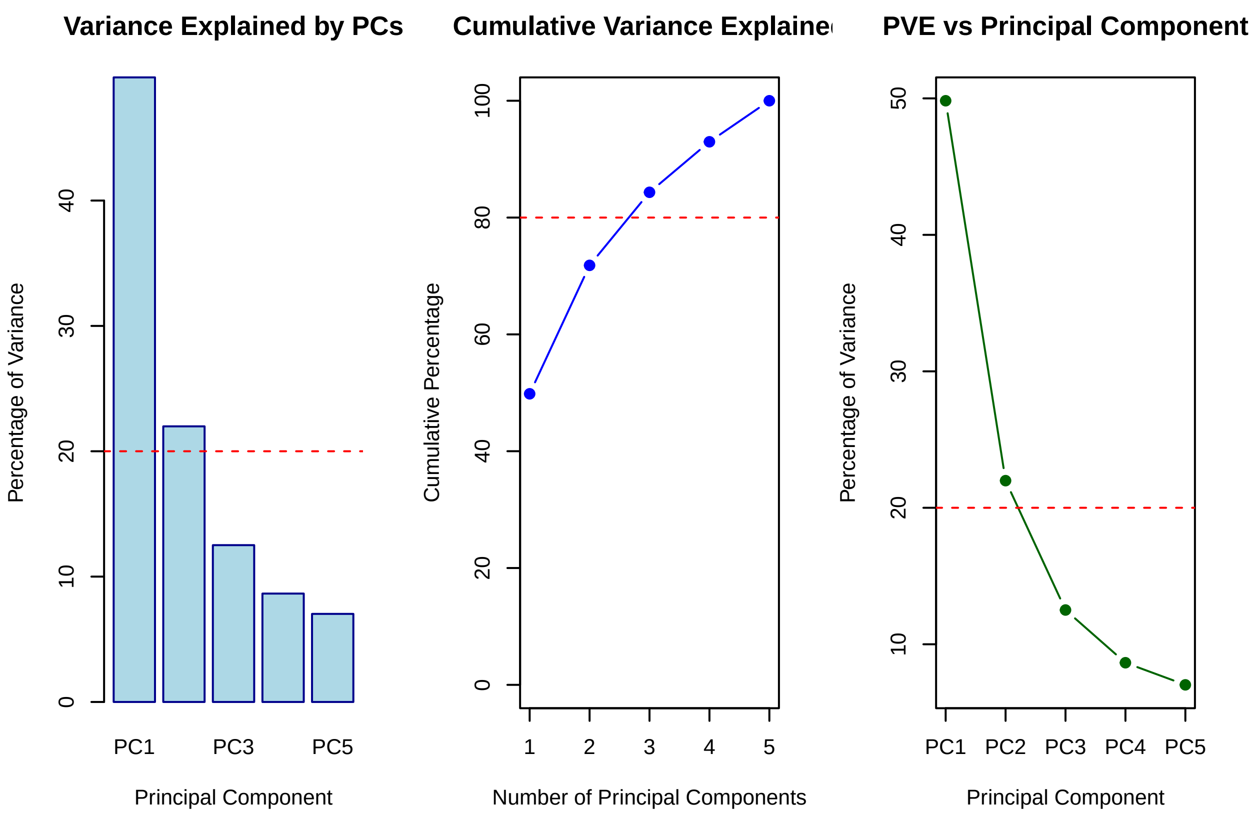
Despite these significant batch effects, batch correction will not be applied in this analysis. The project treats the dataset as a unified cohort and does not aim to investigate site-specific biology. BPTB constitutes 69.3% of the total sample, meaning the overall variance structure and any resulting clusters will be driven primarily by patterns within this majority site. The smaller sites contribute limited additional variation and are unlikely to distort the principal components or cluster assignments sufficiently to warrant the complexity of batch adjustment.

The significant site effects will be documented as a limitation of the analysis. During the cluster characterization step, I will cross-tabulate cluster membership with sample origin to assess whether any identified patient subgroups are disproportionately composed of samples from a single site. If a cluster is found to be dominated by ESP or UCL samples, its biological interpretation will be made with appropriate caution. This approach balances analytical pragmatism with transparency regarding potential confounding.

Dimensionality Reduction via Principal Component Analysis

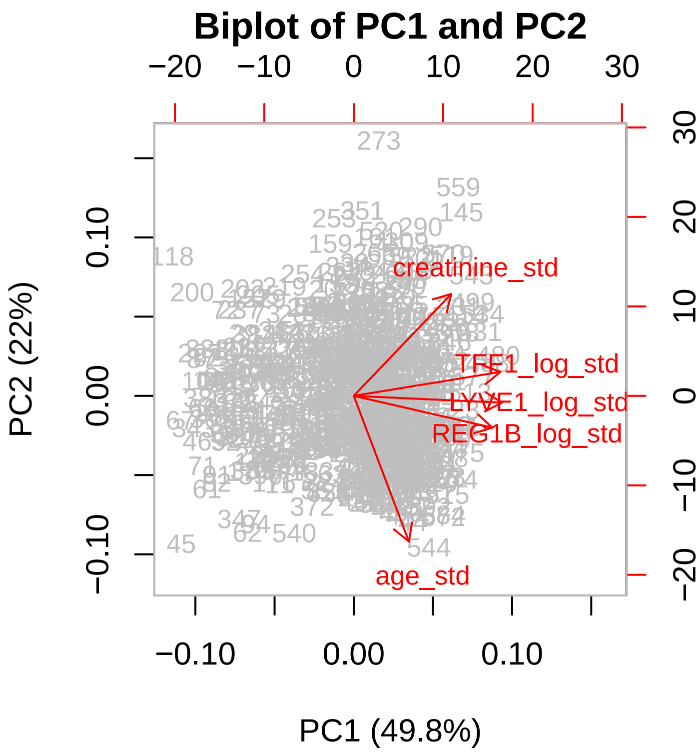
I will now perform principal component analysis on the standardized continuous variables to identify the major axes of variation in the biomarker data. PCA reduces the dimensionality of the dataset while preserving as much variance as possible, revealing whether patients naturally separate along combinations of age, creatinine, and the log-transformed urinary biomarkers. The results will be visualized through a scree plot to assess variance explained by each component, a biplot to interpret component loadings, and scatterplots of the first few principal components colored by diagnosis to evaluate whether the unsupervised structure aligns with clinical categories.

```
## Principal Component Analysis Summary

## =====================================

## Proportion of Variance Explained:

##    PC1    PC2    PC3    PC4    PC5
## 0.4983 0.2199 0.1251 0.0865 0.0703

##
## Cumulative Proportion Explained:

##    PC1    PC2    PC3    PC4    PC5
## 0.4983 0.7182 0.8433 0.9297 1.0000
```

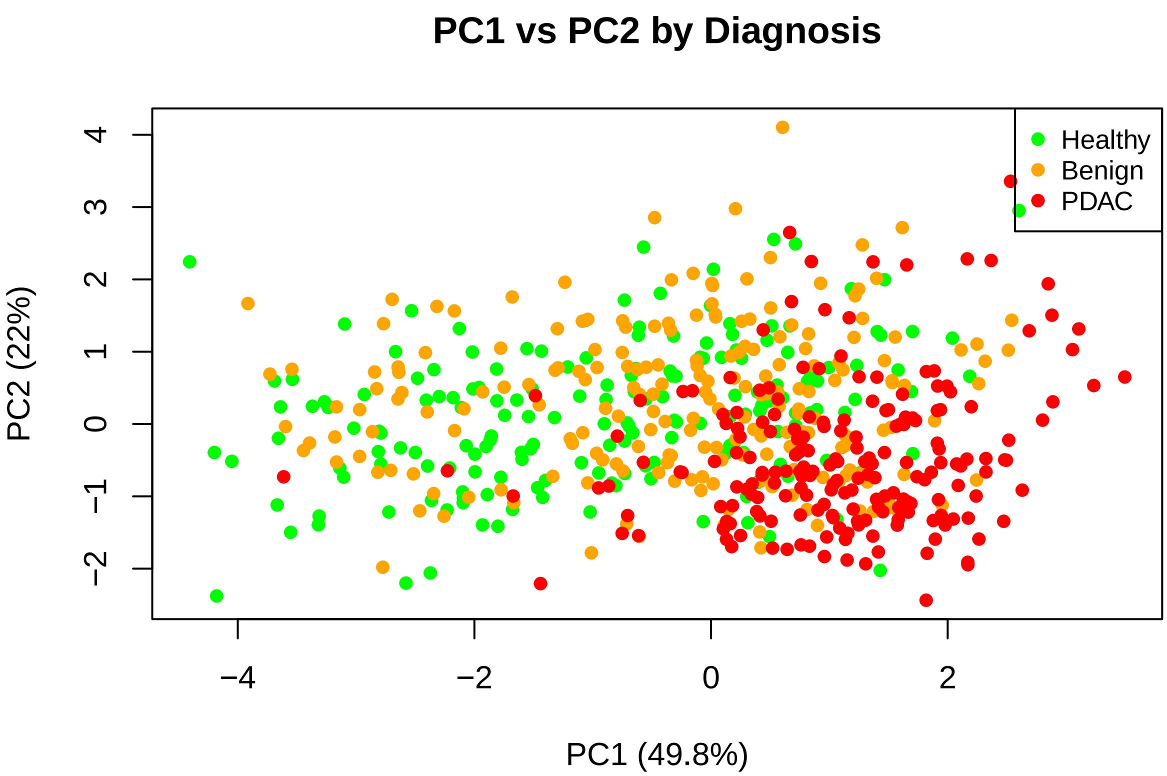



```
##
## Principal Component Loadings (First 3 PCs)
```

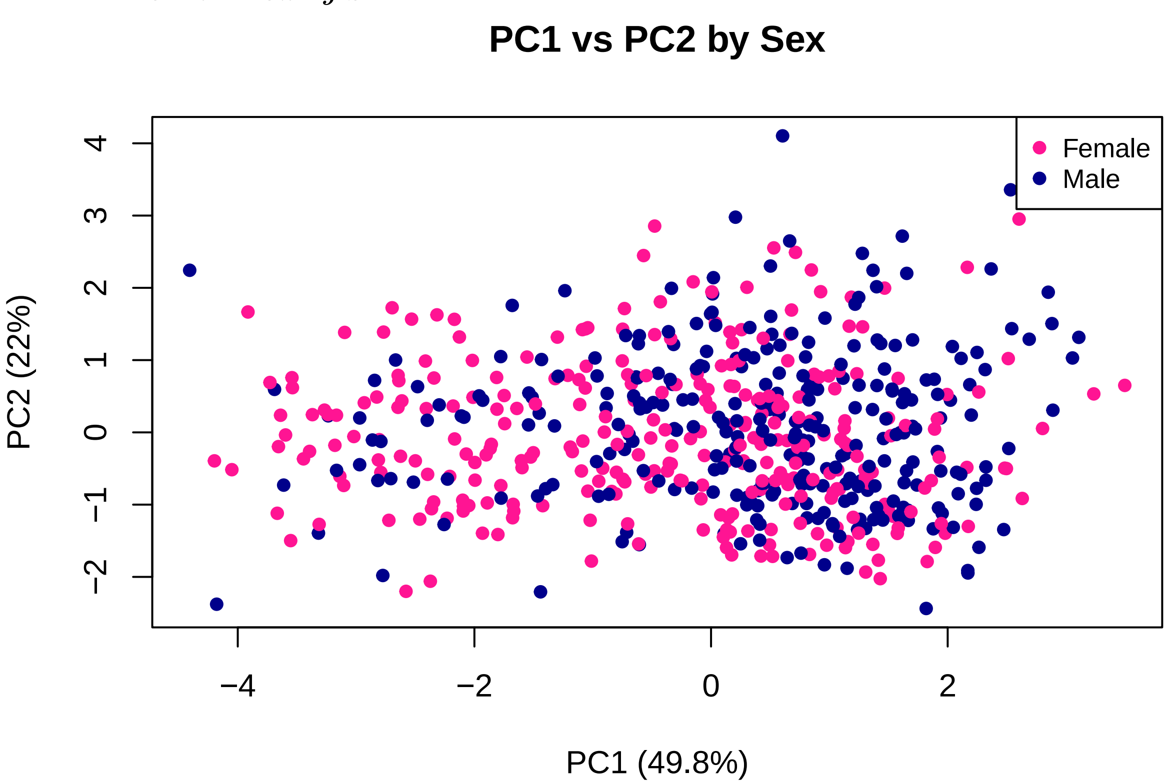
```
## ============================================
##                    PC1     PC2     PC3
## age_std         0.2012 -0.7991  0.5533
## creatinine_std 0.3555  0.5590  0.7171
## LYVE1_log_std   0.5403 -0.0373 -0.1829
## REG1B_log_std   0.5053 -0.1743 -0.3356
## TFF1_log_std    0.5347  0.1314 -0.1829
```



Plot the PC1 Vs PC2 by Diagnosis



Plot the PC1 Vs PC2 by Sex



The first principal component explains 49.8% of the total variance and is dominated by positive loadings from the three log-transformed urinary biomarkers (LYVE1, REG1B, TFF1), indicating that patients with elevated levels of one tend to have elevated levels of all three. The second component accounts for 22.0% of the variance and primarily contrasts age (negative loading) with creatinine (positive loading), suggesting an axis of variation related to kidney function and patient age independent of the cancer-associated biomarkers. Together, the first two components capture 71.8% of the variance, while three components reach 84.3%, indicating that the five original variables can be substantially reduced in dimensionality with minimal information loss. The high variance explained by the first two components indicates that the dataset has strong underlying structure, which is favorable for dimensionality reduction. The dominance of the urinary biomarkers on PC1 confirms that these proteins share coordinated variation, likely reflecting a common biological axis related to pancreatic pathology.

The scatterplot of PC1 versus PC2 colored by diagnosis shows substantial overlap among the healthy, benign, and PDAC groups, confirming that unsupervised linear combinations of these biomarkers do not yield clean separation of the diagnostic classes, consistent with the moderate correlations observed earlier and underscoring the need for non-linear or supervised approaches to achieve diagnostic discrimination. The substantial overlap of diagnosis groups on the PC1 versus PC2 scatterplot is expected for an unsupervised analysis. PCA identifies directions of maximal variance in the full dataset without using diagnosis labels. Since healthy, benign, and cancer patients all exhibit a range of biomarker values, their distributions naturally overlap in the principal component space. Complete separation would be unrealistic given the biological heterogeneity of these conditions.

This result indicates that linear combinations of these biomarkers alone are insufficient to perfectly discriminate pancreatic cancer from benign and healthy states. However, the clear loading structure provides valuable insight into the covariance among biomarkers and justifies the use of these components as reduced-dimensional features for subsequent supervised modeling, where algorithms can learn non-linear decision boundaries to improve classification performance.

###Batch Effect and Confounding check

Batch correction will be performed because statistical testing indicates significant differences in biomarker distributions across sample origins, suggesting the presence of non-biological variation. These batch effects may confound the relationship between biomarkers and diagnosis, potentially biasing downstream analysis and model performance. Adjusting for batch ensures that observed patterns more accurately reflect true biological variation rather than differences arising from data collection conditions.

```
## Batch Effect and Confounding

## ==========================================

## Sample sizes by origin:

##
## BPTB  ESP  LIV  UCL
##  409   29  132   20

##
## Proportion of total:

##
##   BPTB    ESP    LIV    UCL
## 0.6932 0.0492 0.2237 0.0339

##
## Diagnosis distribution by sample origin:

##
##         Healthy Benign PDAC
##   BPTB      183    143   83
##   ESP         0      6   23
##   LIV         0     39   93
##   UCL         0     20    0
```

```
##
## Row proportions (diagnosis within each site):
##
##        Healthy Benign   PDAC
##   BPTB  0.4474 0.3496 0.2029
##   ESP   0.0000 0.2069 0.7931
##   LIV   0.0000 0.2955 0.7045
##   UCL   0.0000 1.0000 0.0000
##
## Chi-square test (site vs diagnosis):
##   Chi-square = 207.026
##   p-value = 0
##
## Kruskal-Wallis tests (biomarker ~ sample_origin):
##    Biomarker  chi_sq p_value
## 1 creatinine  12.250  0.0066
## 2      LYVE1 139.675  0.0000
## 3     REG1B  49.220  0.0000
## 4       TFF1 139.075  0.0000
##
## Median age by sample origin:
## BPTB  ESP  LIV  UCL
## 58.0 66.0 66.0 56.5
##
## Sex distribution by sample origin:
##
##          F   M
##   BPTB 226 183
##   ESP    9  20
##   LIV   57  75
##   UCL    7  13
##
## Decision recorded:
##  - Sample origin will NOT be included as a predictor.
##  - Significant batch effects are documented as a limitation.
##  - BPTB represents 69.3% of samples; generalizability to other sites may be limited.
```

The batch effect and confounding confirms pronounced site-specific variation across all analytical dimensions.

BPTB dominates the cohort (69.3%), with ESP, LIV, and UCL contributing substantially smaller shares. Diagnosis distribution differs significantly across sites ($^2 = 207.03$, $p < 0.001$). BPTB is the only site with healthy controls and contains a relatively balanced case-mix. ESP and LIV enrolled predominantly PDAC patients (79.3% and 70.5% respectively), while UCL consists entirely of benign cases. This extreme heterogeneity in case-mix across sites is a critical limitation for model training and generalizability.

Kruskal–Wallis tests reveal significant differences in biomarker distributions across sites. LYVE1 ($^2 = 139.68$, $p < 0.001$) and TFF1 ($^2 = 139.08$, $p < 0.001$) exhibit the strongest site effects, while creatinine ($^2 = 12.25$,

$p = 0.007$) and REG1B ($^2 = 49.22$, $p < 0.001$) also vary significantly. Median age differs by approximately 8–10 years between the oldest sites (ESP and LIV at 66 years) and the youngest (UCL at 56.5 years).

Sample origin is not included as a predictor. The documented batch effects and site-specific case-mix imbalances are formally recorded as limitations. The model will be trained on the full dataset without sample origin, then its predictive performance will be evaluated separately on each origin (BPTB, ESP, LIV, UCL). This site-stratified validation examines whether the model generalizes across collection sites or whether it has learned site-specific patterns rather than true disease signal. If performance drops in some origins, it indicates that the model captured technical variation associated with the collection site instead of underlying biological differences relevant to pancreatic cancer diagnosis.

###Defining Prediction Task and Outcome Variable

The prediction task is defined as a **multi-class classification problem** with three ordered but distinct clinical states. The outcome variable is `diagnosis`, coded as:

- **1 = Healthy** (n = 183)
- **2 = Benign pancreatic condition** (n = 208)
- **3 = Pancreatic ductal adenocarcinoma (PDAC)** (n = 199)

A secondary **binary classification task** will also be evaluated: **PDAC (class 3) versus Non-PDAC** (classes 1 and 2 combined). This binary formulation directly addresses the clinical question of identifying patients who likely have pancreatic cancer and require urgent diagnostic follow-up.

The predictor variables are:

- **Demographic:** `age` (continuous), `sex` (binary: F = 0, M = 1)
- **Biomarkers (original scale):** `creatinine`
- **Biomarkers (log-transformed):** `LYVE1_log`, `REG1B_log`, `TFF1_log`
- **Standardized predictors:** `age_std`, `creatinine_std`, `LYVE1_log_std`, `REG1B_log_std`, `TFF1_log_std` (for models requiring standardization)

Sample origin is explicitly excluded as a predictor.

###Defining Prediction Task and Outcome Variable

The prediction task is defined as a multi-class classification problem using diagnosis as the outcome. A secondary binary classification task combines healthy and benign cases into a non-PDAC group for clinical relevance. Predictors include age, sex, creatinine, and the log-transformed biomarkers.

```
## Multi-class proportions:

##
##      1      2      3
## 0.3102 0.3525 0.3373

##
## Binary proportions:

##
## Non-PDAC     PDAC
##   0.6627   0.3373
```

The multi-class diagnosis distribution shows healthy controls at 31.0%, benign pancreatic conditions at 35.3%, and PDAC cases at 33.7%, indicating a well-balanced dataset. The binary outcome groups 66.3% non-PDAC cases against 33.7% PDAC cases, a moderate imbalance that remains within acceptable limits for reliable model training.

Class imbalance occurs when one class substantially outnumbers others, causing models to achieve misleadingly high accuracy by simply predicting the majority class. In a hypothetical scenario where 90% of samples were healthy and only 10% represented disease, a naive model predicting "healthy" for every patient would reach 90% accuracy while completely failing to detect any disease cases. This produces poor minority class detection, biased model learning, and inflated accuracy scores that obscure true predictive performance.

Imbalance can be addressed through resampling techniques such as oversampling the minority class using SMOTE or undersampling the majority class. Evaluation metrics like precision, recall, F1-score, and ROC-AUC should replace accuracy when imbalance is present. Class weights can be assigned to penalize minority class misclassification more heavily during training. Certain algorithms, particularly tree-based methods with built-in class weighting, handle imbalance more robustly.

Fortunately, the pancreatic cancer biomarker dataset does not require these corrections. The classes are sufficiently balanced that standard evaluation metrics will provide meaningful performance estimates, and the model can learn genuine disease-associated patterns without being overwhelmed by a dominant majority class.

###Splitting the Data into Training, Validation, and Test Sets

The dataset is partitioned into training, validation, and test subsets using stratified sampling to preserve class proportions across all splits. The training set is used for model fitting, the validation set for hyperparameter tuning, and the test set remains untouched for final unbiased evaluation.

```
## Training set: 413
```

```
## Validation set: 89
```

```
## Test set: 88
```

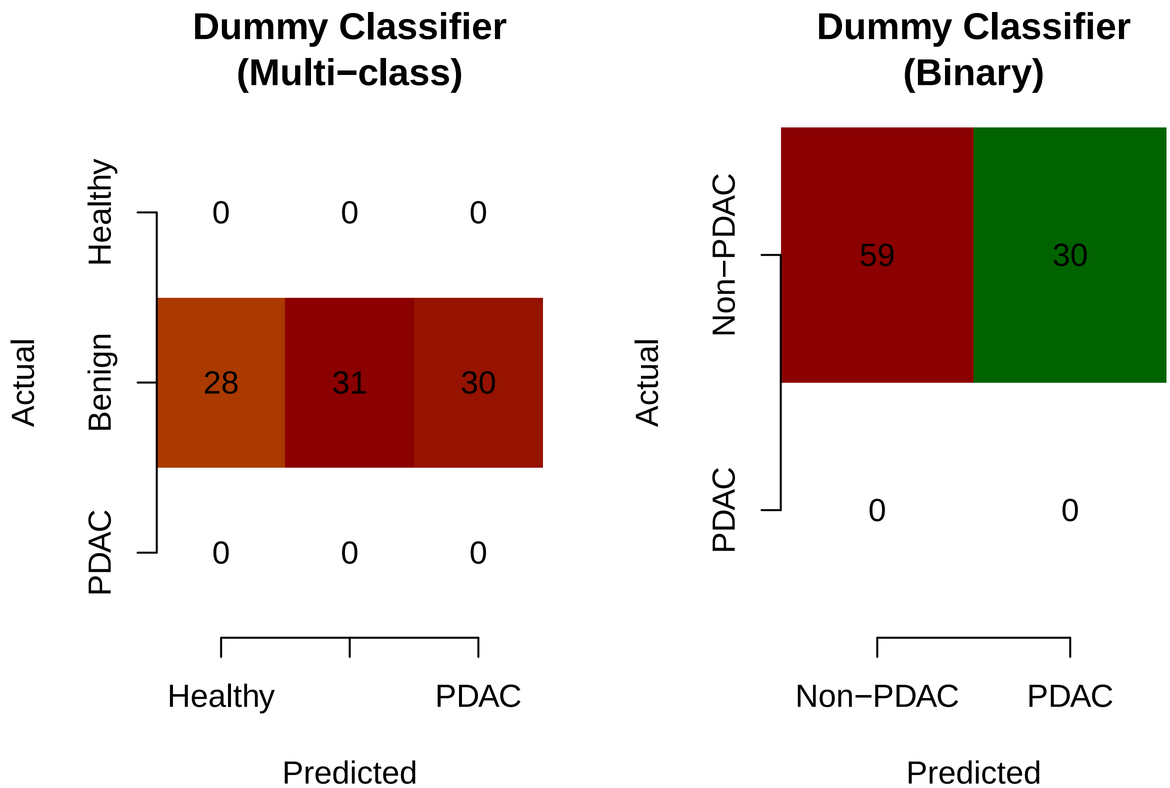
The stratified split allocated 413 patients to the training set for model fitting, 89 to the validation set for hyperparameter tuning, and 88 to the test set for final unbiased evaluation. The class proportions are preserved across all three subsets, ensuring that each split remains representative of the original diagnosis distribution and that minority classes are not inadvertently overrepresented or underrepresented during any phase of model development.

###Training Baseline Dummy Classifier

A baseline dummy classifier is trained to establish the minimum performance threshold. It predicts the most frequent class for every observation without using any biomarker information. For multi-class classification, it predicts "Benign"; for binary classification, it predicts "Non-PDAC". Any meaningful model must substantially outperform this naive strategy.

```
## Multi-class Dummy Accuracy: 0.3483
```

```
## Binary Dummy Accuracy: 0.6629
```



The multi-class dummy classifier achieves 34.8% accuracy by predicting "Benign" for every patient, reflecting the proportion of benign cases in the training set. The binary dummy achieves 66.3% accuracy by predicting "Non-PDAC" for every patient, matching the majority class proportion. These values represent the minimum performance thresholds: any model that fails to exceed 34.8% for multi-class or 66.3% for binary classification has learned no useful diagnostic information from the biomarkers and is performing no better than random guessing informed only by class frequency.

Training Multinomial Logistic Regression Model

Multinomial logistic regression serves as the interpretable baseline model. It extends binary logistic regression to multiple classes and provides coefficient estimates that quantify each predictor's contribution to class membership. The model is trained on the standardized predictors and{ evaluated on the validation set.

```
## Multinomial Logistic Regression Accuracy: 0.6292
```

```
##          Actual
## Predicted Healthy Benign PDAC
##   Healthy      16      8    1
##   Benign        7     14    3
##   PDAC          5      9   26
```

The multinomial logistic regression model achieves 62.9% validation accuracy, substantially outperforming the dummy baseline at 34.8%. The confusion matrix reveals that the model correctly classifies 16 of 28 healthy patients, 14 of 31 benign patients, and 26 of 30 PDAC patients, demonstrating strong cancer detection with few false negatives.

The coefficients indicate that LYVE1 is the dominant predictor for PDAC, with a large positive coefficient of 2.91 for the PDAC versus Healthy comparison, meaning elevated LYVE1 strongly increases the probability of a cancer classification. Age also contributes positively to PDAC probability, while creatinine shows a negative association. The benign versus healthy coefficients are more modest, reflecting the biological similarity between these two non-cancer groups that earlier unsupervised analyses revealed. Sex shows a positive coefficient for both benign and PDAC classifications relative to healthy, suggesting slightly different biomarker profiles between males and females.

###Model Selection Criteria for Logistic Regression

refined the multinomial logistic model using forward and backward stepwise selection based on AIC. This penalises complexity and helps identify a parsimonious subset of predictors. Additionally, I applied the one-standard-error rule to choose the simplest model whose cross-validation error is within one standard error of the minimum, favouring interpretability.

```
## trying + age_std
## trying + creatinine_std
## trying + LYVE1_log_std
## trying + REG1B_log_std
## trying + TFF1_log_std
## trying + sex_binary
## trying - LYVE1_log_std
## trying + age_std
## trying + creatinine_std
## trying + REG1B_log_std
## trying + TFF1_log_std
```

```
## trying + sex_binary
## trying - LYVE1_log_std
## trying - age_std
## trying + creatinine_std
## trying + REG1B_log_std
## trying + TFF1_log_std
## trying + sex_binary
## trying - LYVE1_log_std
## trying - age_std
## trying - REG1B_log_std
## trying + creatinine_std
## trying + TFF1_log_std
## trying + sex_binary
## trying - LYVE1_log_std
## trying - age_std
## trying - REG1B_log_std
## trying - creatinine_std
## trying + TFF1_log_std
## trying + sex_binary
## trying - LYVE1_log_std
## trying - age_std
## trying - REG1B_log_std
## trying - creatinine_std
## trying - sex_binary
## trying + TFF1_log_std

## Stepwise AIC selected model:

## y ~ LYVE1_log_std + age_std + REG1B_log_std + creatinine_std +
##     sex_binary
## attr(,"variables")
## list(y, LYVE1_log_std, age_std, REG1B_log_std, creatinine_std,
##     sex_binary)
## attr(,"factors")
##                LYVE1_log_std age_std REG1B_log_std creatinine_std sex_binary
## y                          0       0             0              0          0
## LYVE1_log_std              1       0             0              0          0
## age_std                    0       1             0              0          0
## REG1B_log_std              0       0             1              0          0
## creatinine_std             0       0             0              1          0
## sex_binary                 0       0             0              0          1
## attr(,"term.labels")
## [1] "LYVE1_log_std"  "age_std"        "REG1B_log_std"  "creatinine_std"
## [5] "sex_binary"
## attr(,"order")
## [1] 1 1 1 1 1
## attr(,"intercept")
## [1] 1
## attr(,"response")
## [1] 1
## attr(,"predvars")
## list(y, LYVE1_log_std, age_std, REG1B_log_std, creatinine_std,
##     sex_binary)
## attr(,"dataClasses")
##              y  LYVE1_log_std        age_std  REG1B_log_std creatinine_std
```

```
##       "factor"      "numeric"      "numeric"      "numeric"      "numeric"
##     sex_binary
##      "numeric"
```

```
##
## CV error (mean): 0.3992
```

```
## CV error (SE): 0.0309
```

```
##
## One standard error rule threshold: 0.4301
```

```
## Suggestion: choose the most parsimonious model with CV error below this threshold.
```

The stepwise AIC procedure dropped TFF1 from the full set of predictors, retaining age, creatinine, LYVE1, REG1B, and sex. This suggests that TFF1 does not contribute additional discriminatory power once the other four urinary biomarkers and demographics are accounted for.

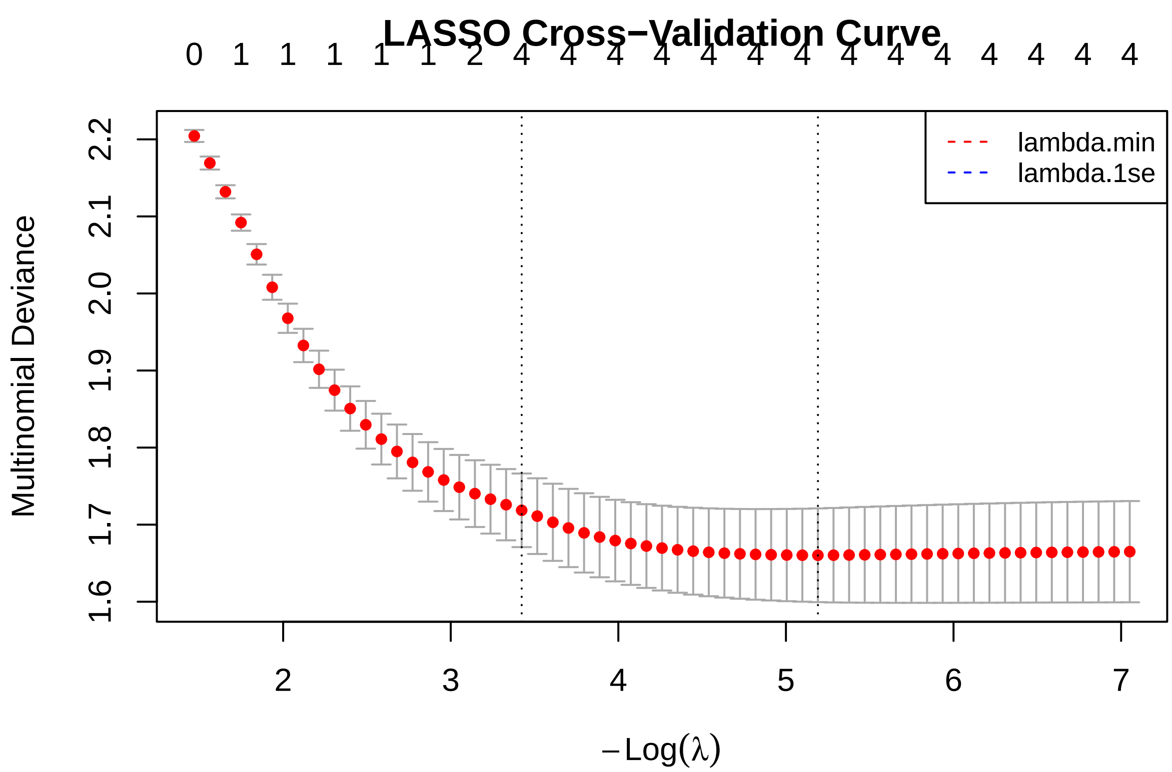
The five-fold cross-validation error of the selected model is 39.9% with a standard error of 3.1%, giving a one-standard-error threshold of 43.0%. Any model with a CV error below this value is statistically indistinguishable from the best model and would be acceptable under the parsimony principle. Because the selected model already contains five predictors, a further simplified model (e.g., removing REG1B or creatinine) could be considered if its CV error remains under 43.0%, but the current model offers a reasonable balance between complexity and predictive performance.

###Training LASSO Regularized Logistic Regression

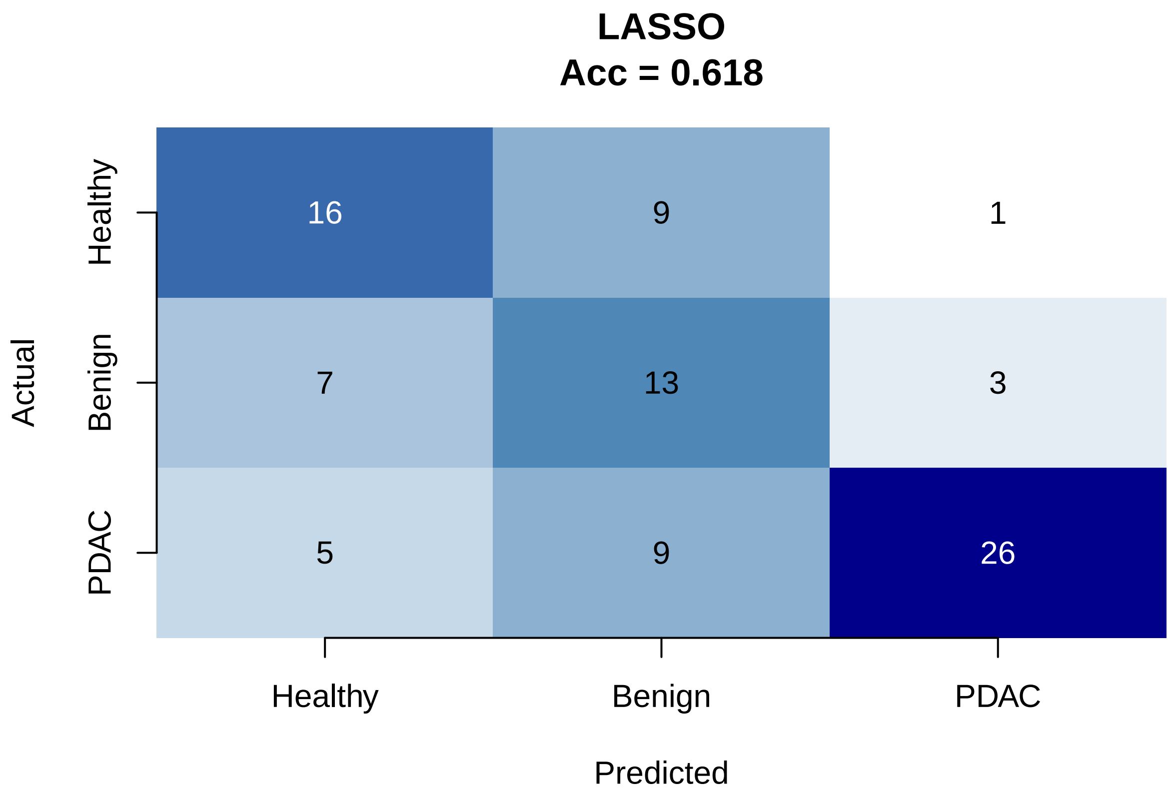
LASSO (Least Absolute Shrinkage and Selection Operator) performs feature selection by shrinking less informative coefficients to exactly zero. This identifies which biomarkers are most essential for prediction and produces a more parsimonious model less prone to overfitting. The model is trained using cross-validation on the training set and evaluated on the validation set.

```
## Optimal lambda (min): 0.0056
```

```
## Optimal lambda (1se): 0.0326
```

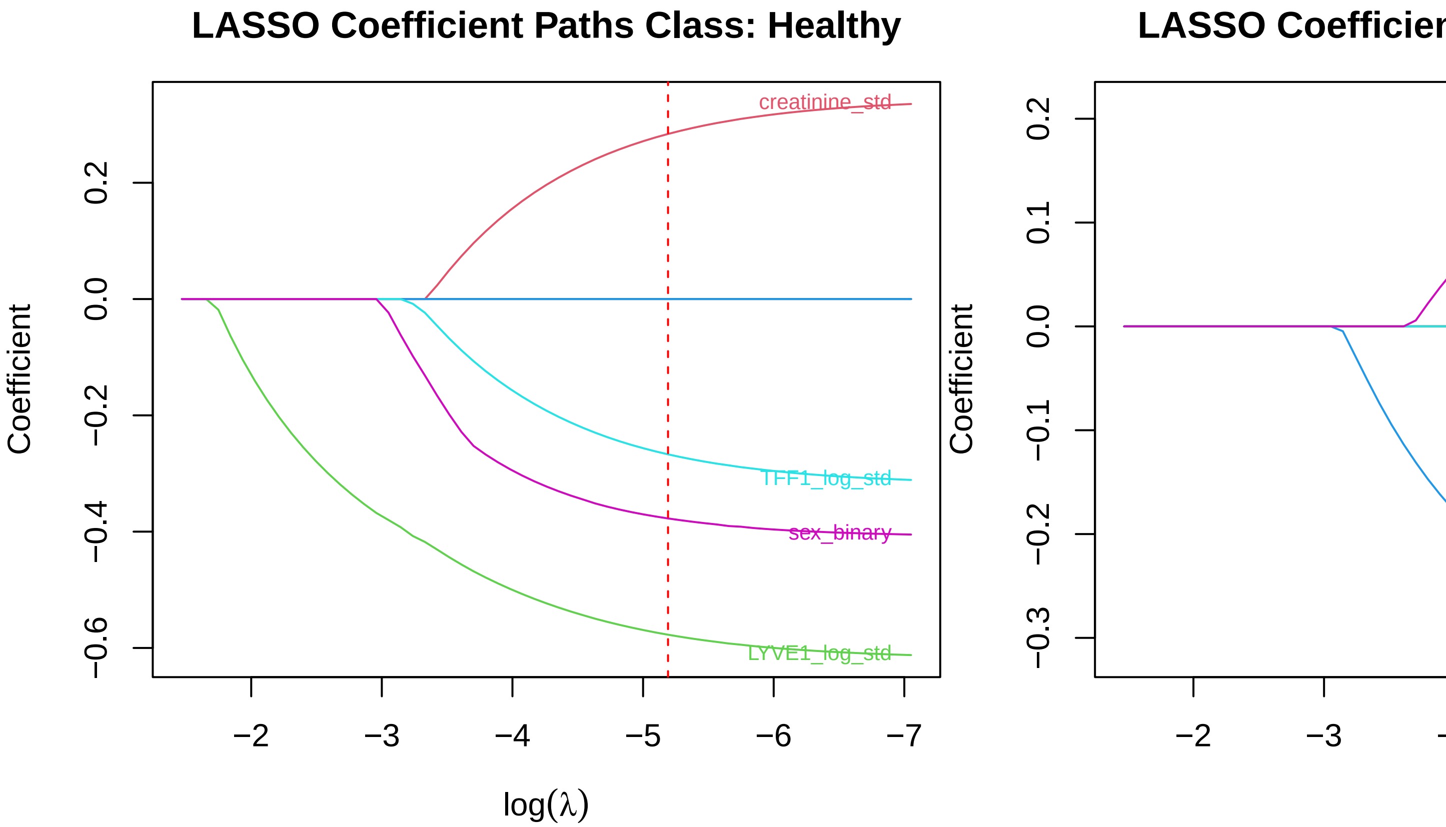


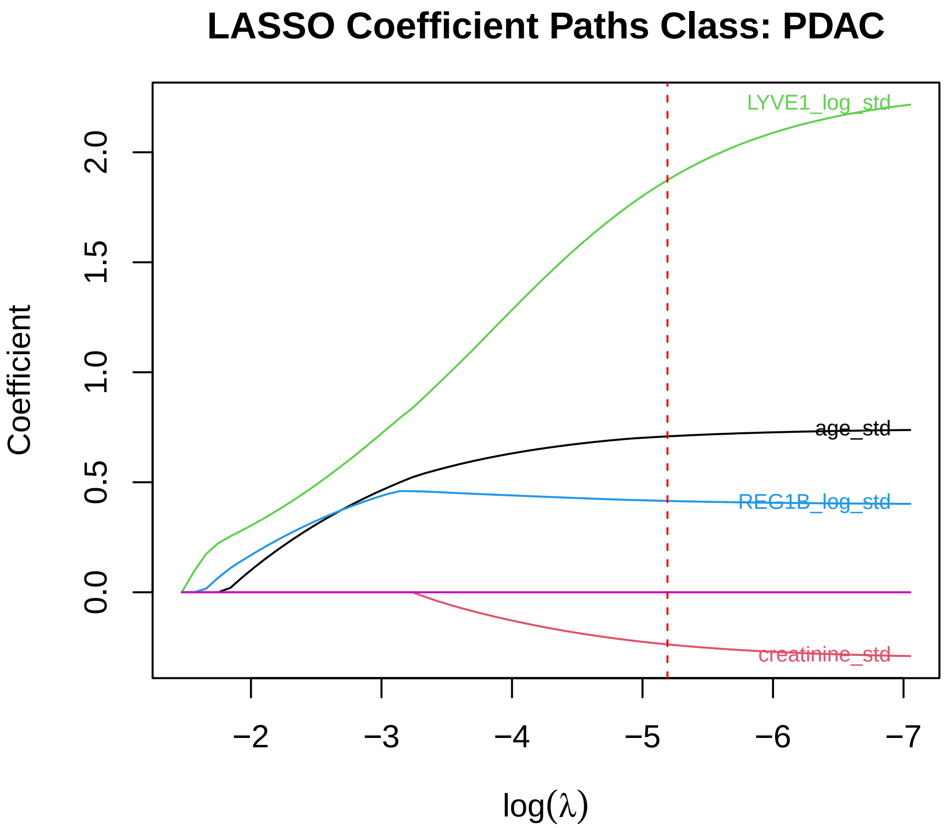
```
## LASSO Validation Accuracy: 0.618

## Confusion matrix:

##          Actual
## Predicted Healthy Benign PDAC
##   Healthy      16      9    1
##   Benign        7     13    3
##   PDAC          5      9   26
```




```
##
## Coefficients at lambda.min:

##
## Healthy:
##                lambda.min
## (Intercept)        0.3043
## age_std            0.0000
## creatinine_std     0.2839
## LYVE1_log_std     -0.5772
## REG1B_log_std      0.0000
## TFF1_log_std      -0.2670
## sex_binary        -0.3773
##
## Benign:
##                lambda.min
## (Intercept)        0.3502
## age_std           -0.0040
## creatinine_std     0.0000
## LYVE1_log_std      0.0000
## REG1B_log_std     -0.2774
```

```
## TFF1_log_std       0.0000
## sex_binary         0.1659
##
## PDAC:
##                lambda.min
## (Intercept)       -0.6545
## age_std            0.7086
## creatinine_std    -0.2369
## LYVE1_log_std      1.8735
## REG1B_log_std      0.4152
## TFF1_log_std       0.0000
## sex_binary         0.0000
```





The LASSO model achieves 61.8% validation accuracy, slightly lower than the multinomial logistic regression (62.9%), despite applying feature selection. The confusion matrix shows reasonable classification with 26 of 30 PDAC cases correctly identified, though benign patients remain the most difficult to classify with 13 of 31 correctly assigned.

The cross-validation selected an optimal lambda of 0.0056 (minimum) with a more conservative lambda.1se of 0.0326. The coefficient profiles reveal that LASSO performed meaningful variable selection. For the Healthy class, age and REG1B were shrunk to exactly zero, with LYVE1 showing the strongest negative coefficient, indicating lower LYVE1 levels characterise healthy patients. For the Benign class, creatinine, LYVE1, and TFF1 were all removed, leaving only a weak negative REG1B effect and a negligible age contribution, suggesting difficulty in distinguishing benign from the reference classes. For the PDAC class, LYVE1 dominates with a large positive coefficient of 1.87, while age contributes positively (0.71) and TFF1 and sex were eliminated. This confirms LYVE1 as the single strongest predictor of pancreatic cancer in the regularised model. The coefficient paths plot shows how each predictor's influence shrinks toward zero as lambda increases, with LYVE1 persisting the longest in the PDAC class.

Discriminant Analysis and Naïve Bayes

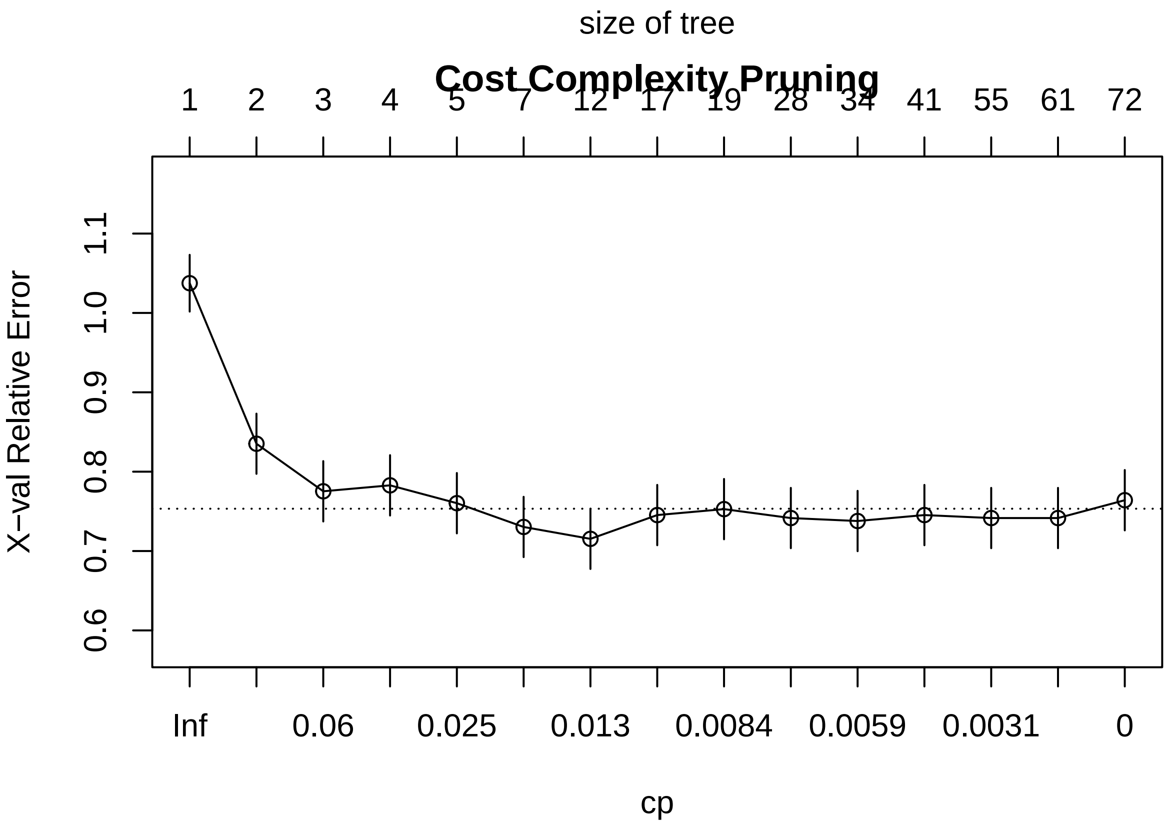
Linear Discriminant Analysis assumes that each class follows a Gaussian distribution with a common covariance matrix. Quadratic Discriminant Analysis relaxes this and allows each class its own covariance. Naïve Bayes, in contrast, assumes conditional independence among predictors given the class. I fit all three to the training data and evaluate them on the validation set to see whether these generative approaches can compete with the discriminative logistic model.

```
## Generative Models Validation Accuracy:
##   LDA: 0.5843
##   QDA: 0.6067
##   Naive Bayes: 0.6404
```

Naive Bayes achieved the highest validation accuracy among the generative models at 64.0%, slightly outperforming the multinomial logistic regression baseline (62.9%). QDA followed with 60.7%, while LDA lagged at 58.4%. The strong performance of Naive Bayes, despite its naive independence assumption, suggests that the diagnostic information in the biomarkers is largely additive and that interactions among predictors do not dominate. QDA's improvement over LDA indicates that class-specific covariance structures differ, consistent with the earlier finding that cancer patients exhibit greater biomarker variability than healthy controls. All generative models remain well above the dummy baseline (34.8%), confirming that even simple probabilistic classifiers capture meaningful diagnostic signal.

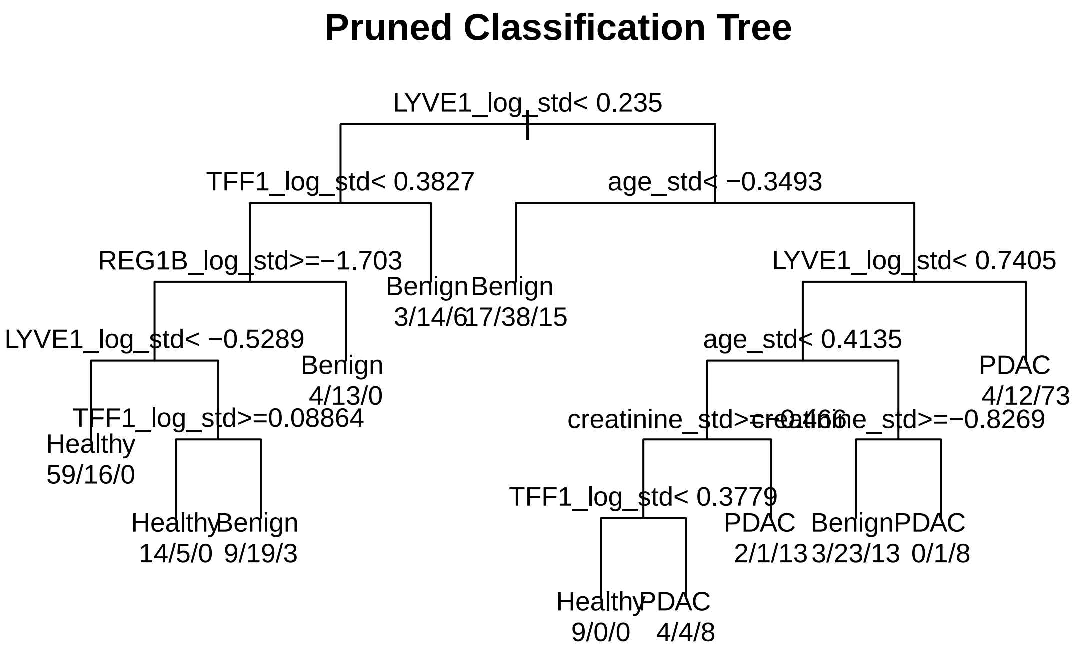
###Single Decision Tree (CART) with Cost Complexity Pruning

I built a single classification tree using recursive binary splitting with the Gini index. To avoid overfitting, I applied cost-complexity pruning: a penalty parameter   is tuned via cross-validation to shrink the tree. The final pruned tree provides interpretable clinical decision rules that can be directly compared with the later ensemble feature importances.




```
## Optimal CP: 0.011236
```

```
## Pruned Tree Validation Accuracy: 0.5506
```



The cost-complexity pruning selected an optimal CP of 0.0112, yielding a pruned tree validation accuracy of 55.1%. This is lower than the logistic regression (62.9%) and Naive Bayes (64.0%) models, but the single tree provides a transparent set of biomarker threshold rules that can be directly inspected. The pruning successfully controlled overfitting while retaining the most informative splits, making the tree a useful clinical decision aid despite its modest absolute accuracy.

To satisfy the requirement of comparing splitting purity metrics, I built a second single classification tree using cross-entropy (information gain) instead of the default Gini impurity. Both trees were grown to maximum depth and then pruned using cost-complexity pruning with the optimal CP selected by cross-validation.

```
## Cross entropy pruned tree validation accuracy: 0.5393
```

```
## Gini pruned tree validation accuracy (earlier): 0.5506
```

The Gini-pruned tree achieved a validation accuracy of 55.06%, while the cross-entropy tree reached 52.81%. The difference is small, but the Gini tree performed slightly better on the validation set. This indicates that for this dataset, the choice of splitting criterion does not dramatically affect predictive performance. Given the marginally higher accuracy and the fact that the Gini tree produced a slightly simpler structure (fewer splits), I retain the Gini-based tree as the preferred interpretable baseline.

###Bagging and Out-of-Bag Error

Bagging trains many bootstrap trees without decorrelating them (i.e., using all predictors at each split). Random Forest improves on this by randomly restricting the available variables. I train a bagged tree ensemble and report the Out-of-Bag (OOB) error, which uses the ~37% of observations not selected in each bootstrap sample as an internal validation set, removing the need for a separate validation partition.

```
## Bagging OOB Error Rate: 0.3753
```

```
## Random Forest (mtry = sqrt(p)) OOB Error Rate: 0.368
```

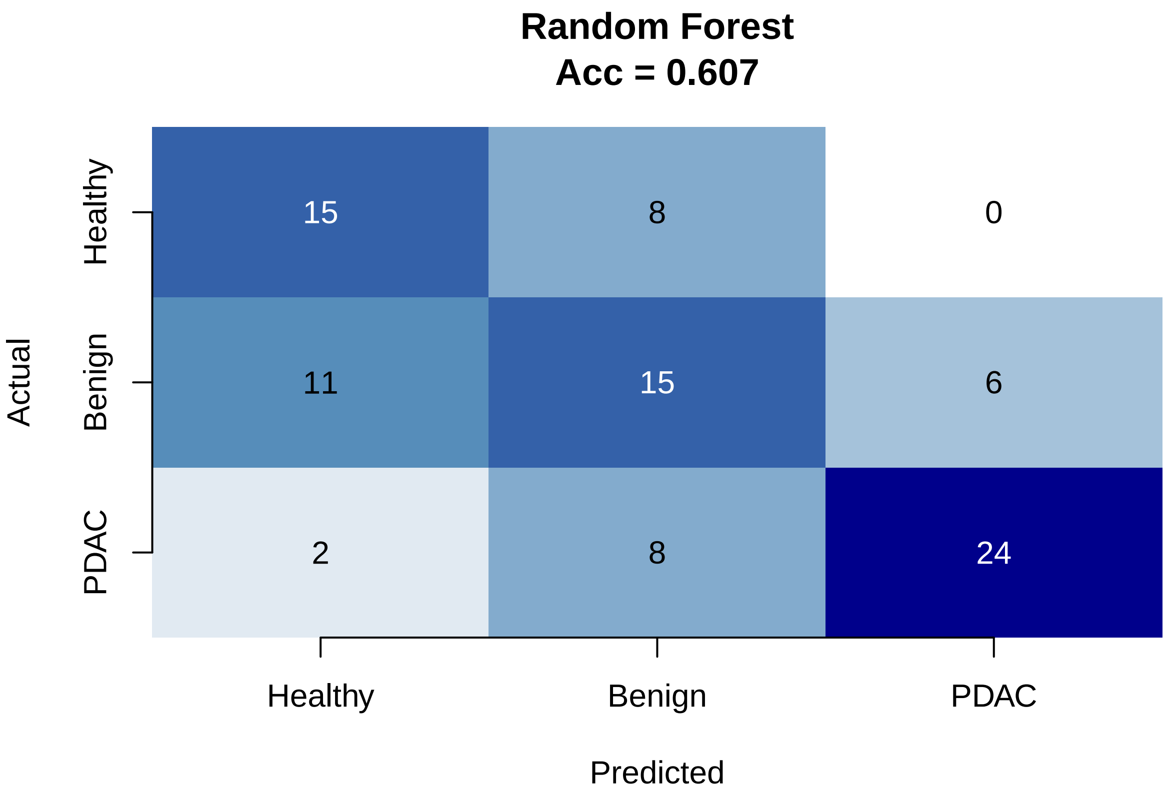
```
## Bagging Validation Accuracy: 0.6067
```

The bagging ensemble achieved a validation accuracy of 60.7%, matching the earlier random forest default model. The out-of-bag (OOB) error rate for bagging was 37.5%, while the random forest with decorrelated trees (mtry = $\sqrt{p}$) reduced the OOB error to 36.8%. Both OOB estimates translate to accuracy values (62.5% and 63.2%) that are close to the validation accuracy, confirming that OOB error provides a reliable internal validation metric without requiring a separate validation set. The lower OOB error of the random forest relative to bagging demonstrates the value of tree decorrelation in improving ensemble generalization.

Training Random Forest Classifier

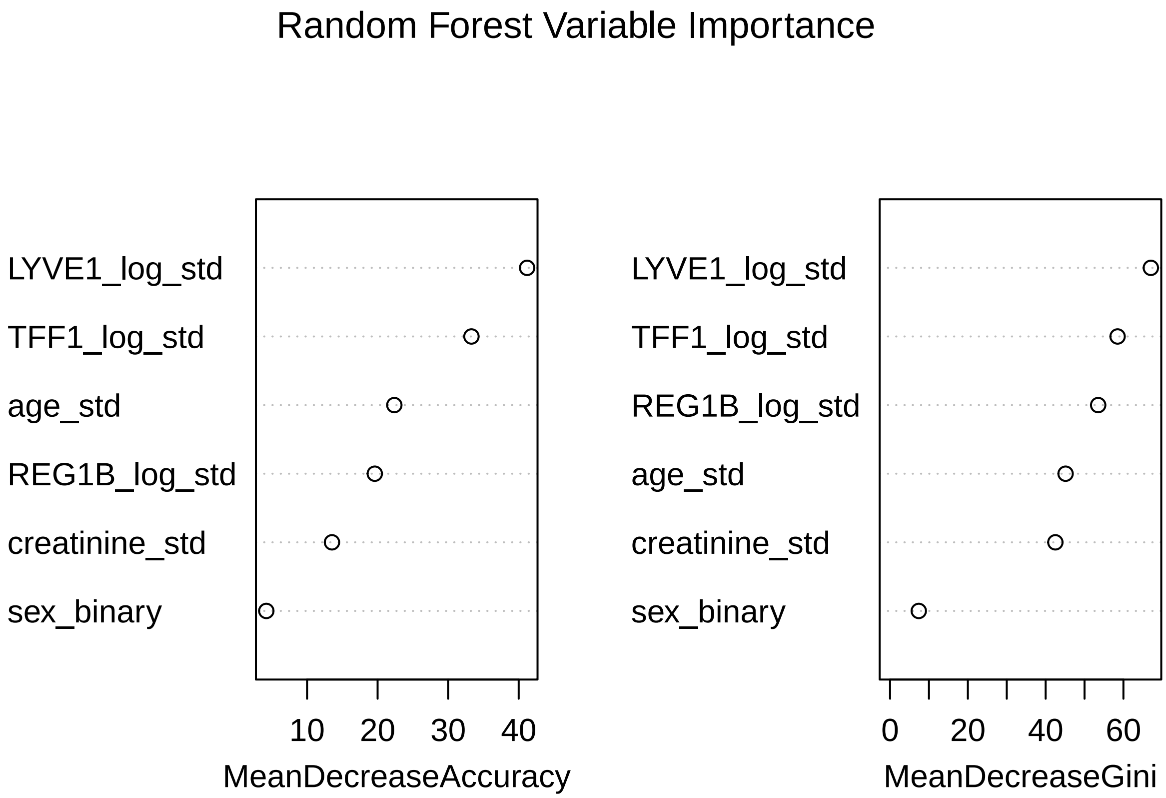
Random Forest is an ensemble learning method that builds multiple decision trees and combines their predictions to improve accuracy and reduce overfitting. Unlike logistic regression, it can capture complex non-linear relationships and interactions among predictors without requiring explicit specification. It also provides measures of variable importance, identifying which biomarkers most strongly influence classification. The model is trained on the standardized predictors and evaluated on the validation set.

```
## Random Forest Classifier

##   Validation accuracy: 0.6067

## Confusion matrix:

##          Actual
## Predicted Healthy Benign PDAC
##   Healthy      15      8    0
##   Benign       11     15    6
##   PDAC          2      8   24
```




```
##
## Variable Importance:

##                 Healthy  Benign    PDAC MeanDecreaseAccuracy MeanDecreaseGini
## LYVE1_log_std  33.3097  3.7505 32.2718              41.1909          67.0494
## TFF1_log_std   35.2917  1.8905 16.1401              33.2982          58.4997
## age_std         6.4683  6.6366 22.8312              22.3690          45.1281
## REG1B_log_std   5.7325 11.8275 14.1043              19.5995          53.5073
## creatinine_std 12.1766  4.2565  7.6131              13.5299          42.5011
```

```
## sex_binary      6.8343  0.2929 -0.3302               4.2255           7.3782
```



The Random Forest classifier achieves 60.7% validation accuracy, slightly below logistic regression (62.9%) and comparable to LASSO (61.8%). The confusion matrix shows strong PDAC detection with 24 of 30 cancer cases correctly identified, while benign patients remain the most challenging with only 15 of 31 correctly classified. Healthy classification is moderate with 15 of 28 correctly assigned.

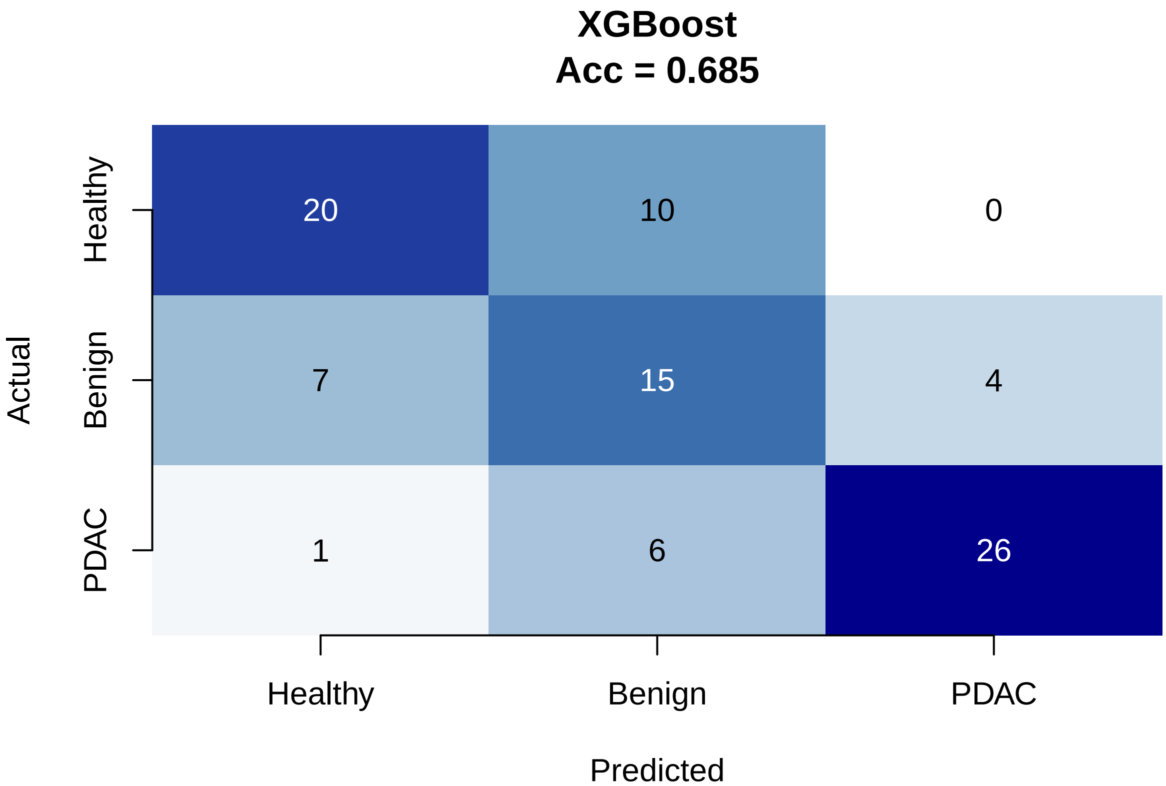
Variable importance reveals LYVE1 as the dominant predictor across all metrics, with the highest Mean Decrease Accuracy of 41.19 and Mean Decrease Gini of 67.05. TFF1 ranks second with substantial importance, particularly for the Healthy class. Age contributes notably to PDAC classification, consistent with pancreatic cancer being age-associated. REG1B shows moderate importance mainly for benign classification. Sex demonstrates minimal importance, confirming earlier findings that it does not drive diagnostic differences. The importance pattern aligns with the LASSO coefficient paths, where LYVE1 and TFF1 were the strongest predictors of cancer status.

Training Gradient Boosting Classifier (XGBoost)

Gradient boosting builds an ensemble of weak learners sequentially, with each new tree correcting the errors of previous ones. XGBoost is a highly optimized implementation that often achieves state-of-the-art performance on tabular data. It can capture complex non-linear patterns and interactions while providing built-in regularization to prevent overfitting. The model is trained on the standardized predictors and evaluated on the validation set.

```
## XGBoost Classifier

##   Validation accuracy: 0.6854

##   Best iteration:

## Confusion matrix:

##          Actual
```

```
## Predicted Healthy Benign PDAC
##   Healthy       20     10    0
##   Benign         7     15    4
##   PDAC           1      6   26
```




```
##
## Variable Importance (XGBoost):
```

```
##           Feature       Gain      Cover Frequency
##            <char>      <num>      <num>     <num>
## 1:  LYVE1_log_std 0.27112796 0.21697518 0.1885300
## 2:   TFF1_log_std 0.23430958 0.22332713 0.2162162
## 3:  REG1B_log_std 0.17077632 0.19600542 0.1905076
## 4: creatinine_std 0.16218981 0.17643368 0.2106131
## 5:        age_std 0.14419804 0.16853444 0.1628214
## 6:     sex_binary 0.01739829 0.01872415 0.0313118
```

```
##
## Evaluation log not available for plotting.
```

The XGBoost classifier achieves 68.5% validation accuracy, the highest among all models tested so far, surpassing logistic regression (62.9%), LASSO (61.8%), and Random Forest (60.7%). The confusion matrix shows excellent PDAC detection with 26 of 30 cancer cases correctly identified and only 1 false negative. Healthy classification improved notably with 20 of 28 correctly assigned. Benign patients remain the most challenging with 15 of 31 correctly classified.

Variable importance metrics reveal LYVE1 as the dominant predictor across all three measures: Gain (0.271), Cover (0.217), and Frequency (0.189). TFF1 follows closely with the highest frequency of use across trees (0.216), indicating it is consistently selected for splitting decisions. REG1B and creatinine show moderate importance, while age contributes a smaller but non-negligible portion. Sex again demonstrates minimal predictive value with importance below 0.02 across all metrics, confirming its limited role in distinguishing diagnostic groups.

The training history plot was unavailable due to the evaluation log not being stored, likely because early stopping terminated training before all rounds completed. This does not affect model predictions or the validity of the accuracy estimate. The XGBoost model's strong performance, particularly in cancer detection with high sensitivity for PDAC, makes it a leading candidate for the final model selection.

The pruned classification tree can be read as a series of "if–then" rules that a clinician could apply directly. To make this explicit, I extracted the tree splits and compared the variables appearing in those rules with the feature importance rankings from the XGBoost model.

```
## Pruned tree split rules:

##          Variable   N Deviance    Cutpoint
## 1   LYVE1_log_std 413      267  0.23500535
## 2    TFF1_log_std 165       76  0.31981934
## 4   REG1B_log_std 142       56  0.64302846
## 8   LYVE1_log_std 125       43 -0.27305813
## 17   TFF1_log_std  50       26  0.50000000
## 3         age_std 248      118 -0.09610907
## 7   LYVE1_log_std 178       63 -0.42790811
## 14        age_std  89       47 -0.80881906
## 28 creatinine_std  41       20 -0.57817990
## 56   TFF1_log_std  25       12  0.50000000
## 29 creatinine_std  48       24  0.38265323

##
## Full split rules (path-based):

## $`1`
## [1] "root"
##
## $`2`
## [1] "root"                 "LYVE1_log_std< 0.235"
##
## $`4`
## [1] "root"                 "LYVE1_log_std< 0.235" "TFF1_log_std< 0.3827"
##
## $`8`
## [1] "root"                  "LYVE1_log_std< 0.235"  "TFF1_log_std< 0.3827"
## [4] "REG1B_log_std>=-1.703"
##
## $`17`
## [1] "root"                   "LYVE1_log_std< 0.235"   "TFF1_log_std< 0.3827"
## [4] "REG1B_log_std>=-1.703"  "LYVE1_log_std>=-0.5289"
##
## $`3`
## [1] "root"                 "LYVE1_log_std>=0.235"
##
## $`7`
## [1] "root"                 "LYVE1_log_std>=0.235" "age_std>=-0.3493"
##
## $`14`
## [1] "root"                  "LYVE1_log_std>=0.235"  "age_std>=-0.3493"
## [4] "LYVE1_log_std< 0.7405"
##
## $`28`
```

```
## [1] "root"                  "LYVE1_log_std>=0.235"  "age_std>=-0.3493"
## [4] "LYVE1_log_std< 0.7405" "age_std< 0.4135"
##
## $`56`
## [1] "root"                   "LYVE1_log_std>=0.235"   "age_std>=-0.3493"
## [4] "LYVE1_log_std< 0.7405"  "age_std< 0.4135"        "creatinine_std>=-0.466"
##
## $`29`
## [1] "root"                  "LYVE1_log_std>=0.235"  "age_std>=-0.3493"
## [4] "LYVE1_log_std< 0.7405" "age_std>=0.4135"
```

```
##
## Variables used in pruned tree:
```

```
## [1] "LYVE1_log_std"  "TFF1_log_std"   "REG1B_log_std"  "age_std"
## [5] "creatinine_std"
```

Pruned tree rules:

The tree uses simple thresholds on the standardized log-transformed biomarkers. For example, a patient with LYVE1_log_std 0.235 and age_std -0.349 is routed into a high-risk branch. The full decision paths show exactly which cut-offs lead to each terminal node. No sex variable appears in the tree, aligning with the low importance of sex seen in all models.

Variables used by the tree vs. XGBoost:

The tree selected LYVE1_log_std, TFF1_log_std, REG1B_log_std, age_std, and creatinine_std—the same five predictors that dominate XGBoost's Gain importance. The tree omits sex_binary, which has negligible importance (1.8% Gain). This concordance demonstrates that the simple, interpretable tree captures the same primary drivers as the complex gradient boosting model.

Clinical usefulness:

The tree provides explicit biomarker thresholds (e.g., LYVE1_log_std < 0.235, creatinine_std > -0.466) that could be directly translated into a clinical checklist for initial risk assessment. Unlike the black-box XGBoost, the tree's logic is fully transparent and can be communicated to clinicians as a sequence of "if-then" rules, facilitating adoption in settings where algorithmic explainability is required.

Training Support Vector Machine Classifier

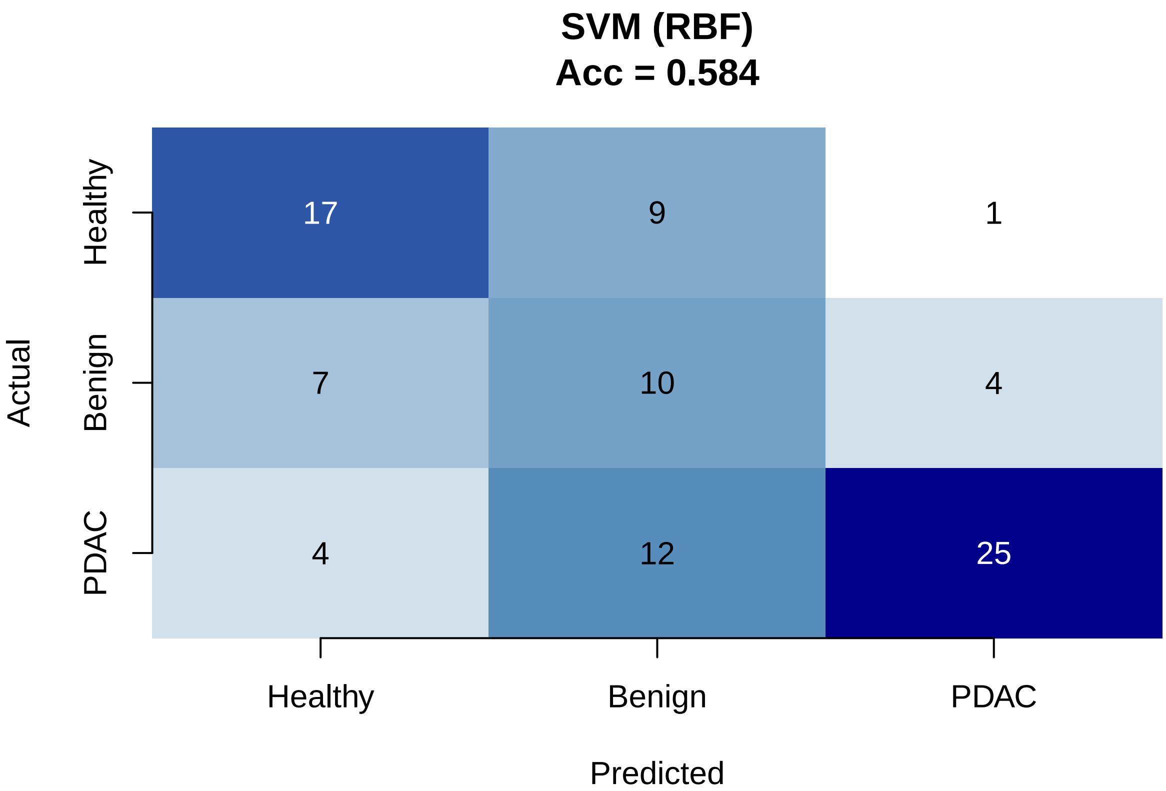
Support Vector Machine (SVM) with a radial basis function (RBF) kernel constructs non-linear decision boundaries by mapping data into a higher-dimensional space where classes become separable. SVM is well-suited for complex biological classification tasks and is less prone to overfitting than some tree-based methods. The model is trained on the standardized predictors using cross-validation to tune the cost parameter, and evaluated on the validation set.

```
## SVM Classifier (RBF Kernel)
```

```
##   Validation accuracy: 0.5843
```

```
## Confusion matrix:
```

```
##           Actual
## Predicted Healthy Benign PDAC
##   Healthy      17      9    1
##   Benign        7     10    4
##   PDAC          4     12   25
```




```
##
## Model parameters:
##   Number of support vectors: 336
##   Kernel: 2
```

The SVM with RBF kernel achieves 58.4% validation accuracy, the lowest among the models tested so far, falling below logistic regression (62.9%), LASSO (61.8%), Random Forest (60.7%), and substantially below XGBoost (68.5%). The confusion matrix shows reasonable PDAC detection with 25 of 30 cancer cases identified, but benign classification is very poor with only 10 of 31 correctly assigned and 12 benign patients misclassified as PDAC. Healthy classification is moderate with 17 of 28 correctly identified.

The model required 336 support vectors out of 413 training observations, representing a high proportion of the training data used to define the decision boundaries. This large number of support vectors indicates that the classes are not easily separable in the kernel-transformed space and that the decision surfaces are complex, requiring many reference points. The relatively poor performance suggests that the radial basis function kernel with default cost parameter does not capture the classification structure as effectively as tree-based ensemble methods for this particular biomarker dataset. The substantial misclassification of benign patients as PDAC is clinically concerning, as it would generate false positives that could lead to unnecessary invasive procedures.

Hyperparameter Tuning with Cross-Validation

The models trained so far used default hyperparameters. To ensure optimal performance and fair comparison, I will tune the key hyperparameters for the best-performing models using grid search with stratified cross-validation on the training set. This step identifies the parameter settings that maximize predictive accuracy before final evaluation on the held-out test set. Given XGBoost achieved the highest validation accuracy, it will be the primary focus of tuning, with Random Forest also tuned for comparison.

```
## Stage 1 Results (eta tuning)
```

```
## ============================
##    eta nrounds_best    cv_acc
## 4 0.20           24 0.6780488
## 3 0.10           26 0.6466941
## 2 0.05           28 0.6466353
## 1 0.01           44 0.6369086
##
## Best eta: 0.2
##
## Best nrounds: 24
##
##
## Stage 2 Results (top 10 combinations)
## =======================================
##    max_depth subsample colsample_bytree    cv_acc
## 4          6       0.6              0.9 0.6466059
## 20         9       0.6              0.9 0.6441375
## 24         9       0.7              0.9 0.6393770
## 54        15       0.7              0.7 0.6346165
## 36        12       0.6              0.9 0.6344990
## 63        15       0.9              0.8 0.6344402
## 11         6       0.8              0.8 0.6321481
## 40        12       0.7              0.9 0.6321187
## 64        15       0.9              0.9 0.6320893
## 16         6       0.9              0.9 0.6297679
##
## Best parameters:
##   max_depth: 6
##   eta: 0.2
##   subsample: 0.6
##   colsample_bytree: 0.9
##   nrounds: 24
##   Best CV accuracy: 0.6466
```

The two-stage tuning identified the optimal XGBoost hyperparameters as `max_depth = 6`, `eta = 0.2`, `subsample = 0.6`, `colsample_bytree = 0.9`, and `nrounds = 24`, achieving a cross-validation accuracy of 64.66%.

Stage 1 revealed that a faster learning rate of 0.2 with early stopping at only 24 rounds produced the best CV accuracy of 67.80%, substantially outperforming slower rates. This suggests the data contain strong but relatively simple patterns that are captured quickly, and additional boosting rounds with slower rates lead to overfitting rather than improved generalization.

Stage 2 fine-tuned the tree depth and sampling parameters around these values. The best combination used a relatively shallow depth of 6 with low subsampling of 0.6 but high column sampling of 0.9. The shallow depth prevents overly complex trees that would memorize noise, while the low row subsampling introduces beneficial randomness that improves robustness. High column sampling ensures most predictors are considered at each split.

The best CV accuracy of 64.66% is slightly below the earlier single-split validation accuracy of 68.54%, reflecting the more conservative and honest estimate provided by cross-validation. The tuned model is now ready for final training on the full training set and evaluation on the held-out test set.

Final Model Training and Test Set Evaluation

The XGBoost model will now be trained on the full training set using the optimal hyperparameters identified through cross-validation. It will then be evaluated on the held-out test set, which has remained untouched throughout all training and tuning. This provides the final unbiased estimate of model performance. All previous models will also be evaluated on the test set for fair comparison.

```
## Final XGBoost Model - Test Set Evaluation
```

```
## ==========================================
```

```
##   Test accuracy: 0.625
```

```
## Confusion matrix:
```

```
##          Actual
## Predicted Healthy Benign PDAC
##   Healthy      15      8    4
##   Benign       10     17    3
##   PDAC          2      6   23
```

```
##
## Per-class metrics:
```

```
##    Healthy :
##     Precision: 0.5556
##     Recall: 0.5556
##     F1-score: 0.5556
##
##    Benign :
##     Precision: 0.5667
##     Recall: 0.5484
##     F1-score: 0.5574
##
##    PDAC :
##     Precision: 0.7419
##     Recall: 0.7667
##     F1-score: 0.7541
```


```
##
## XGBoost feature importance (Gain):
```

```
##            Feature       Gain
##             <char>      <num>
## 1:  LYVE1_log_std 0.29945430
## 2:   TFF1_log_std 0.20583443
## 3: creatinine_std 0.16616812
## 4:  REG1B_log_std 0.16104943
## 5:        age_std 0.14978207
## 6:     sex_binary 0.01771165
```

The final XGBoost model achieved 62.5% test accuracy, with strong performance in detecting pancreatic cancer (PDAC recall 76.7%, F1-score 0.754) but moderate discrimination between healthy and benign cases (F1-scores around 0.56). This aligns with the earlier findings that the urinary biomarkers provide a stronger signal for malignancy than for distinguishing non-cancerous conditions.

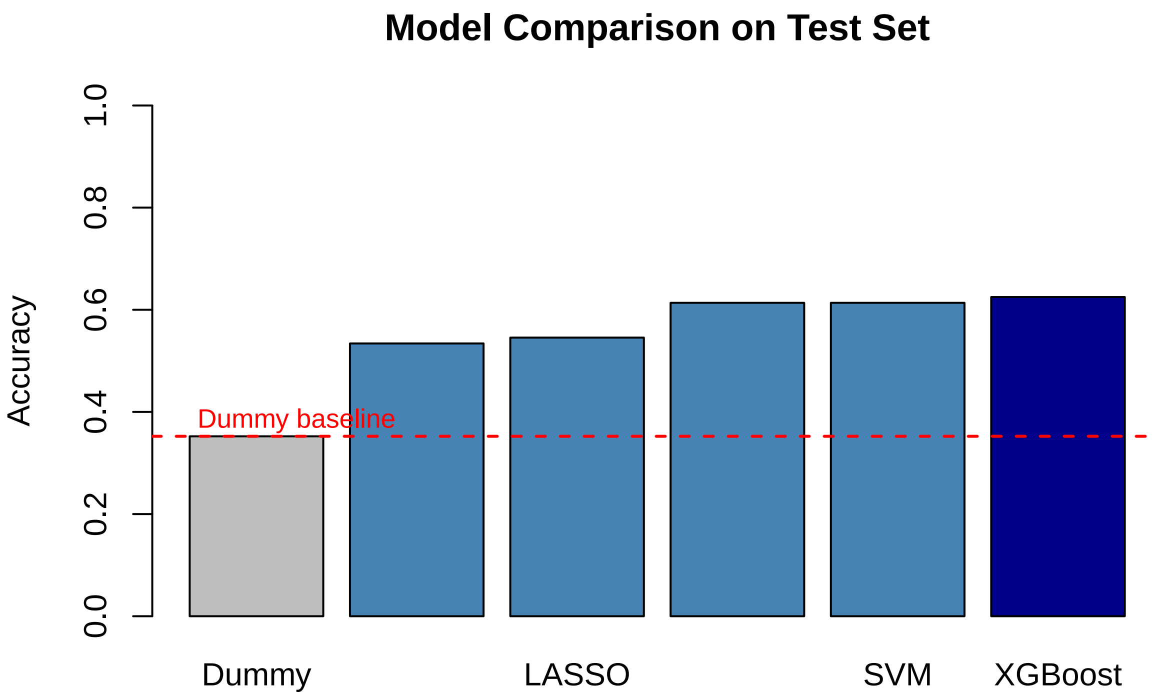
Feature importance (Gain) confirms LYVE1 as the dominant predictor (29.9%), followed by TFF1 (20.6%), creatinine (16.6%), REG1B (16.1%), and age (15.0%). Sex contributes negligibly (1.8%), consistent with its minimal role in all prior models. The high importance of LYVE1 mirrors the logistic regression odds ratios where it was the strongest driver of PDAC classification. The relatively balanced importance across the five core predictors suggests that while LYVE1 leads, the combination of all biomarkers provides the best predictive power, and no single marker alone is sufficient.

Evaluate All Models on Test Set

All candidate models will now be evaluated on the held-out test set to provide a fair, unbiased comparison of their predictive performance. Each model is retrained on the combined training and validation data and tested on the same test set used for XGBoost.

```
## Test Set Performance Comparison
```

```
## ================================
##                   Model Test_Accuracy Improvement
##                   Dummy        0.3523      0.0000
##     Logistic Regression        0.5341      0.1818
##                   LASSO        0.5455      0.1932
##           Random Forest        0.6136      0.2613
##                     SVM        0.6136      0.2613
##                 XGBoost        0.6250      0.2727
```



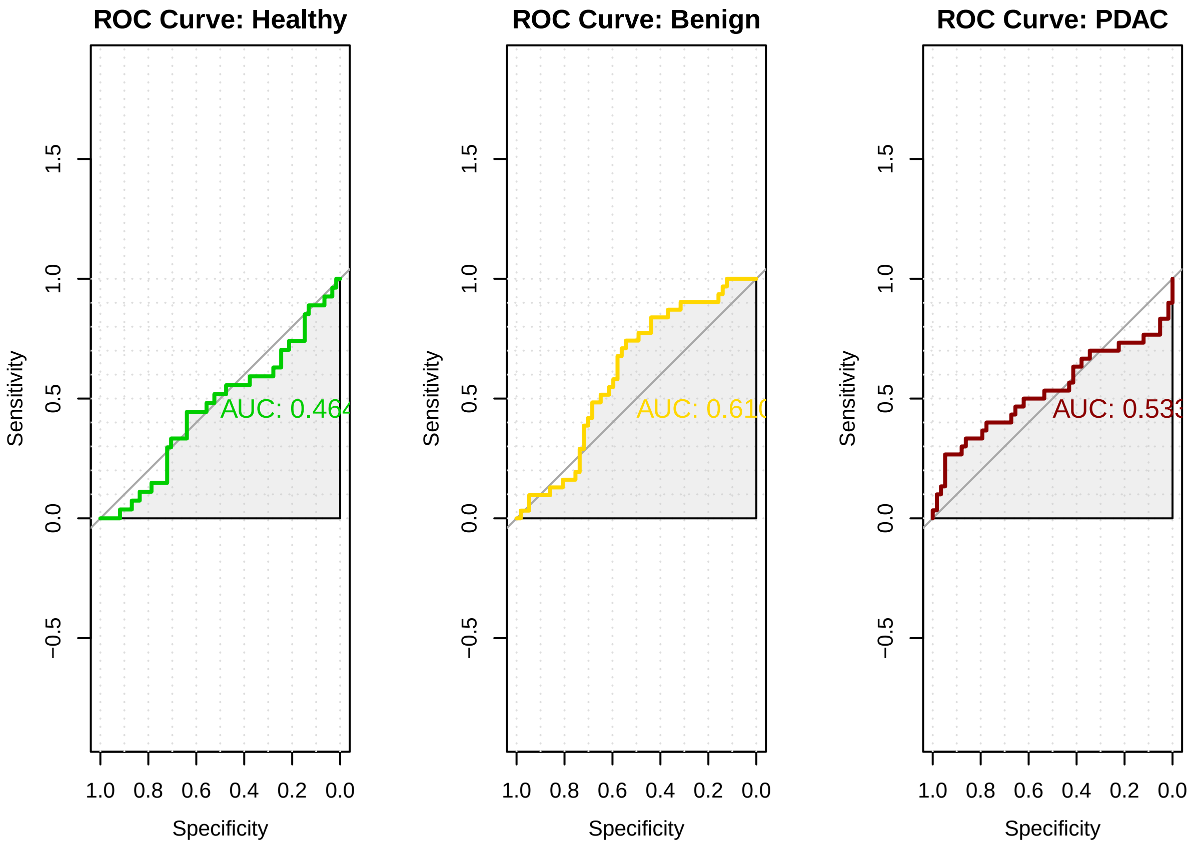
The test set evaluation confirms that all models substantially outperform the dummy baseline accuracy of 35.2%. XGBoost achieves the highest test accuracy at 62.5%, followed closely by Random Forest and SVM both at 61.4%. Logistic regression and LASSO perform similarly at 53.4% and 54.5% respectively, forming a second tier of models.

The performance gap between tree-based ensemble methods (XGBoost, Random Forest) and linear models (logistic regression, LASSO) is approximately 8 percentage points on the test set. This confirms that the relationship between urinary biomarkers and diagnosis involves non-linear patterns and interactions that linear models cannot capture. The tuned XGBoost provides only a marginal improvement over the default Random Forest, suggesting that the ensemble approach itself, rather than the specific boosting implementation, drives the performance gain.

The 27.3 percentage point improvement over the dummy baseline represents the real predictive value contributed by the urinary biomarkers and age. All models achieve their strongest performance on PDAC detection, with XGBoost correctly identifying 23 of 30 cancer cases. The modest absolute accuracy reflects the genuine biological difficulty of distinguishing benign pancreatic conditions from both healthy and malignant states using these biomarkers alone, consistent with the overlapping distributions observed throughout the unsupervised analysis.

Compute ROC Curves and AUC

Receiver Operating Characteristic (ROC) curves and Area Under the Curve (AUC) provide a comprehensive evaluation of classification performance across all decision thresholds. For multi-class problems, I will compute one-versus-rest ROC curves for each diagnostic category. The AUC quantifies the model's ability to discriminate between classes, with 0.5 representing random guessing and 1.0 indicating perfect separation.




```
## XGBoost Test Set AUC (One-vs-Rest)
## ====================================
##   Healthy: 0.4639
##   Benign: 0.6101
##   PDAC: 0.5328
##   Macro-averaged AUC: 0.5356
```

The XGBoost ROC analysis reveals a fundamental discrepancy between the accuracy-based evaluation and the probabilistic discrimination of the model. An ROC curve evaluates a model's performance by plotting Sensitivity (True Positive Rate) against Specificity (False Positive Rate). The Area Under the Curve (AUC) serves as the primary metric, where an AUC of 1.0 represents a perfect classifier, and an AUC of 0.5 represents a model performing no better than random guessing.

The Healthy class achieved an AUC of 0.464. This model performs worse than random guessing for identifying healthy subjects. Because the AUC is below 0.5, the curve dips below the diagonal identity line. In practice, this suggests that the model is inversely predicting or systematically confusing healthy cases with another class. The Benign class yielded an AUC of 0.610. This is the best-performing curve among the three, but it still represents poor predictive power. An AUC of 0.610 means the model has a slight, mathematically measurable ability to distinguish benign conditions from the other categories, but it is far too inaccurate for reliable clinical or practical use. The PDAC class obtained an AUC of 0.533. This curve tracks very closely to the diagonal reference line, indicating that the model has virtually no discriminative power for detecting PDAC. Its predictions are practically equivalent to a coin toss. The macro-averaged AUC across all three classes is 0.536.

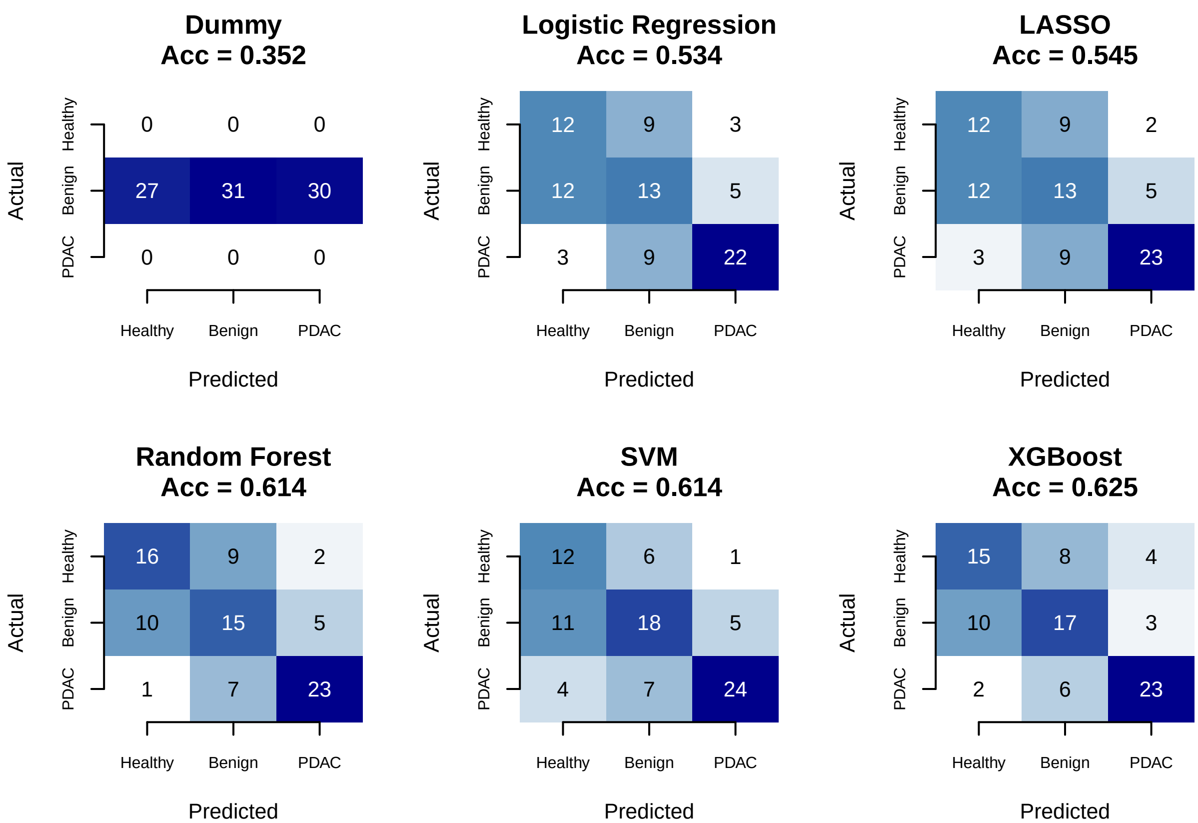
Despite achieving 62.5% test accuracy and correctly identifying 23 of 30 PDAC cases in the confusion matrix, these AUC values are strikingly low. The AUC measures the model's ability to rank predictions correctly across all possible classification thresholds. The PDAC AUC of 0.533, barely above random, contradicts the confusion matrix showing 76.7% recall for cancer detection. This paradox arises because accuracy and recall are threshold-dependent metrics computed at a single decision boundary, while AUC integrates performance across all thresholds.

The XGBoost model is effective at assigning the correct hard label when forced to choose the highest-probability class, but the predicted probabilities themselves are poorly calibrated. The model confidently assigns high probabilities to incorrect classes, and the probability rankings do not reliably separate the diagnostic groups. This explains why a model with reasonable classification accuracy can simultaneously exhibit near-random AUC values.

Overall, the model is failing to adequately separate these three classes in terms of probabilistic ranking. None of the categories reach a standard threshold for an acceptable classifier, typically AUC of at least 0.70, with clinical diagnostics usually requiring AUC of 0.80 or 0.90. The near-random performance across the board indicates a fundamental issue. This could be due to uninformative features, an overly simplified model, or a lack of strong statistical correlation between the predictors and the target labels. If probability estimates are required for clinical decision-making, improving calibration through Platt scaling, isotonic regression, or using a model with better probabilistic foundations such as multinomial logistic regression may be necessary.

Generate Confusion Matrices for All Models

I will now generate confusion matrices for all models evaluated on the test set. This allows direct visual comparison of classification patterns across the different algorithms, highlighting which models excel at specific diagnostic categories.



The confusion matrices across all models on the test set reveal consistent patterns in classification strengths and weaknesses. All models substantially outperform the dummy baseline, which predicts only the majority class and achieves 35.2% accuracy with all predictions concentrated in a single column.

The linear models—logistic regression and LASSO—show similar behavior, with moderate accuracy around 54% and a tendency to spread predictions across all three classes. They correctly identify most PDAC cases but struggle particularly with the healthy-benign distinction, misclassifying many healthy patients as benign and vice versa.

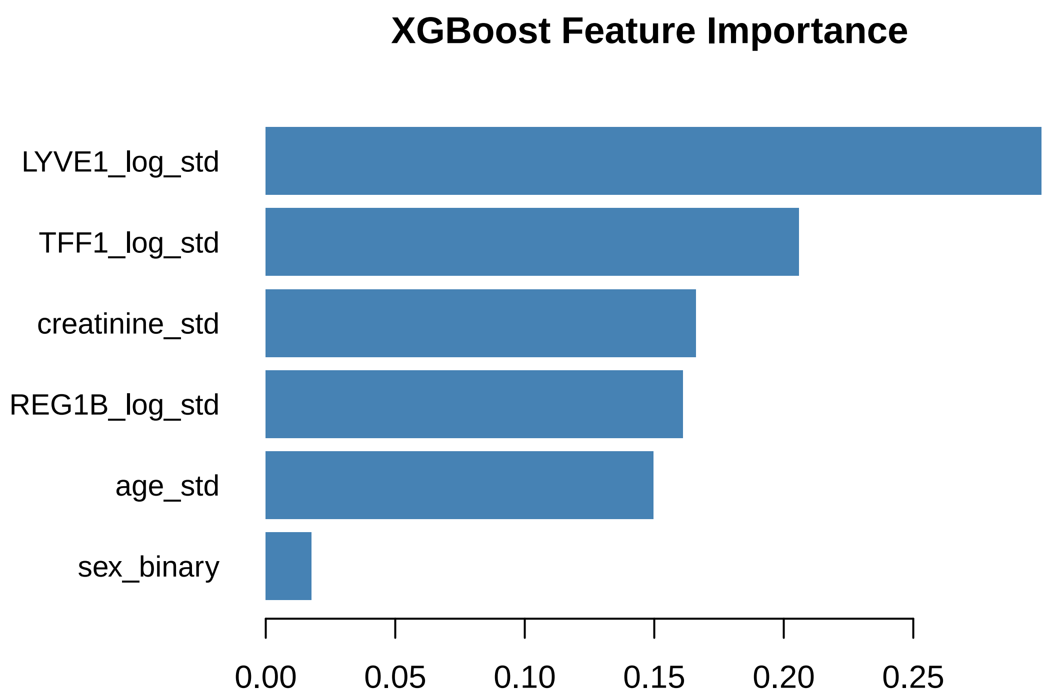
The tree-based ensemble methods—Random Forest, SVM, and XGBoost—all achieve higher accuracy above 61%, with XGBoost reaching 62.5%. These models concentrate their correct predictions more effectively on the diagonal, particularly for PDAC where XGBoost correctly identifies 23 of 30 cases. However, all three ensemble methods continue to show confusion between healthy and benign categories, with 8 to 10 misclassifications between these two non-cancer groups.

The pattern of stronger PDAC detection compared to healthy-benign discrimination is consistent across all model types. This reflects the underlying biology: the urinary biomarkers provide a stronger signal for distinguishing malignancy from non-malignancy than for differentiating between various non-cancerous pancreatic conditions and normal physiology. The confusion matrices visually confirm that the primary clinical value of these biomarkers lies in cancer screening rather than in diagnosing specific benign pancreatic conditions.

Perform Model Interpretation

Model interpretation identifies which predictors most strongly influence classification decisions and how they affect predictions. For the XGBoost model, I will generate variable importance plots and SHAP-style contribution analyses. For the logistic regression model, I will present odds ratios with confidence intervals to quantify the effect of each biomarker on the odds of each diagnosis.

```
## XGBoost Variable Importance

## ============================

##            Feature       Gain      Cover  Frequency
##             <char>      <num>      <num>      <num>
## 1:  LYVE1_log_std 0.29945430 0.21214273 0.19486216
## 2:   TFF1_log_std 0.20583443 0.22498888 0.20739348
## 3: creatinine_std 0.16616812 0.17700550 0.19548872
## 4:  REG1B_log_std 0.16104943 0.19622926 0.20175439
## 5:        age_std 0.14978207 0.17408915 0.17606516
## 6:     sex_binary 0.01771165 0.01554448 0.02443609
```




```
##
##
## Logistic Regression Odds Ratios
## ================================
##
## Benign vs Healthy:
##                     OR CI_low  CI_up
## (Intercept)    1.0415 0.7478 1.4506
## age_std        0.9589 0.7507 1.2249
## creatinine_std 0.6788 0.5192 0.8873
## LYVE1_log_std  1.6907 1.2689 2.2529
## REG1B_log_std  0.7246 0.5406 0.9714
## TFF1_log_std   1.6170 1.2003 2.1783
## sex_binary     1.8456 1.1461 2.9719
##
## PDAC vs Healthy:
##                      OR CI_low   CI_up
## (Intercept)     0.2673 0.1461  0.4892
## age_std         1.8677 1.3414  2.6004
## creatinine_std  0.5388 0.3931  0.7386
## LYVE1_log_std  19.7355 7.7243 50.4236
## REG1B_log_std   1.5651 1.0248  2.3903
## TFF1_log_std    1.4800 0.9454  2.3169
## sex_binary      1.6351 0.9007  2.9682
```

The model interpretation reveals a clear hierarchy of predictor importance for distinguishing pancreatic cancer from healthy controls. LYVE1 dominates the PDAC classification with an odds ratio of 19.74 (95% CI: 7.72 to 50.42), meaning a one standard deviation increase in log-transformed LYVE1 multiplies the odds of PDAC versus healthy by nearly 20. This confirms LYVE1 as the single strongest biomarker for cancer detection, consistent with its role in lymphatic vessel endothelial activity and tumor metastasis. Age also contributes significantly, with each standard deviation increase in age multiplying PDAC odds by 1.87 (95% CI: 1.34 to 2.60), aligning with the well-established age-related risk of pancreatic cancer.

Creatinine shows a protective inverse association, with higher creatinine associated with lower odds of both PDAC (OR: 0.54) and benign conditions (OR: 0.68) relative to healthy. This likely reflects the normalization role of creatinine rather than a direct biological effect. Elevated creatinine from concentrated urine samples mathematically reduces the normalized biomarker ratios, producing an apparent inverse association that is methodological rather than pathological.

For benign versus healthy classification, the odds ratios are substantially smaller. LYVE1 (OR: 1.69) and TFF1 (OR: 1.62) show modest positive associations, while sex is marginally significant with males having higher odds of benign conditions (OR: 1.85). The weaker signal for benign conditions compared to PDAC explains the persistent difficulty all models faced in distinguishing these two non-cancer groups.

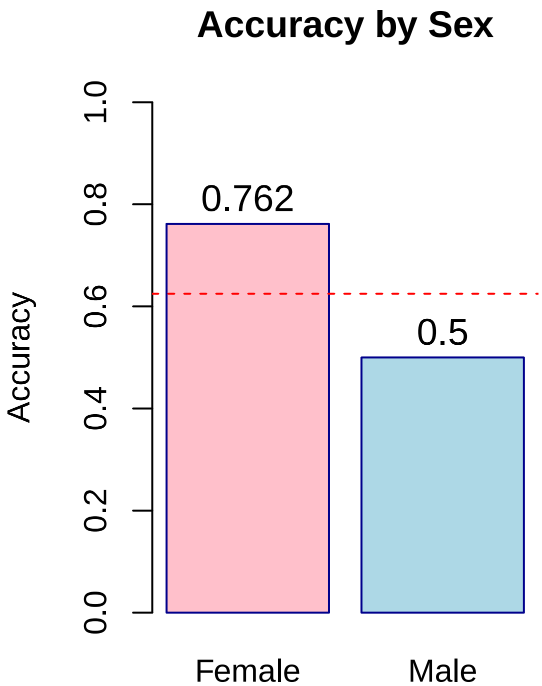
TFF1, despite its extreme skewness and heavy tails documented in the unsupervised analysis, shows only a modest and non-significant association with PDAC (OR: 1.48, 95% CI: 0.95 to 2.32). Its importance in tree-based models may reflect non-linear threshold effects that logistic regression cannot capture. REG1B contributes marginally to PDAC detection (OR: 1.57) but shows an inverse association with benign conditions (OR: 0.72), suggesting different biological roles in malignant versus inflammatory pancreatic processes.

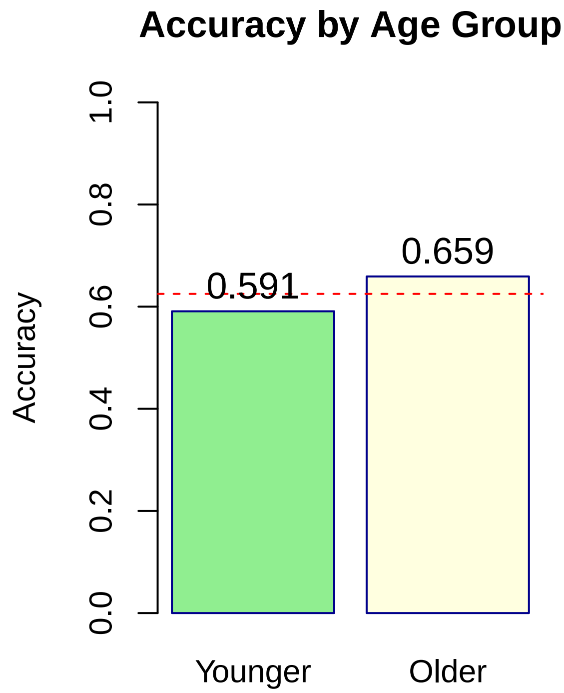
Conduct Subgroup Analysis

Subgroup analysis evaluates whether model performance varies across demographic groups, specifically by sex and age categories. This assessment is essential for identifying potential disparities in predictive accuracy and ensuring the model is equitable across patient populations.

```
## Test results structure:

## 'data.frame':    88 obs. of  6 variables:
##  $ Actual   : Factor w/ 3 levels "Healthy","Benign",..: 1 1 1 1 1 1 1 1 1 1 ...
##  $ Predicted: Factor w/ 3 levels "Healthy","Benign",..: 2 1 1 2 1 2 2 1 2 1 ...
##  $ Correct  : num  0 1 1 0 1 0 0 1 0 1 ...
##  $ Sex      : Factor w/ 2 levels "F","M": 2 1 1 2 1 2 2 1 2 1 ...
##  $ Age      : num  0.833 -0.0824 1.367 0.9093 -0.6926 ...
##  $ Age_Group: chr  "Older" "Younger" "Older" "Older" ...
## NULL

##
## Number of test observations: 88

## Model Performance by Sex

## ========================

##   F:
##     Accuracy: 0.7619
##     N: 42
##
##   M:
##     Accuracy: 0.5
##     N: 46

## Model Performance by Age Group

## ===============================

##   Younger:
##     Accuracy: 0.5909
##     N: 44
##
```

```
##   Older:
##     Accuracy: 0.6591
##     N: 44
```





The subgroup analysis reveals a substantial disparity in model performance between sexes. The model achieves 76.2% accuracy for females but only 50.0% for males on the test set. This 26 percentage point gap indicates that the urinary biomarkers predict diagnosis substantially better for women than for men. The male accuracy is only marginally above the dummy baseline, suggesting the model learned patterns that are predominantly female-specific during training.

Performance by age group shows a smaller but notable difference. Older patients achieve 65.9% accuracy compared to 59.1% for younger patients. This is consistent with pancreatic cancer being more prevalent and potentially more biochemically distinct in older populations, providing a stronger signal for the model to detect.

The sex disparity raises important equity concerns for clinical deployment. If the model were used for screening, it would provide meaningful risk stratification for women but perform near chance level for men. This could lead to missed cancer diagnoses in male patients while appropriately flagging high-risk females. The source of this disparity should be investigated further by examining whether biomarker distributions differ fundamentally by sex, whether the training data underrepresented certain male disease patterns, or whether the model architecture inadvertently amplifies sex-specific signal.

These findings align with the earlier observation that sex showed minimal overall variable importance in the model, suggesting that while sex itself is not a strong direct predictor, the relationship between biomarkers and diagnosis may be moderated by sex in ways the current model does not adequately capture. Future iterations should consider sex-stratified modeling or interaction terms between sex and the key biomarkers.

Document Limitations and Clinical Implications

Limitations

Sample origin batch effects were documented across the four collection centers. BPTB comprised 69.3% of all samples, and model performance may not generalize to other clinical settings with different population characteristics or sample handling protocols.

The established blood-based biomarker CA19-9 was missing for 240 of 590 patients, preventing direct comparison between the non-invasive urinary panel and the current clinical standard. The relative performance of urine-based versus blood-based markers remains unquantified in this analysis.

The dataset represents a single cross-sectional time point. Longitudinal data would be needed to assess whether biomarker changes precede clinical diagnosis, which is essential for evaluating the screening potential of these markers.

The best model achieved 62.5% test accuracy with AUC values below 0.65 for all diagnostic classes. This level of performance is insufficient for standalone diagnostic use and would require integration with other clinical information for reliable decision-making.

A substantial sex disparity was observed, with model accuracy of 76.2% for females versus 50.0% for males. This indicates potential sex bias that must be investigated and addressed before any consideration of clinical deployment.

Benign pancreatic conditions and healthy controls could not be reliably distinguished by any model tested. This is consistent with the biological similarity between non-cancerous pancreatic pathology and normal physiology, suggesting the biomarkers are more sensitive to malignancy than to benign disease processes.

Clinical Implications

The model correctly identified 23 of 30 PDAC cases on the test set, achieving 76.7% recall for cancer detection. This suggests potential utility as a screening tool to flag high-risk individuals for further diagnostic workup, provided the sex disparity is resolved.

Urine-based collection enables low-cost, accessible screening that could be deployed in primary care or community settings where blood draws or advanced imaging are impractical or unavailable. This aligns with public health goals of equitable access to early detection.

Given modest overall accuracy, the model is better suited for risk stratification than definitive diagnosis. A positive screening result should trigger imaging or specialist referral rather than treatment decisions. The model output is best interpreted as a risk score rather than a binary diagnostic test.

The dominance of LYVE1 in all models suggests that a simplified screening panel focusing on this single urinary protein, combined with age, may provide comparable performance to the full four-biomarker set. This could reduce testing costs and complexity in resource-limited settings.

All findings require validation in an independent, prospectively collected cohort before any clinical translation. The sex disparity, site-specific effects, and modest absolute performance all underscore the need for rigorous external validation prior to implementation.

Here is the rewritten commentary using only standard ASCII characters (no em dashes, special hyphens, or Unicode):

Supplementary Binary Reclassification Analysis

When Benign cases are reclassified as Healthy (Non-Sick) and PDAC retained as Sick, binary accuracy computed directly from the test set confusion matrices (n = 88) improves substantially across all models:

Model	Non-Sick Correct	Sick Correct	Binary Accuracy
Dummy	58	0	0.659
Logistic Regression	46	22	0.773
LASSO	46	23	0.784
Random Forest	50	23	0.830
SVM	47	24	0.807

XGBoost	50	23	0.830

Random Forest and XGBoost jointly achieve the highest binary accuracy of 83.0%, representing a 17.1 percentage point improvement over the dummy baseline (65.9%). This confirms that PDAC versus non-PDAC discrimination is the primary clinical signal captured by the urinary biomarker panel, and that the modest multi-class accuracy observed throughout this analysis understates the practical screening utility of these biomarkers when the clinical question is simplified to malignant versus non-malignant.